



WINDOWS FORENSICS

MODULE-6

Aniket Sunil Pagare

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-

What is Windows Forensics?

Windows forensics involves the analysis and investigation of Windows operating systems to uncover digital evidence related to security incidents, crimes, or system misuse. It is widely used in incident response, cybercrime investigations, and malware analysis.

Core Areas of Windows Forensics

1. File System Forensics (NTFS, FAT32, exFAT)

- **NTFS features:**
 - MFT (Master File Table)
 - Alternate Data Streams (ADS)
 - File timestamps (MAC times: Modified, Accessed, Created)
 - USN Journal (record of changes)
 - **Tools:** FTK Imager, Autopsy, Sleuth Kit
-

2. Registry Forensics

- Windows Registry stores system and user configuration.
 - Key hives:
 - NTUSER.DAT (user activity)
 - SYSTEM, SOFTWARE, SAM, SECURITY
 - Tracks USB insertions, user logins, recently opened files, etc.
 - **Tools:** Registry Explorer, RegRipper, RECcmd
-

3. User Activity Artifacts

- Recent files: Recent, Jumplists, ShellBags
- Browser history: Chrome, Edge, Firefox
- Prefetch files: help determine app execution (C:\Windows\Prefetch)

- LNK (shortcut) files
 - Event Logs (.evtx)
-

4. Memory (RAM) Forensics

- Volatile data (running processes, network connections, etc.)
 - **Tools:**
 - **LiME:** RAM acquisition on Linux
 - **WinPMEM:** Windows memory acquisition
 - **Volatility / Rekall:** Analysis
-

5. Event Log Analysis

- Stored in .evtx files under C:\Windows\System32\winevt\Logs
 - Event IDs:
 - 4624: Successful login
 - 4625: Failed login
 - 4672: Admin privileges used
 - 1102: Audit log cleared
 - **Tools:** Event Viewer, LogParser, Windows Event Log Explorer
-

6. Network Forensics

- Analyzing Windows network activity:
 - IP addresses, active connections
 - Firewall logs
 - ARP cache
 - **Tools:** Wireshark, NetworkMiner, Netstat
-

7. Malware Analysis & Detection

- Detect persistence mechanisms:
 - Registry Run Keys
 - Scheduled tasks
 - Services
 - Unpack PE files using **Detect It Easy, PEStudio, Exeinfo PE**
-

8. Email Forensics

- Outlook PST/OST files
 - Thunderbird MBOX
 - Email headers, attachments, spoofing detection
-

9. USB Forensics

- Registry and event log entries to detect USB connections
 - Key Paths:
 - SYSTEM\CurrentControlSet\Enum\USBSTOR
 - SYSTEM\MountedDevices
 - **Tools:** USBDeview, USB Detective
-

10. Forensic Imaging & Analysis

- Tools: FTK Imager, EnCase, X-Ways, Magnet AXIOM
 - Disk imaging formats: E01, DD/RAW
-

11. Password & Encryption Artifacts

- Hashes stored in SAM file (can be extracted using pwdump)
 - Application password cracking: PDF, ZIP, Word files using **John the Ripper, Passware Toolkit**
-

12. Anti-Forensics Detection

- Detect data obfuscation, timestamp manipulation
 - Hidden files via ADS or steganography
 - Tools: ADS Scanner, Stegdetect, ExifTool
-

13. Timeline Analysis

- Combine MAC times, registry, logs, and prefetch into a timeline
 - Tools: log2timeline, Plaso, Autopsy timeline view
-

14. Tools Summary

Category	Tools
Disk Imaging	FTK Imager, X-Ways, EnCase
Memory Analysis	Volatility, Rekall
Registry	Registry Explorer, RegRipper
Logs	Event Viewer, LogParser
Artifacts	Autopsy, KAPE, Magnet AXIOM
Network	Wireshark, NetworkMiner
Anti-Forensics	ADS Scanner, ExifTool



Common Use Cases

- Investigating ransomware attacks
 - Insider threat analysis
 - Data theft (via USB or network)
 - Malicious persistence detection
 - Application misuse tracking
-

Practical Labs to Try

- Use FTK Imager to acquire a disk image
- Use Volatility to analyze a memory dump
- Extract USB history via registry
- Analyze Prefetch & Jumplists for executed programs

Lab-1-: Acquire Volatile Information from Live windows system

Acquiring volatile information from a live Windows system is a critical step in **incident response** and **digital forensics**. It involves collecting data that resides in **RAM or other temporary memory**, which would be lost upon shutdown or reboot.

What is Volatile Information?

Volatile data is information that:

- Exists only while the system is powered on
- Disappears when the machine is shut down
- Cannot be recovered after reboot

Examples of Volatile Information

Volatile Data	Description
RAM (Memory)	Running processes, passwords in memory, malware
Network Connections	Active TCP/UDP connections, open ports
Logged-in Users	Who is currently logged in

Volatile Data	Description
Running Processes	Services, executables, command-line args
Open Files	Files currently being accessed
Clipboard Contents	Recently copied data
System Time	May be tampered in case of attacks
ARP Cache	List of recently resolved IP/MAC addresses
Routing Table	Shows system's network paths
Unloaded Drivers	Useful for detecting rootkits
Command Line History	Recent user actions (if accessible)

A) Acquire Volatile Information from Live windows system Using Ps tool

PsTools is a suite of command-line utilities developed by **Microsoft Sysinternals**. It is widely used in **Windows forensics, incident response, and volatile memory collection** from live systems.

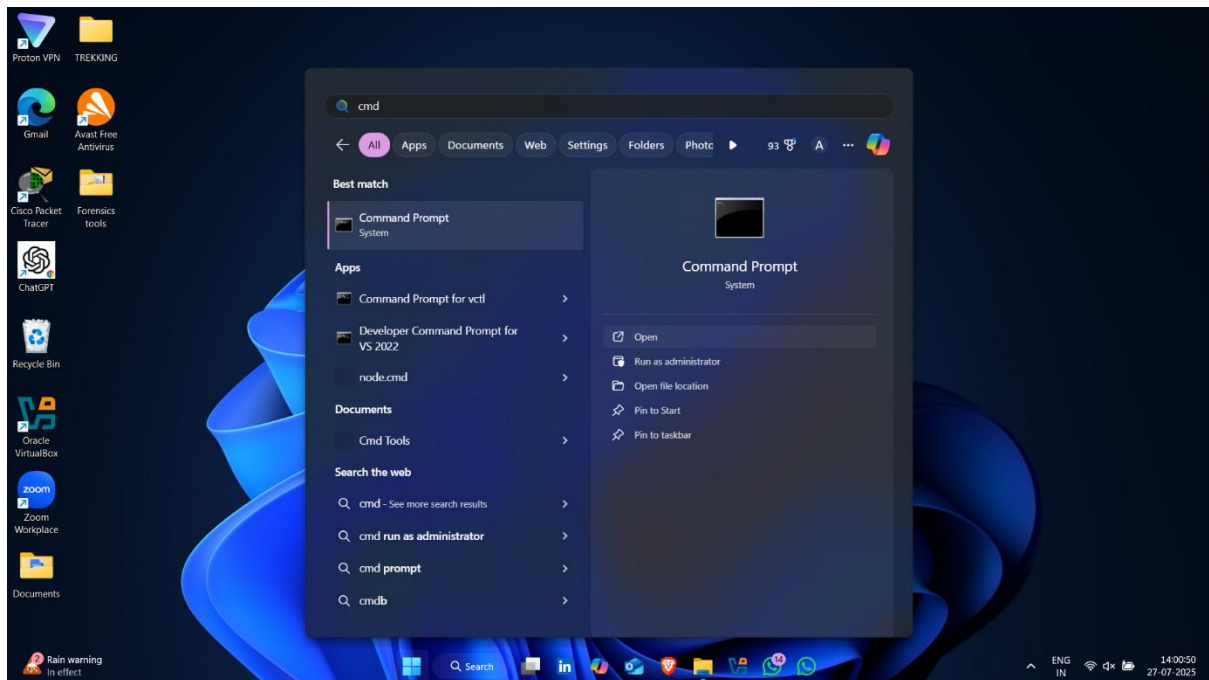
What is PsTools?

PsTools is a powerful collection of tools for system administration, **remote system interaction, and live data acquisition** from Windows systems without requiring GUI access.

Download link -: <https://learn.microsoft.com/en-us/sysinternals/downloads/pstools>

How to use it -:

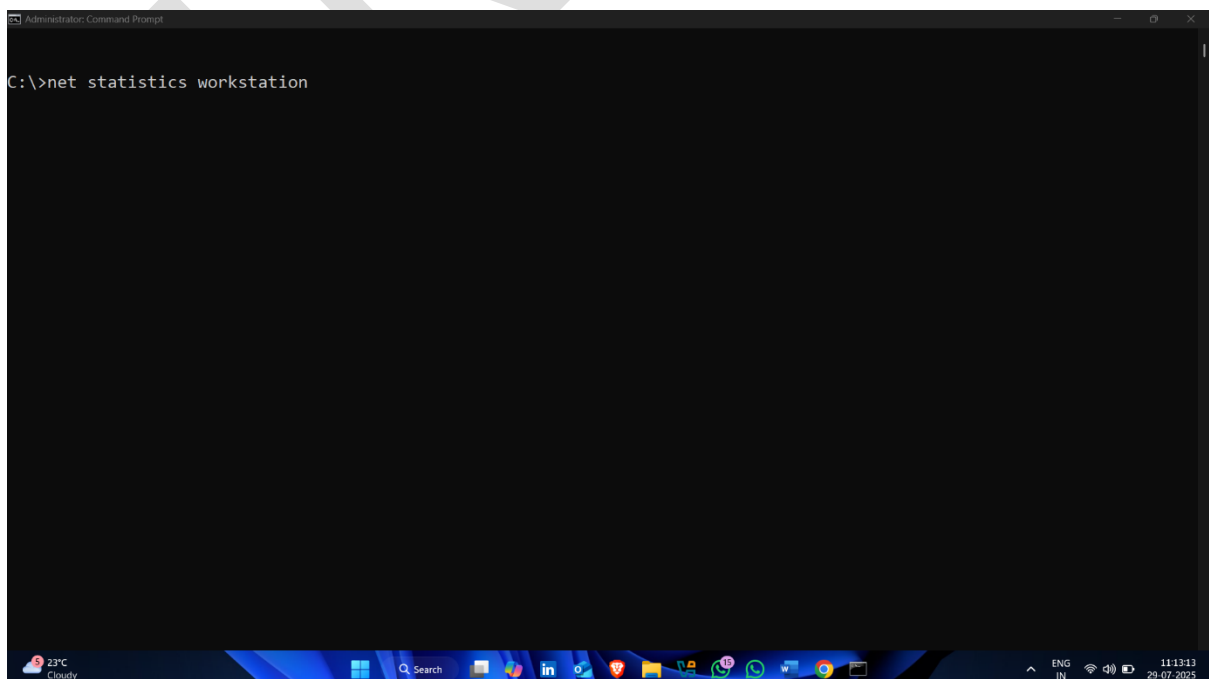
- Open command line interface and **run as a administrator**



- type following command

command -: net statistics workstation

Explanation -: Uptime of the Workstation service (often used to estimate system uptime).



- System Uptime 🖐

```
Select Administrator: Command Prompt

C:\>net statistics workstation
Workstation Statistics for \\HP

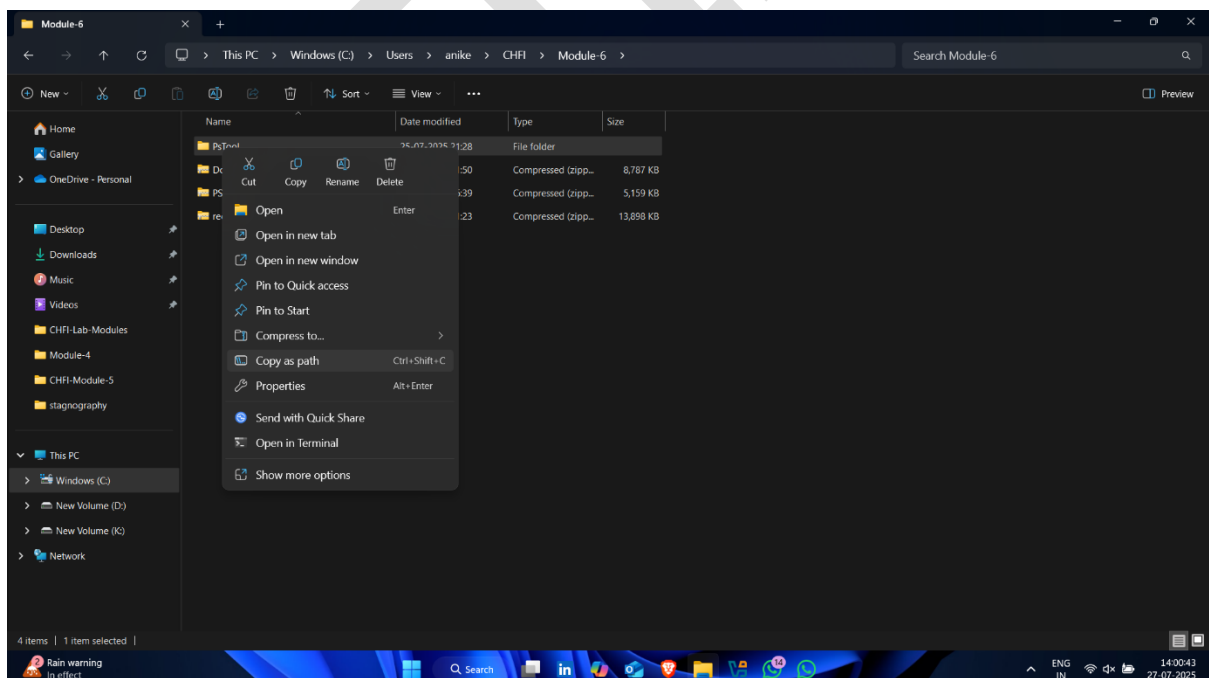
Statistics since 29-07-2025 10:37:44

Bytes received                                0
Server Message Blocks (SMBs) received         4
Bytes transmitted                             0
Server Message Blocks (SMBs) transmitted      0
Read operations                              0
Write operations                              0
Raw reads denied                             0
Raw writes denied                             0

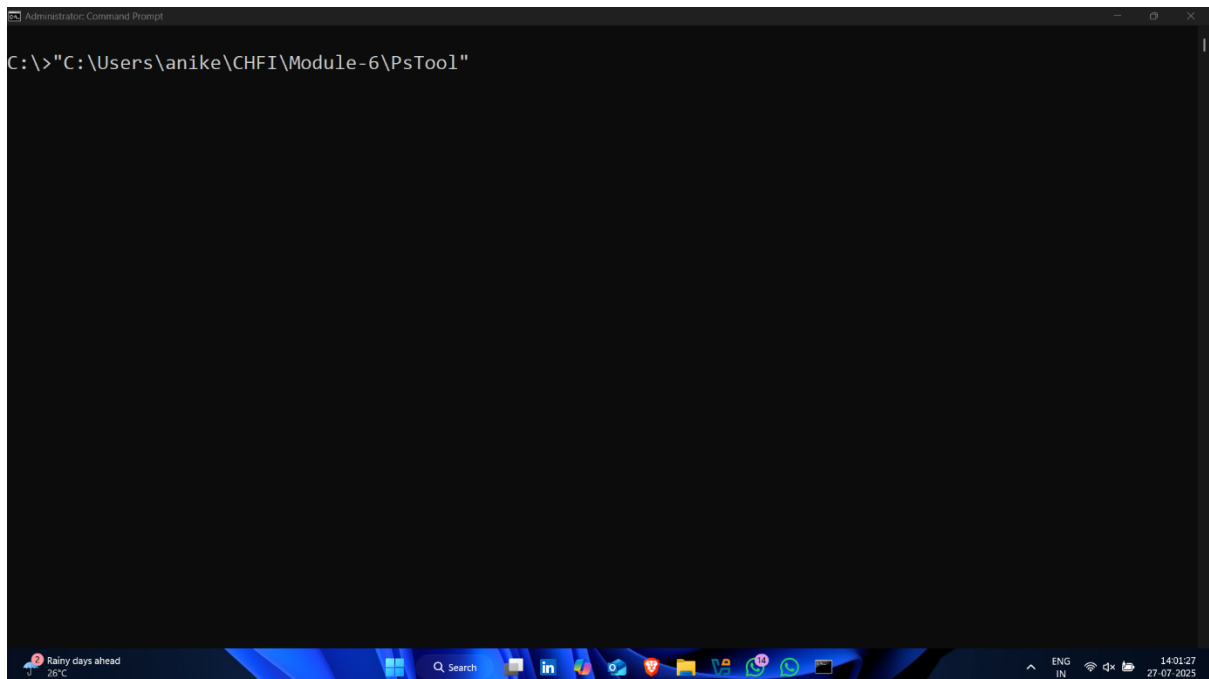
Network errors                                0
Connections made                             0
Reconnections made                           0
Server disconnects                           0

Sessions started                             0
Hung sessions                                0
Failed sessions                              0
Failed operations                             0
Use count                                     0
Failed use count                             0
```

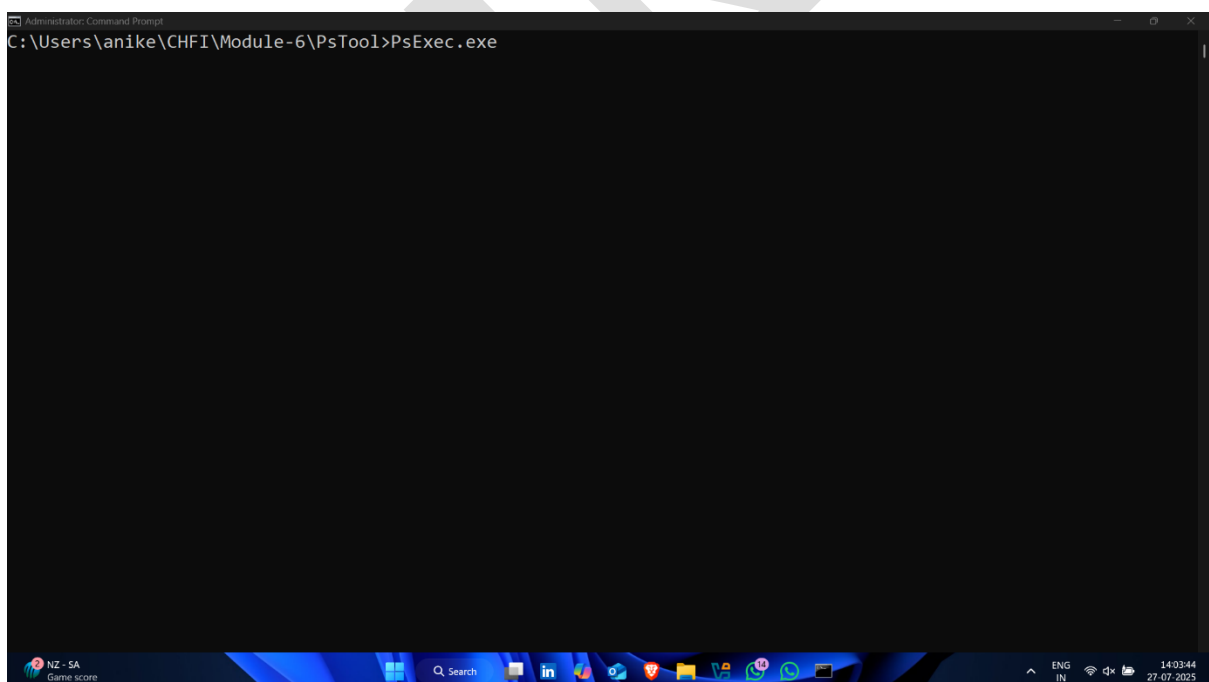
- Now go to the **ps tool folder** , right click on folder and click on **copy as a path**



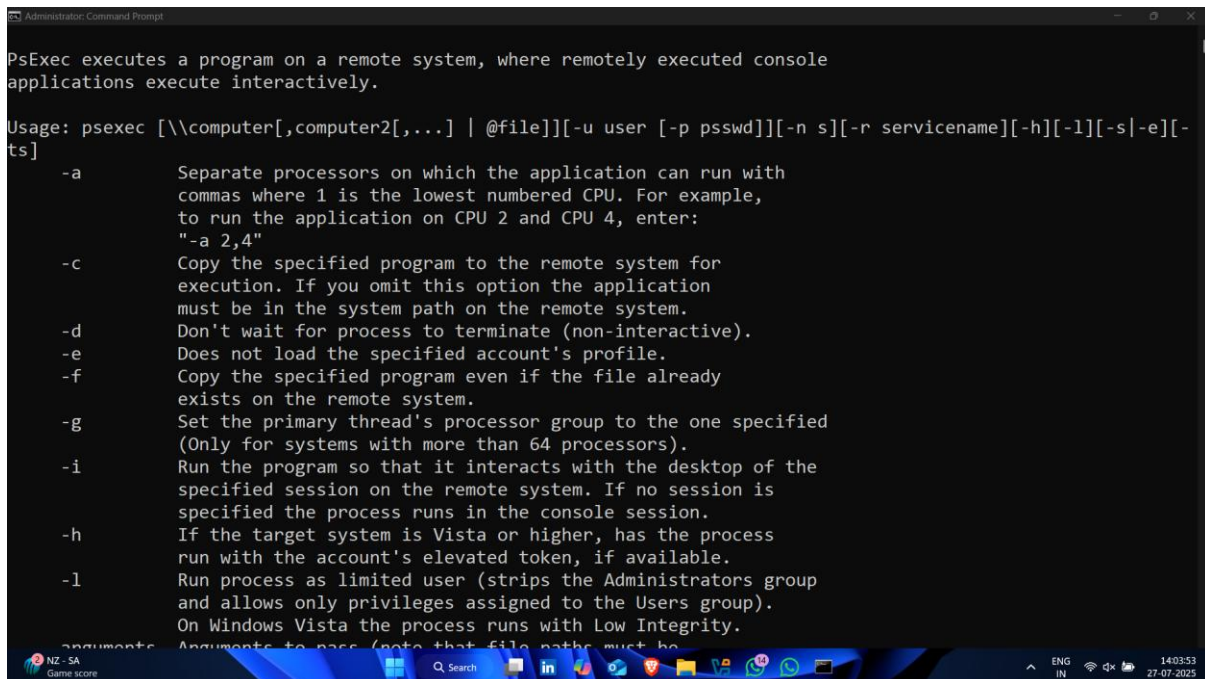
- **And paste on CMD**

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt shows the command `C:\>"C:\Users\anike\CHFI\Module-6\Pstool"` entered. The taskbar at the bottom shows the Windows logo, search bar, and several application icons. The system tray on the right indicates the language is "ENG IN", the date is "27-07-2025", and the time is "14:01:27". A weather widget on the left shows "Rainy days ahead" and "26°C".

- **Now run **PsExec.exe** file**

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt shows the command `C:\Users\anike\CHFI\Module-6\Pstool>PsExec.exe` entered. The taskbar at the bottom shows the Windows logo, search bar, and several application icons. The system tray on the right indicates the language is "ENG IN", the date is "27-07-2025", and the time is "14:03:44". A weather widget on the left shows "NZ - SA" and "Game score".

- All about Pstool



```
Administrator: Command Prompt

PsExec executes a program on a remote system, where remotely executed console
applications execute interactively.

Usage: psexec [\\computer[,computer2[,...]] @file][ -u user [-p psswd]][ -n s][ -r servicename][ -h][ -l][ -s][ -e][ -
ts]

-a          Separate processors on which the application can run with
            commas where 1 is the lowest numbered CPU. For example,
            to run the application on CPU 2 and CPU 4, enter:
            "-a 2,4"

-c          Copy the specified program to the remote system for
            execution. If you omit this option the application
            must be in the system path on the remote system.

-d          Don't wait for process to terminate (non-interactive).

-e          Does not load the specified account's profile.

-f          Copy the specified program even if the file already
            exists on the remote system.

-g          Set the primary thread's processor group to the one specified
            (Only for systems with more than 64 processors).

-i          Run the program so that it interacts with the desktop of the
            specified session on the remote system. If no session is
            specified the process runs in the console session.

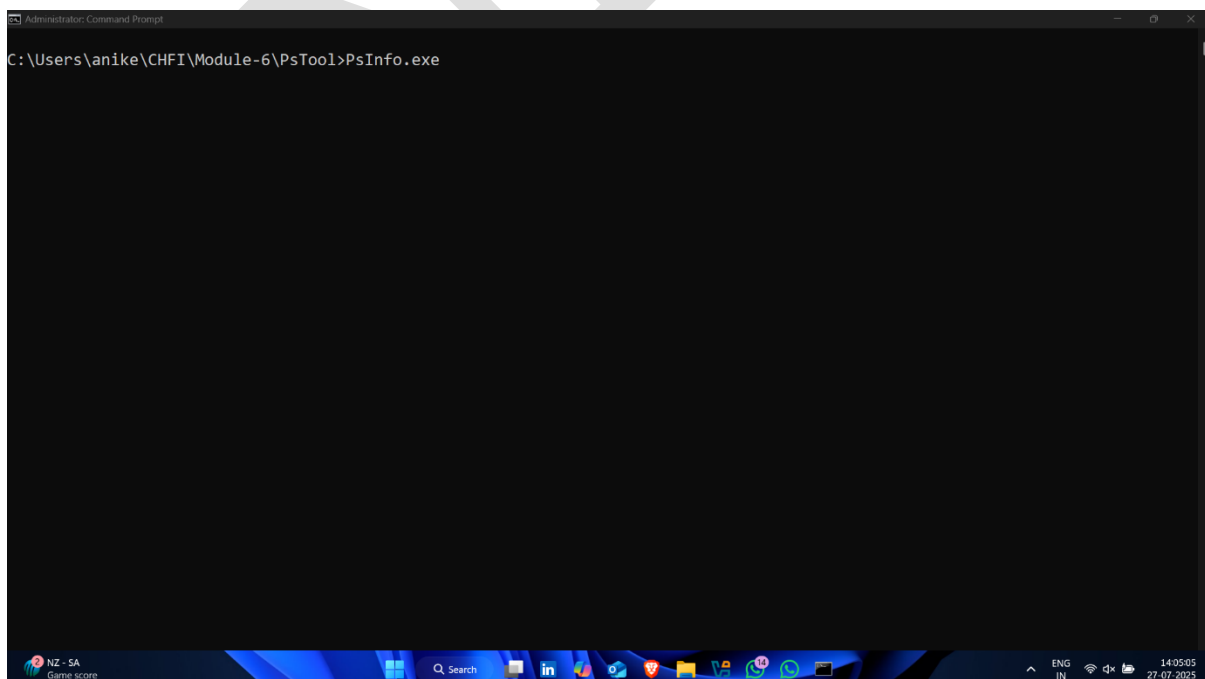
-h          If the target system is Vista or higher, has the process
            run with the account's elevated token, if available.

-l          Run process as limited user (strips the Administrators group
            and allows only privileges assigned to the Users group).
            On Windows Vista the process runs with Low Integrity.

Arguments: Arguments to pass (note that file paths must be
```

- Enter next command – PsInfo.exe

Explanation :- PsInfo.exe is a command-line tool from Microsoft's Sysinternals Suite that gathers and **displays detailed system information** about a Windows machine — both local and remote.



```
Administrator: Command Prompt

C:\Users\anike\CHFI\Module-6\PsTool>PsInfo.exe
```

- System information ✓👉

```
C:\Users\anike\CHFI\Module-6\PsTool>PsInfo.exe

PsInfo v1.79 - Local and remote system information viewer
Copyright (C) 2001-2023 Mark Russinovich
Sysinternals - www.sysinternals.com

System information for \\HP:
Uptime:                0 days 0 hours 8 minutes 35 seconds
Kernel version:        Windows 10 Home, Professional x64
Product type:          Professional
Product version:        6.3
Service pack:          0
Kernel build number:    26100
Registered organization: HP Inc.
Registered owner:       ani...
IE version:             9.00
System root:            C:\Windows
Processors:             8
Processor speed:        2.5 GHz
Processor type:         12th Gen Intel(R) Core(TM) i7-12700H
Physical memory:        32 GB
Video driver:           Intel(R) Iris Xe Graphics

C:\Users\anike\CHFI\Module-6\PsTool>
```

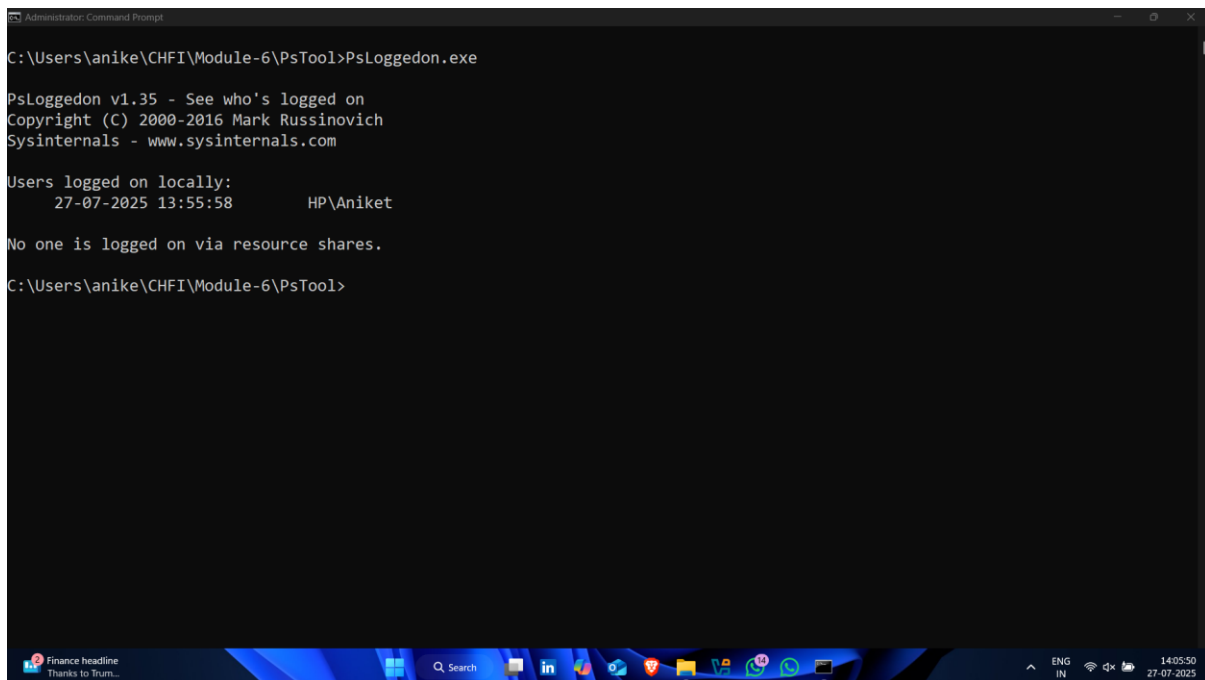
- Enter Next command – **PsLoggedon.exe**

Explanation -: PsLoggedOn.exe is a Sysinternals tool (from Microsoft) used to determine who is currently logged on to a local or remote Windows system — either locally or through network sessions (like SMB or RDP).

```
Administrator: Command Prompt

C:\Users\anike\CHFI\Module-6\PsTool>PsLoggedon.exe
```

- **Logged on Time** 📌



```
Administrator: Command Prompt
C:\Users\anike\CHFI\Module-6\Pstool>PsLoggedon.exe

PsLoggedon v1.35 - See who's logged on
Copyright (C) 2000-2016 Mark Russinovich
Sysinternals - www.sysinternals.com

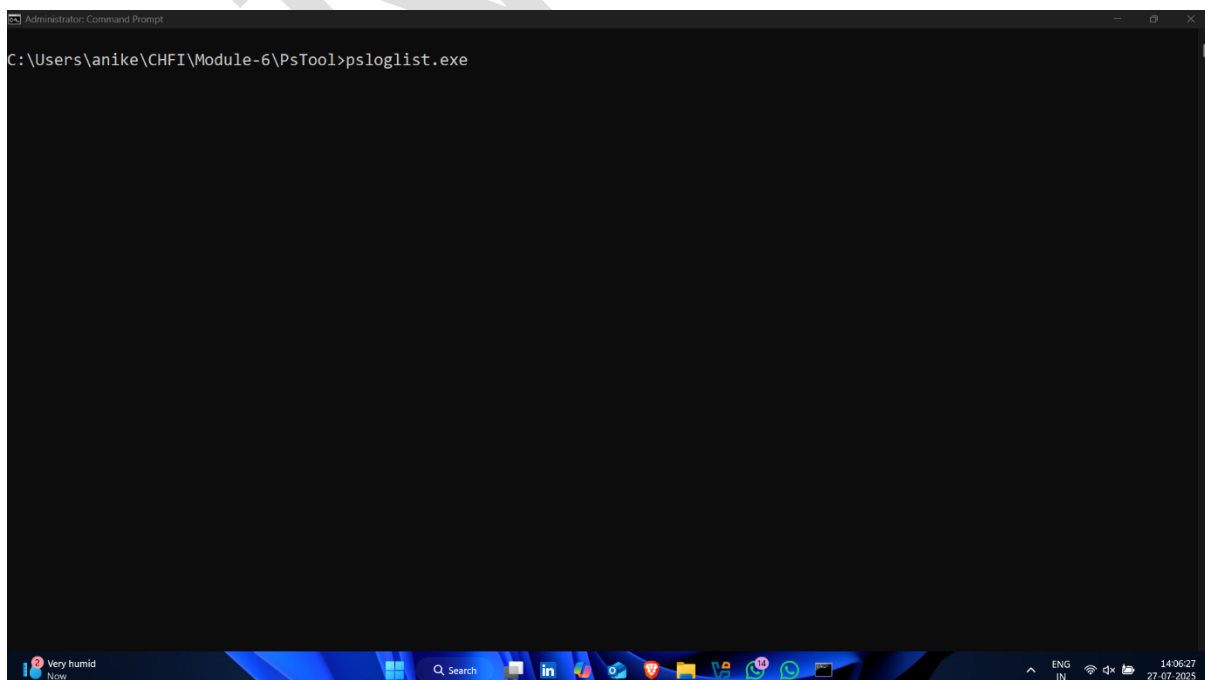
Users logged on locally:
    27-07-2025 13:55:58      HP\Aniket

No one is logged on via resource shares.

C:\Users\anike\CHFI\Module-6\Pstool>
```

- Next command – **psloglist.exe**

Explanation -: PsLogList.exe is a Sysinternals command-line tool used to **view and export Windows Event Logs** (like Application, System, and Security logs) from local or remote systems. It helps in incident response, system monitoring, and Windows forensics.



```
Administrator: Command Prompt
C:\Users\anike\CHFI\Module-6\Pstool>psloglist.exe
```

- Windows Logs

```
Administrator: Command Prompt
User: SYSTEM\NT AUTHORITY
Message text not available. Insertion strings:
165 \??\C:\ProgramData\Packages\Microsoft.WindowsNotepad_8wekyb3d8bbwe\S-1-5-21-581738745-1260712542-106455126-1001\Sy

[149463] Microsoft-Windows-Kernel-General
Type: INFORMATION
Computer: HP
Time: 27-07-2025 13:59:33 ID: 16
User: SYSTEM\NT AUTHORITY
Message text not available. Insertion strings:
97 \??\C:\Users\anike\AppData\Local\Packages\Microsoft.ZuneMusic_8wekyb3d8bbwe\Settings\settings.dat 2 1

[149462] Service Control Manager
Type: INFORMATION
Computer: HP
Time: 27-07-2025 13:59:07 ID: 7040
User: SYSTEM\NT AUTHORITY
The start type of the Background Intelligent Transfer Service service was changed from demand start to auto start.

[149461] Microsoft-Windows-IsolatedUserMode
Type: INFORMATION
Computer: HP
Time: 27-07-2025 13:58:37 ID: 5
User: Aniket\HP
Secure Trustlet NULL Id 0 and Pid 0 started with status 0.

[149460] Microsoft-Windows-IsolatedUserMode
Type: INFORMATION
Computer: HP
Time: 27-07-2025 13:58:37 ID: 2
```

Lab-2-: Investigate Forensics Image Of Windows Ram

Investigating the forensic image of Windows RAM refers to **analyzing a memory dump** (RAM image) from a live or compromised system to identify malicious activity, user actions, running processes, and other volatile data that disappears after shutdown.

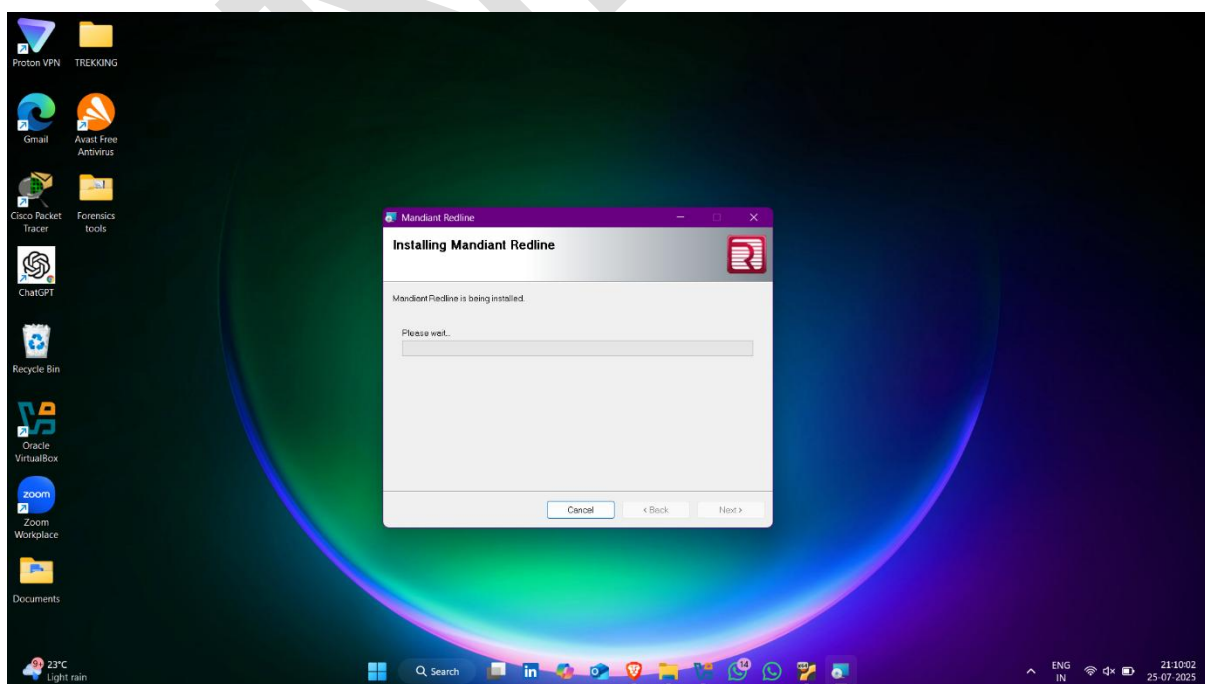
A) Investigate Forensics Image Of Windows Ram Using Redline

Redline is a **free memory forensic tool** developed by **Mandiant** (now part of **Google Cloud**). It is primarily used for **collecting and analyzing volatile data (RAM)** from Windows systems to detect signs of compromise, malware, or unauthorized access.

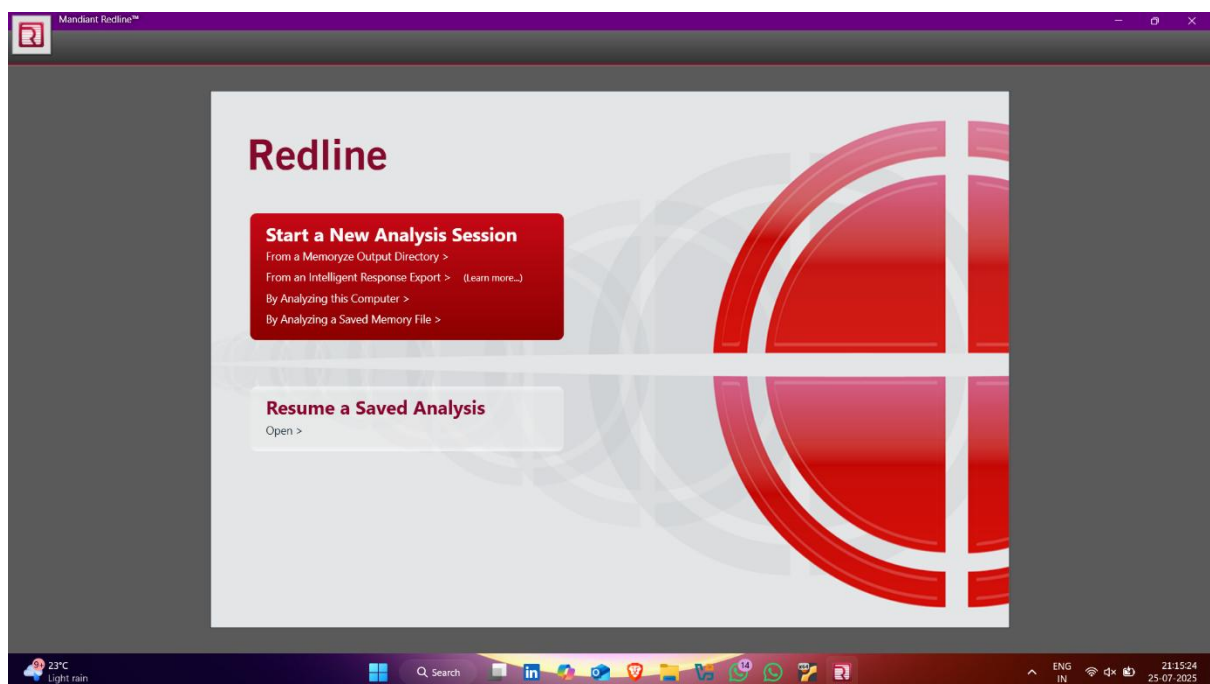
Download Link -: <https://mandiant-redline.software.informer.com/download/#downloading>

How to use it -:

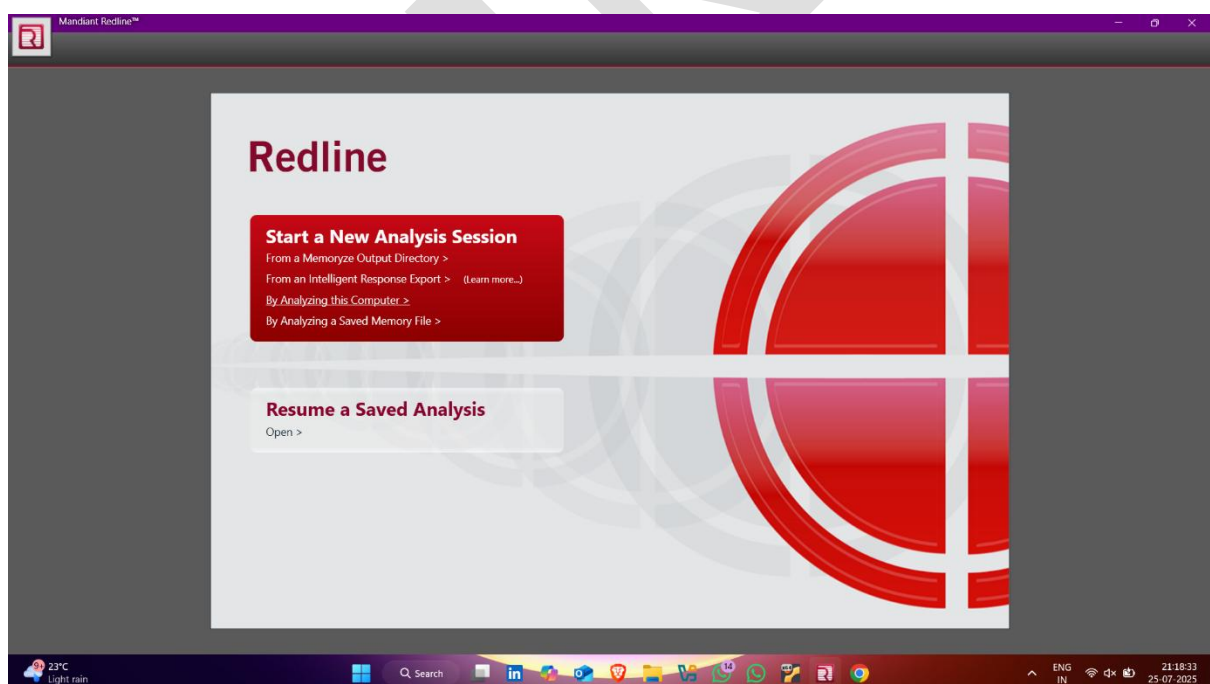
- Open application



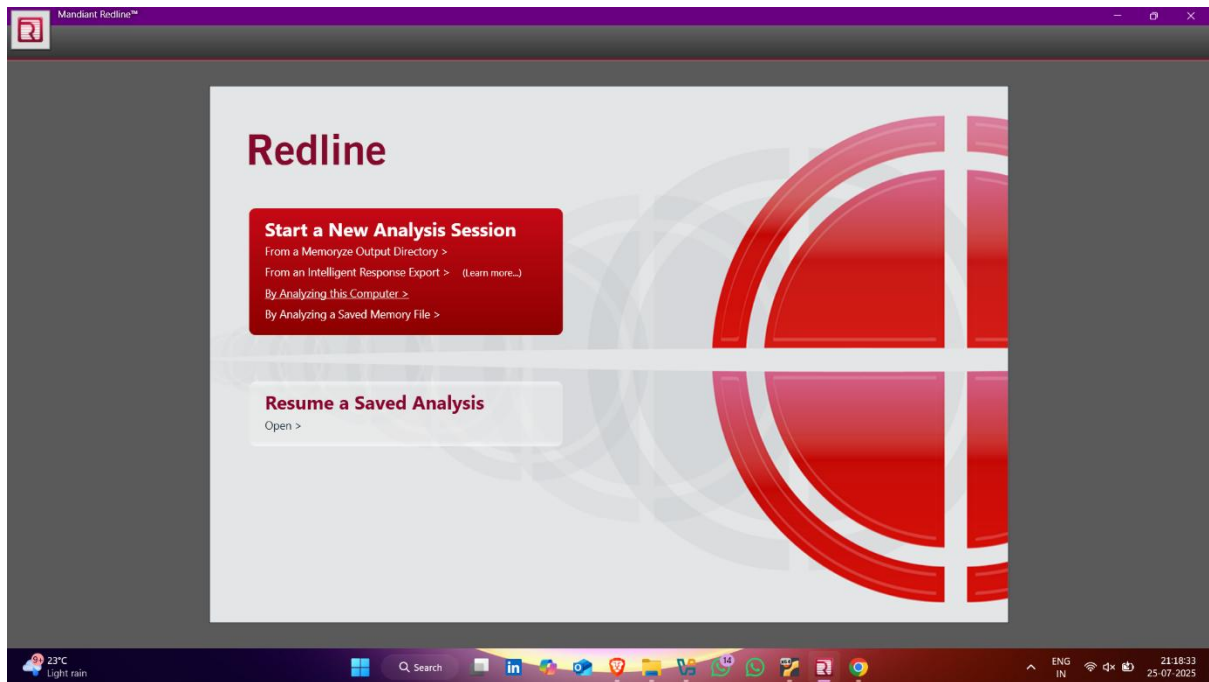
- Redline interface



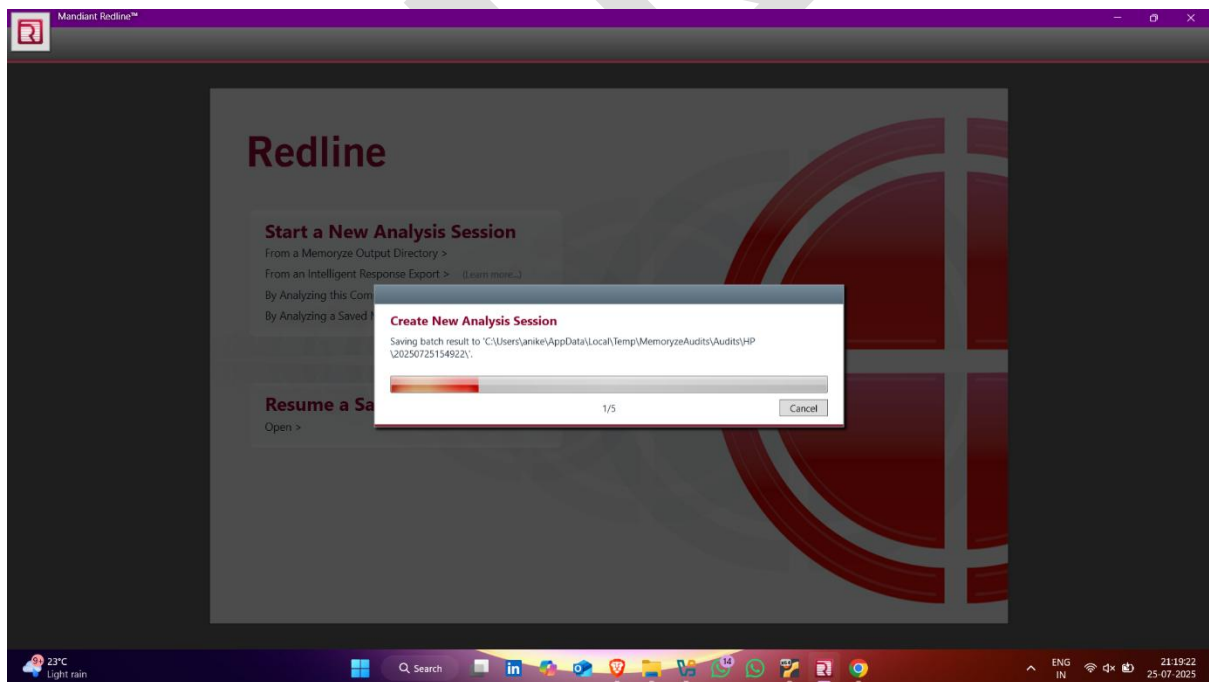
- Click on – By Analyzing computer



- Wait for few seconds



- Ram capturing started 🖱️✅



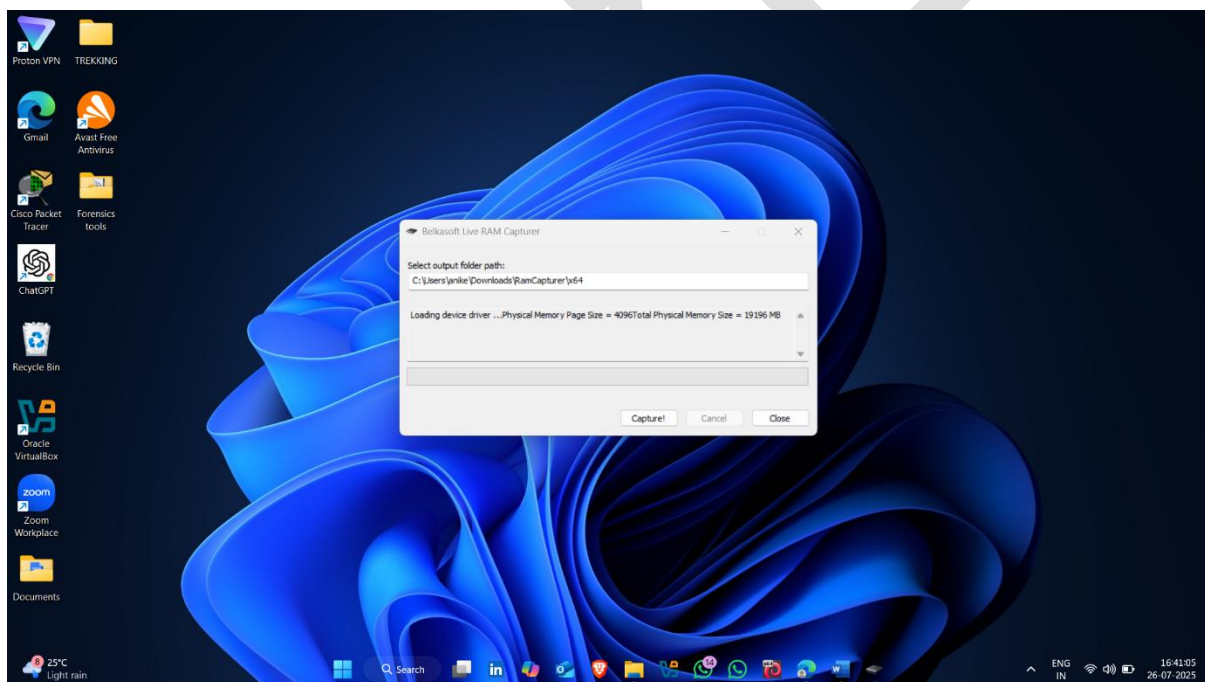
B) Investigate Forensics Image Of Windows Ram Using Belkasoft Ram capturer

Belkasoft RAM Capturer is a **free tool** designed to **capture the volatile memory (RAM)** of a live Windows system. It's widely used by **digital forensics professionals, incident responders, and malware analysts** to acquire memory dumps for post-investigation.

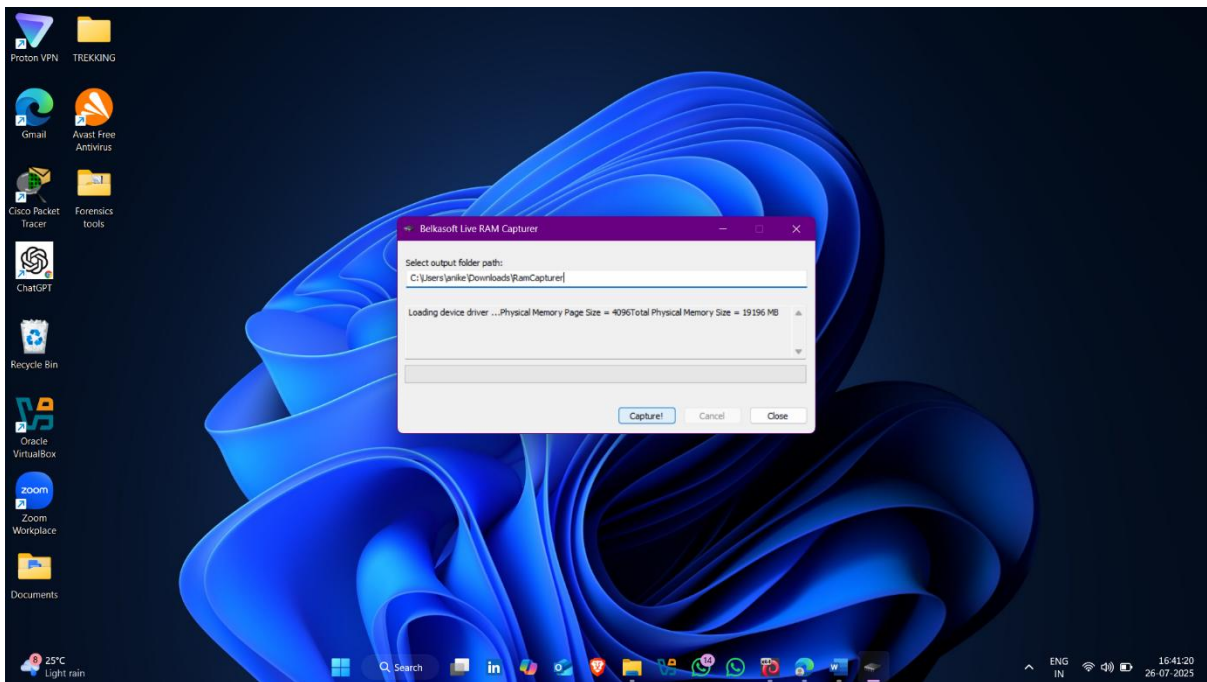
Download link -: <https://www.softpedia.com/dyn-postdownload.php>

How to use it -:

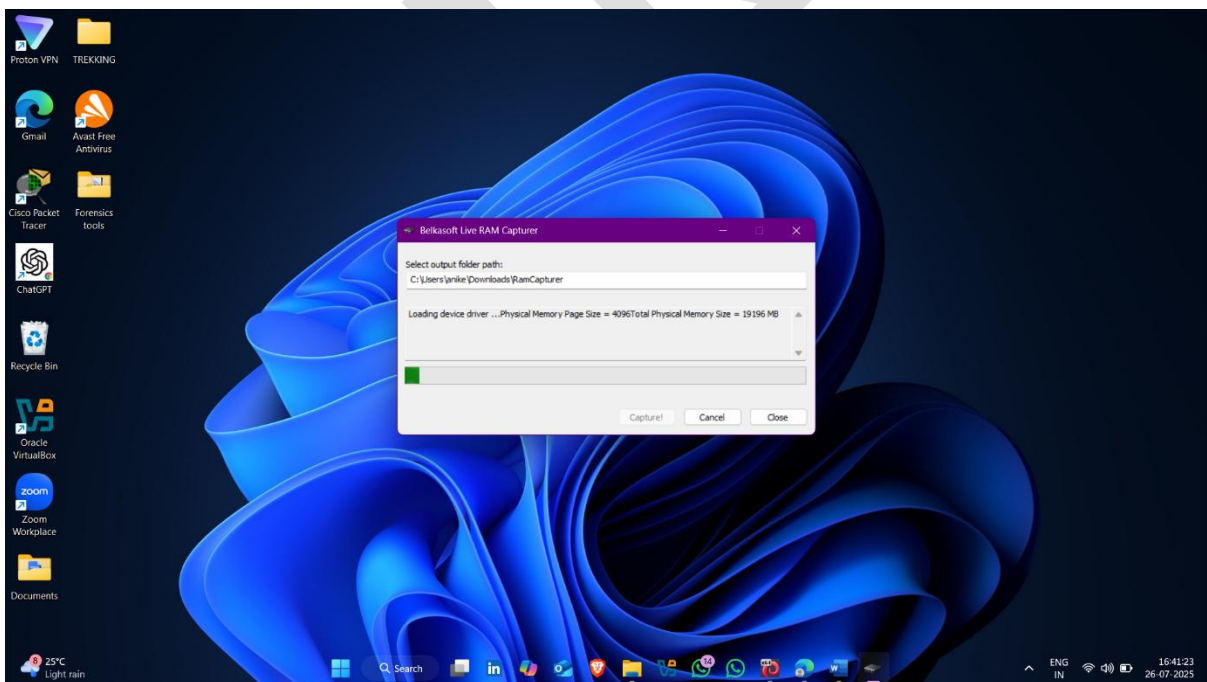
- Open ram capture by belkasoft



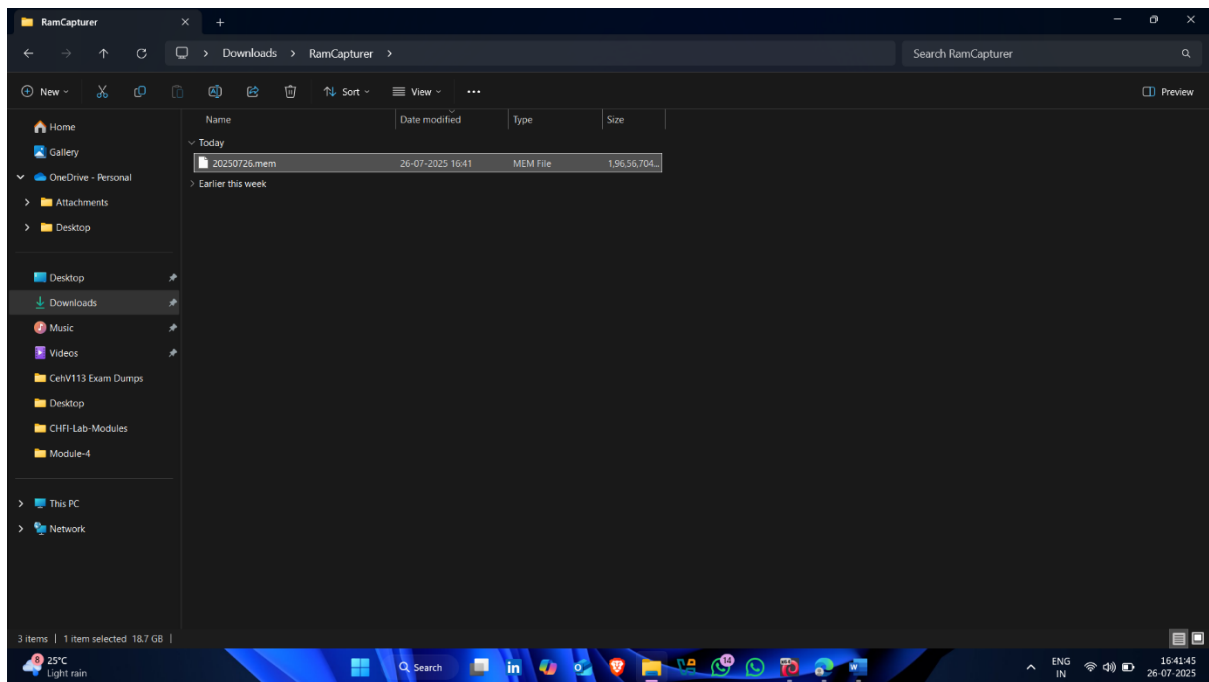
- Set destination for where ram data stored and click on capture



- Ram capture started ✓👉



- Ram capture 🖱️ ✅

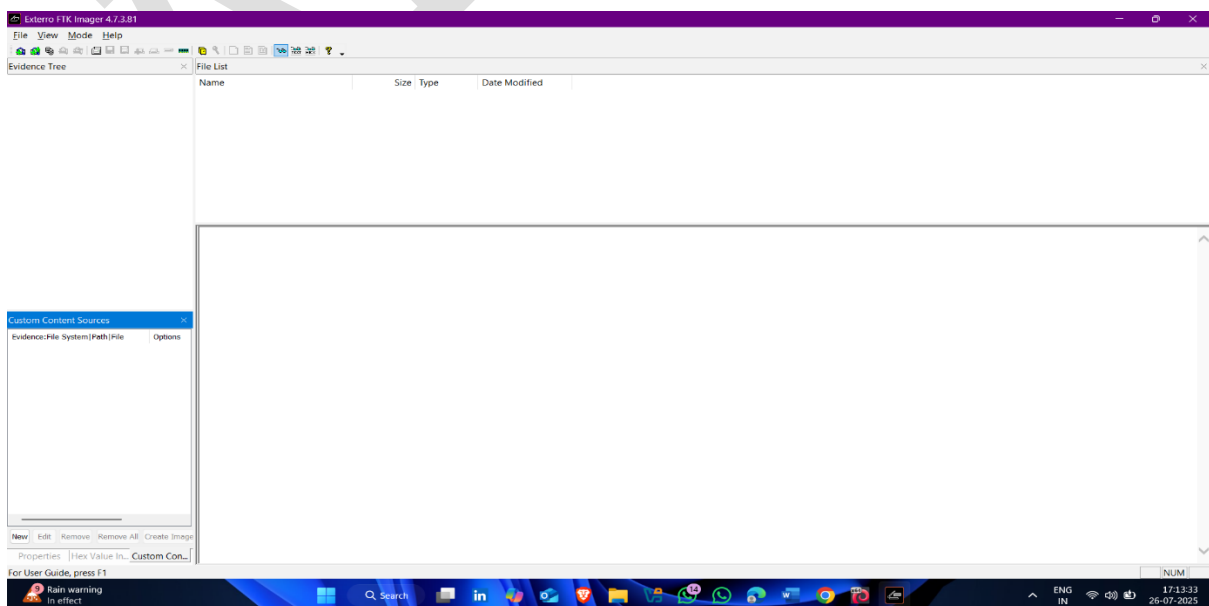


C) Investigate Forensics Image Of Windows Ram Using FTK Imager

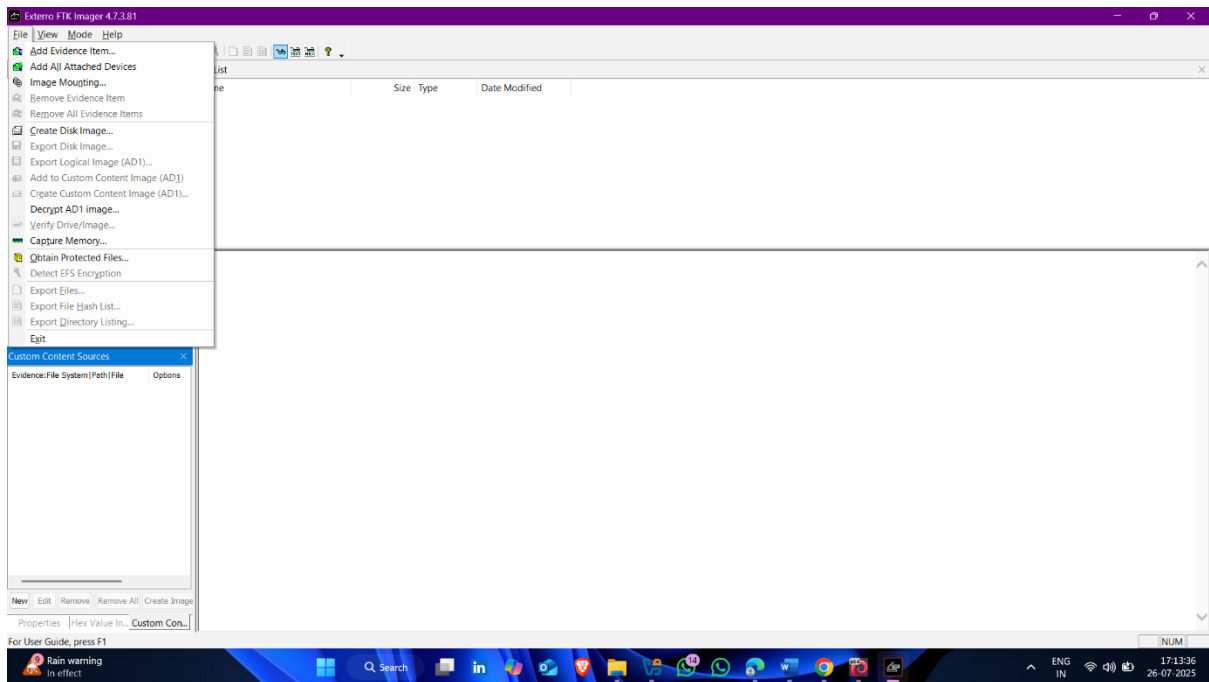
RAM capture with FTK Imager means creating a **memory dump** (.mem file) of a **live running Windows system**,

How to use it -:

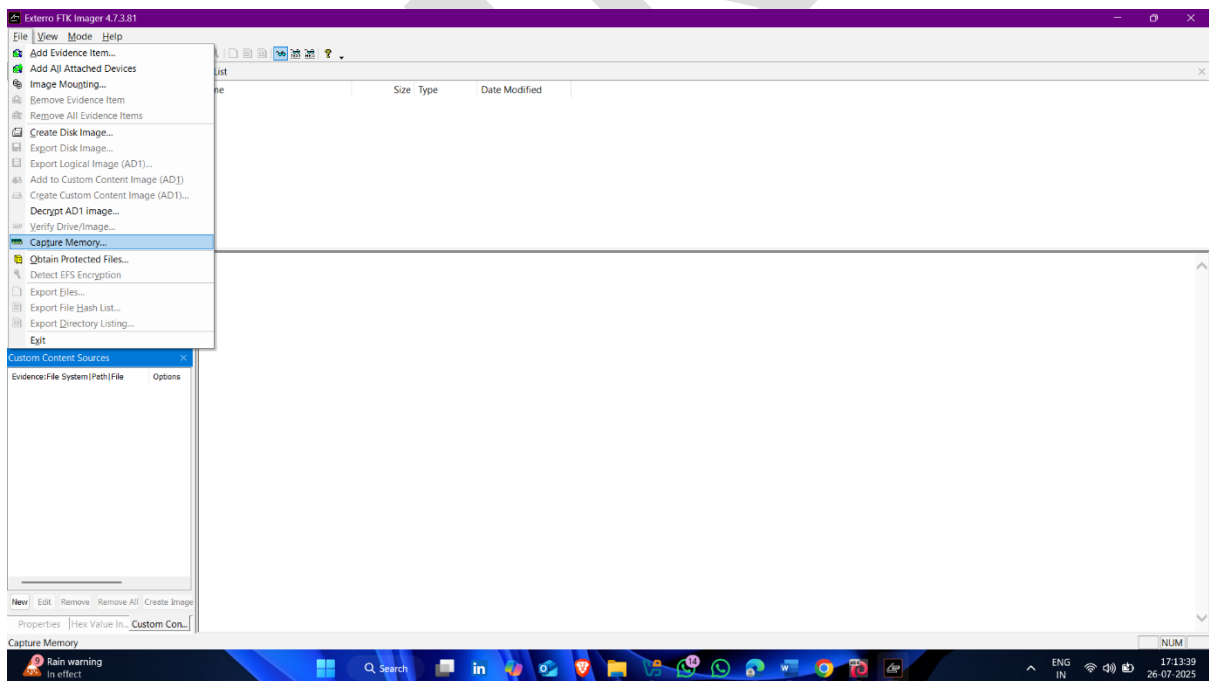
- Open FTK Imager



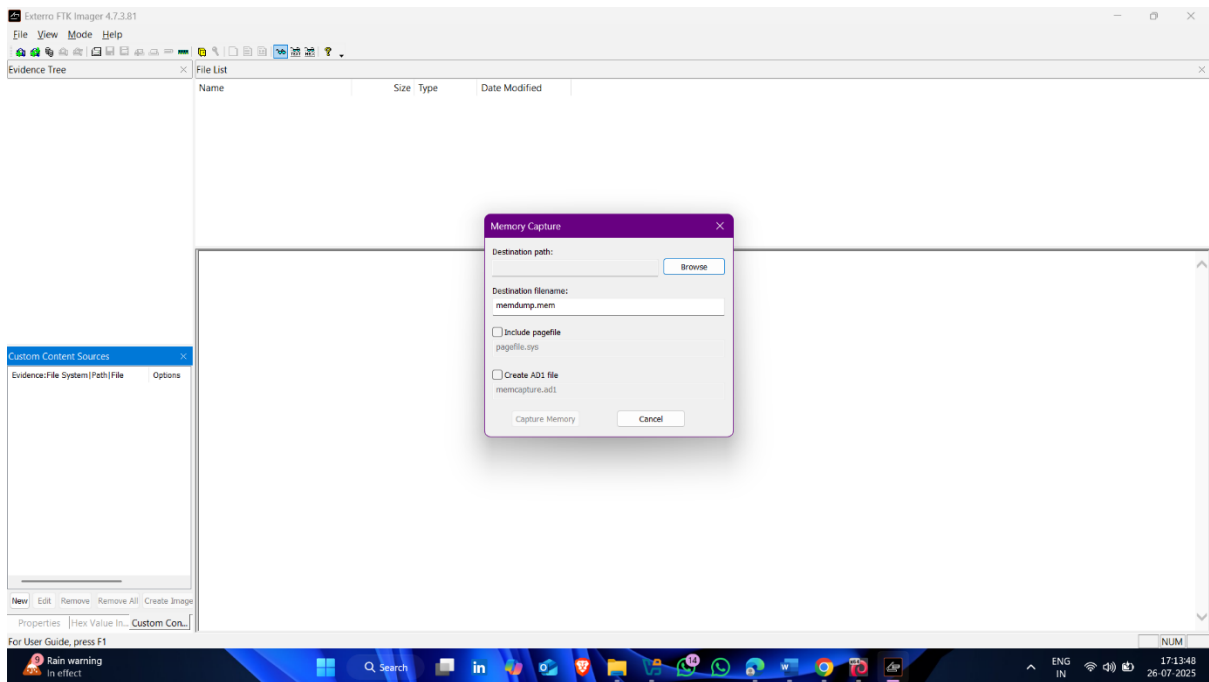
- Click on file section



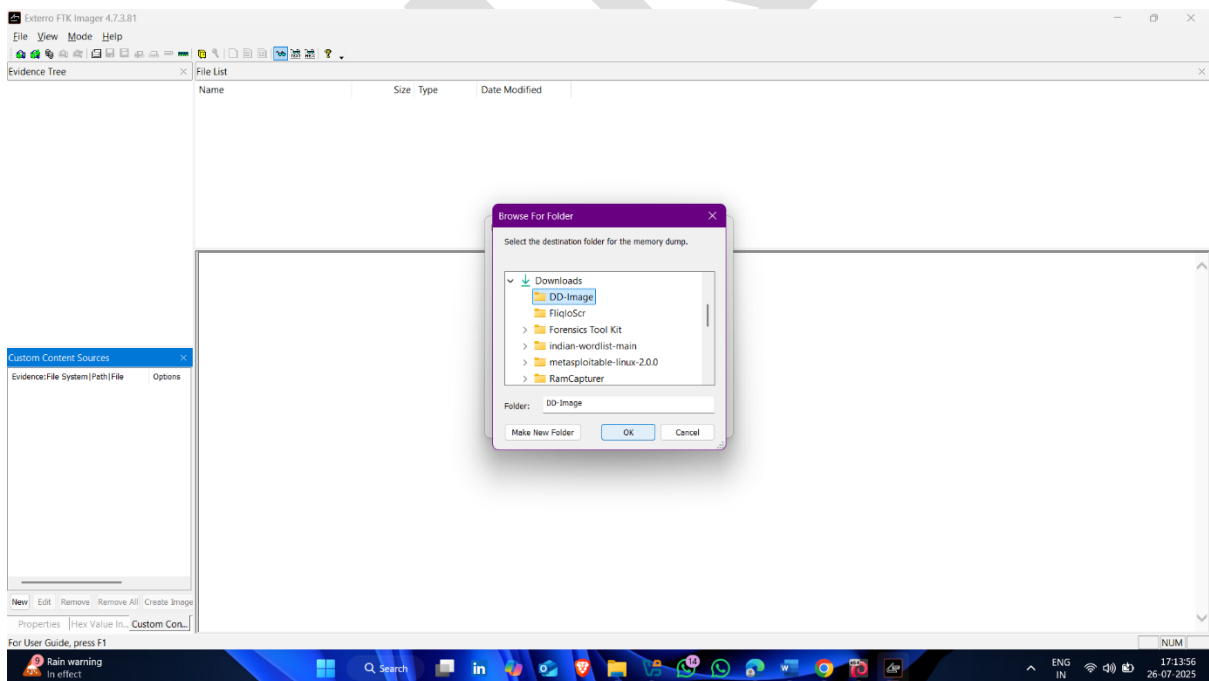
- Then click on **Capture Memory**



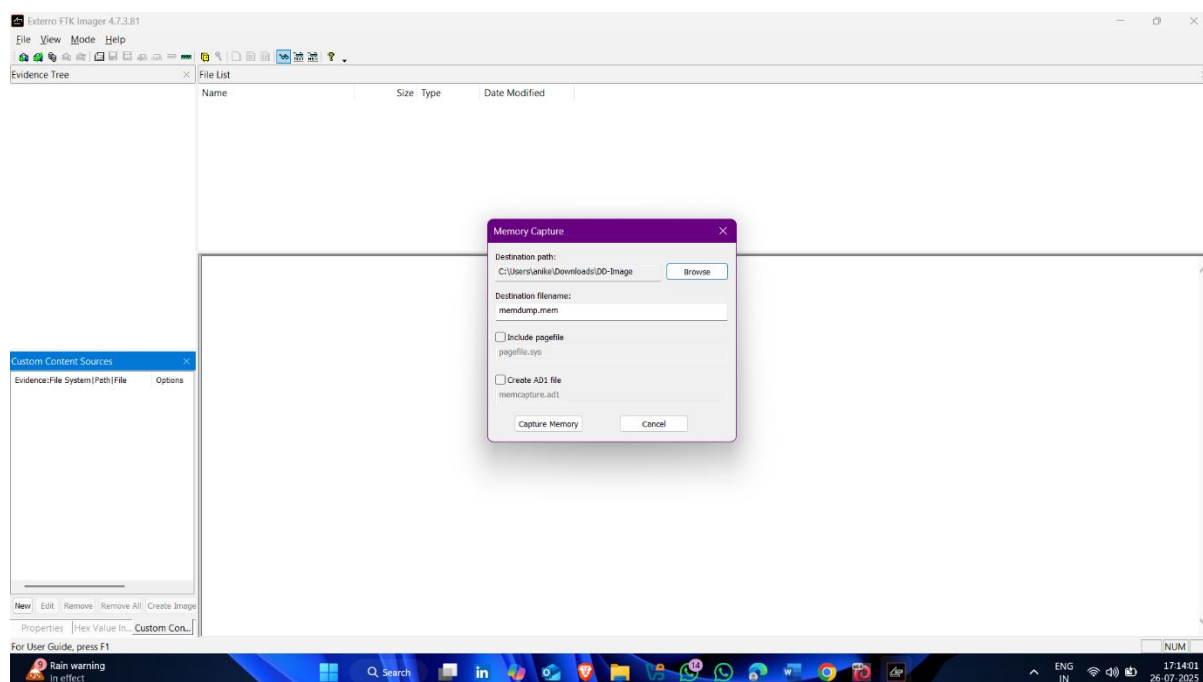
- Click on **Browse** to set destination path



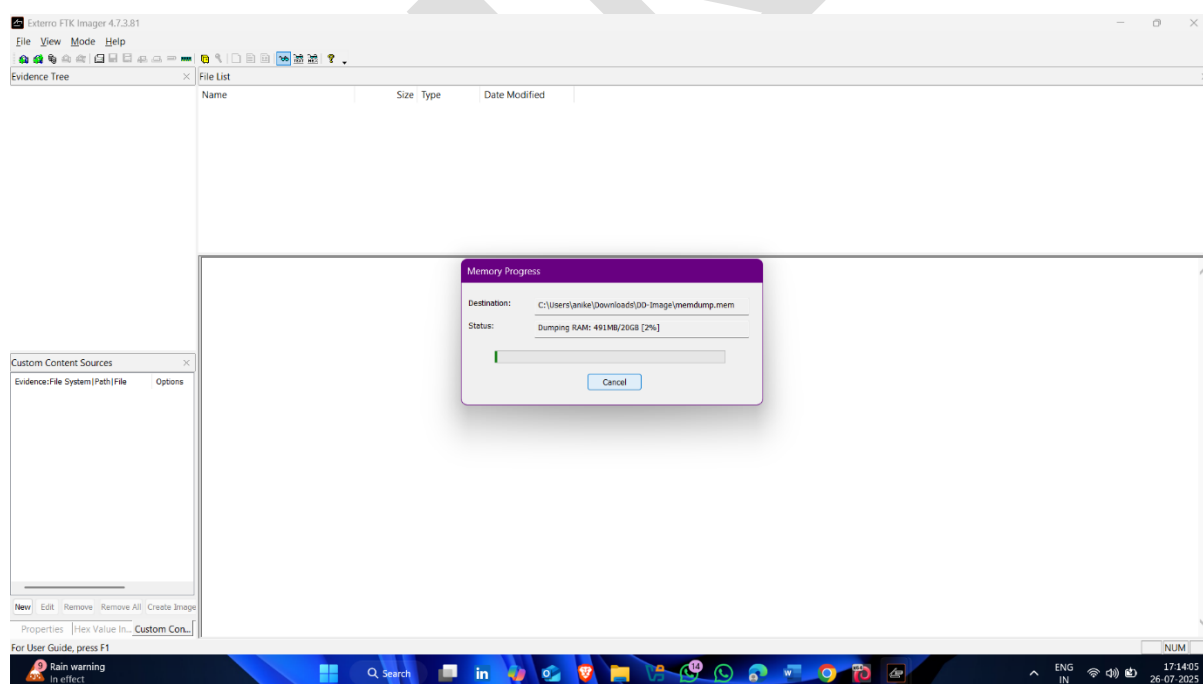
- Select destination and click on ok



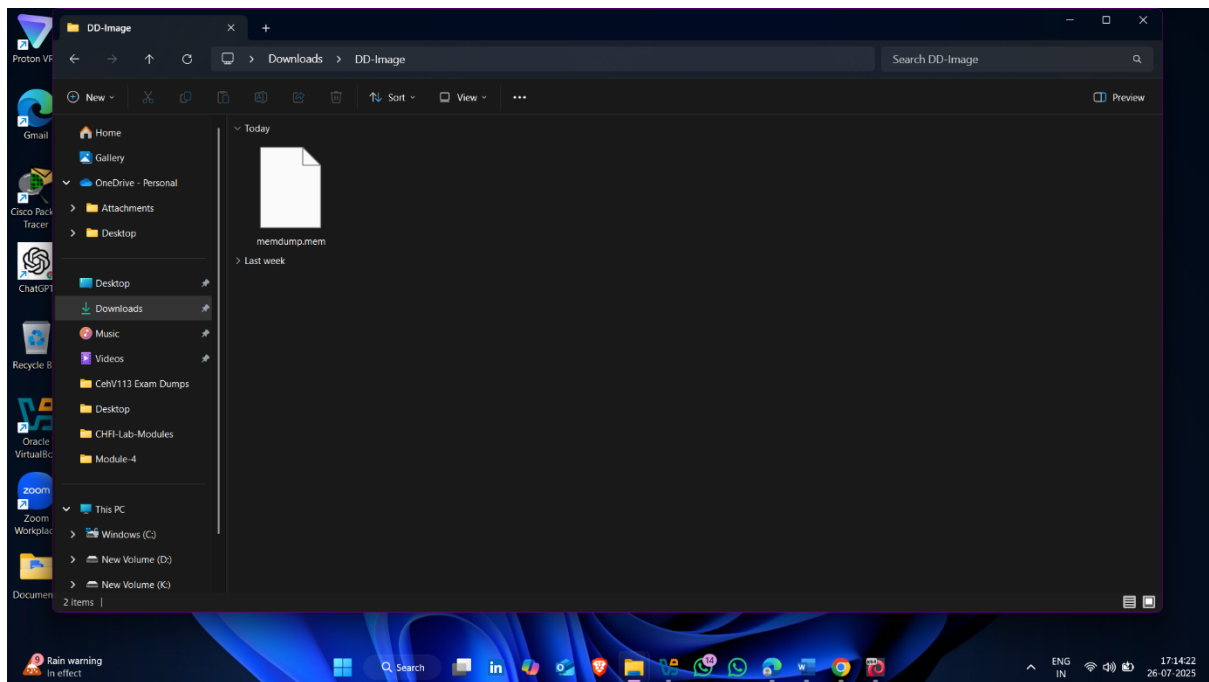
- Click on **capture memory**



- **Memory Capture Started**



- **Memory capture successfully** 🙌 ✅



Lab-3 -: Extract And Analyze Windows Ram Dump Files

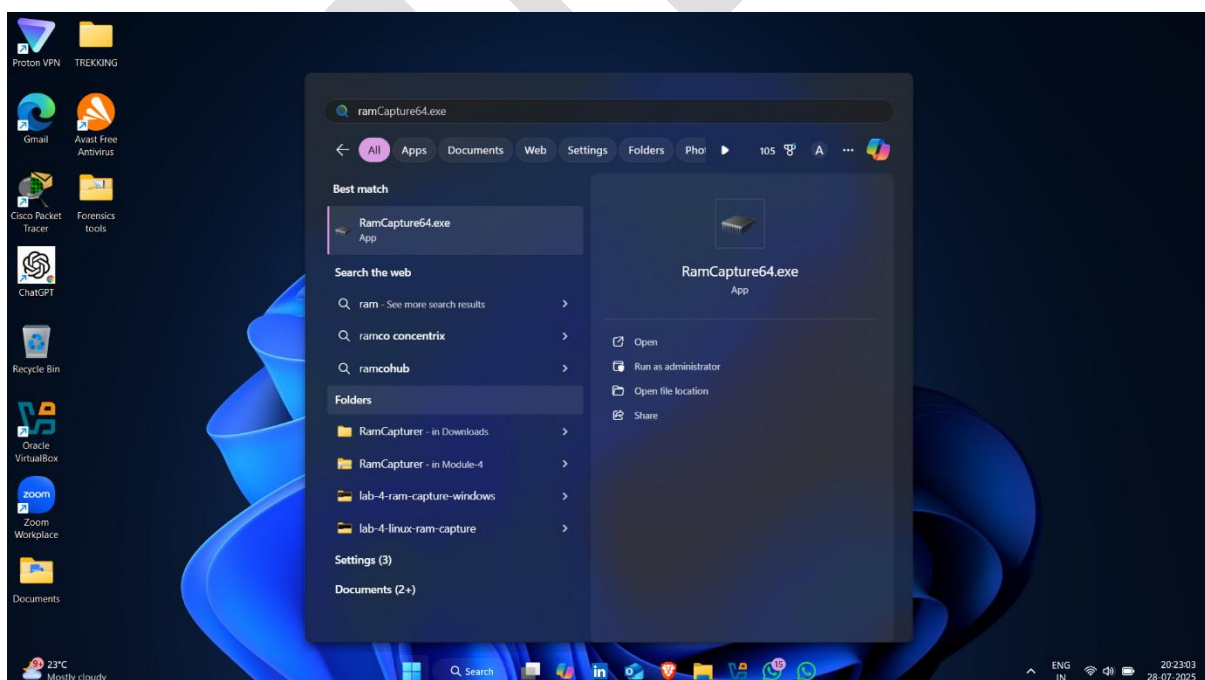
RAM dump analysis is a **key part of digital forensics**, especially for malware detection, incident response, and extracting volatile artifacts like running processes, network connections, loaded DLLs, and encryption keys.

A) Extract And Analyze Windows Ram Dump Files Using MemProcfs

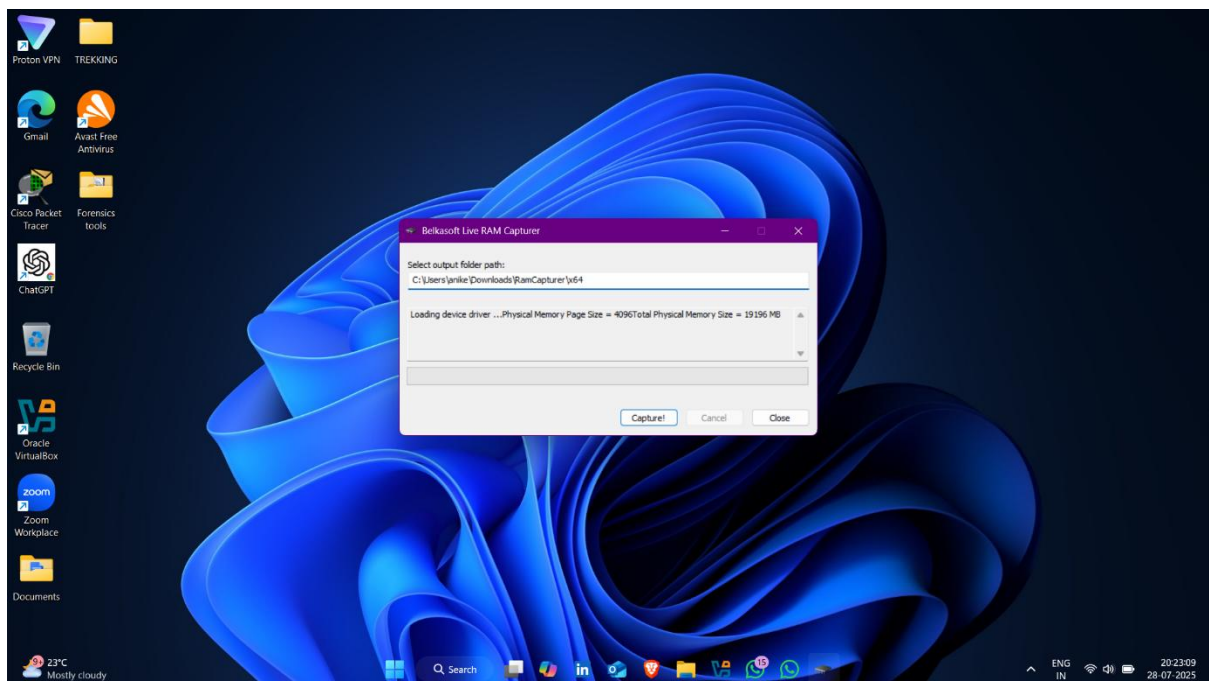
memprocfs (Memory Process File System) is an **advanced memory forensics tool** developed by [Velocidex](#). It lets you mount a **memory dump as a virtual file system** in Linux, so you can **explore the memory contents like files and folders**, just as if it were a mounted drive.

How to use it -:

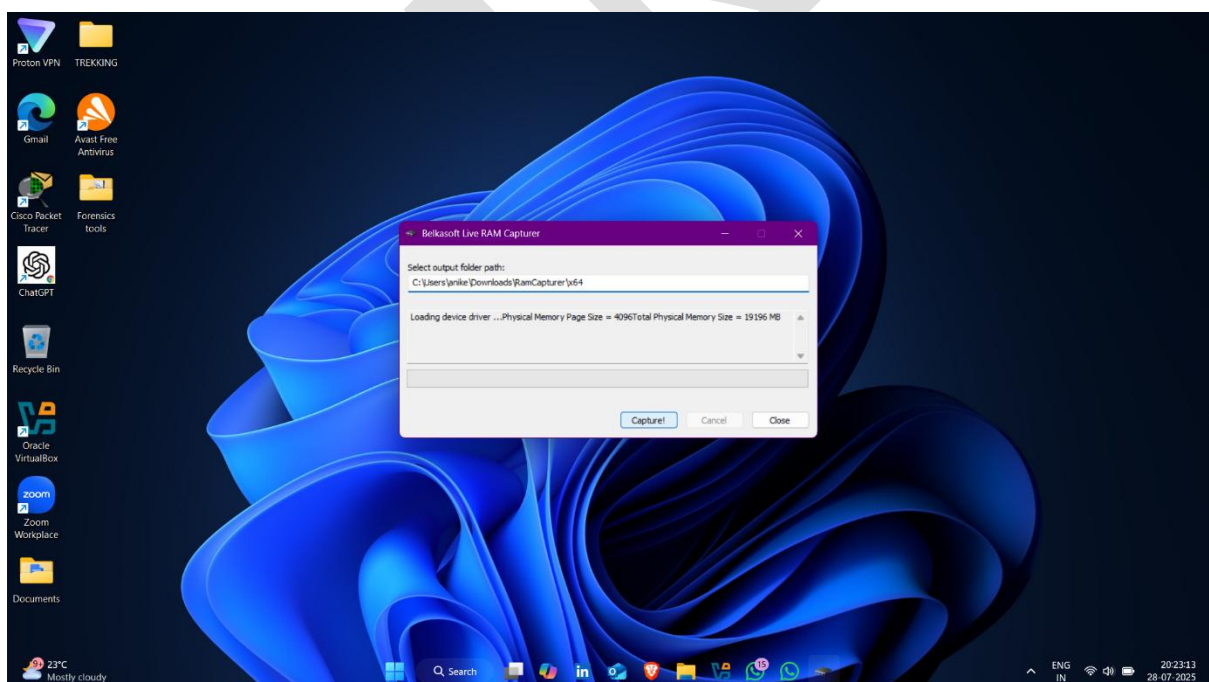
- Open RamCapture application



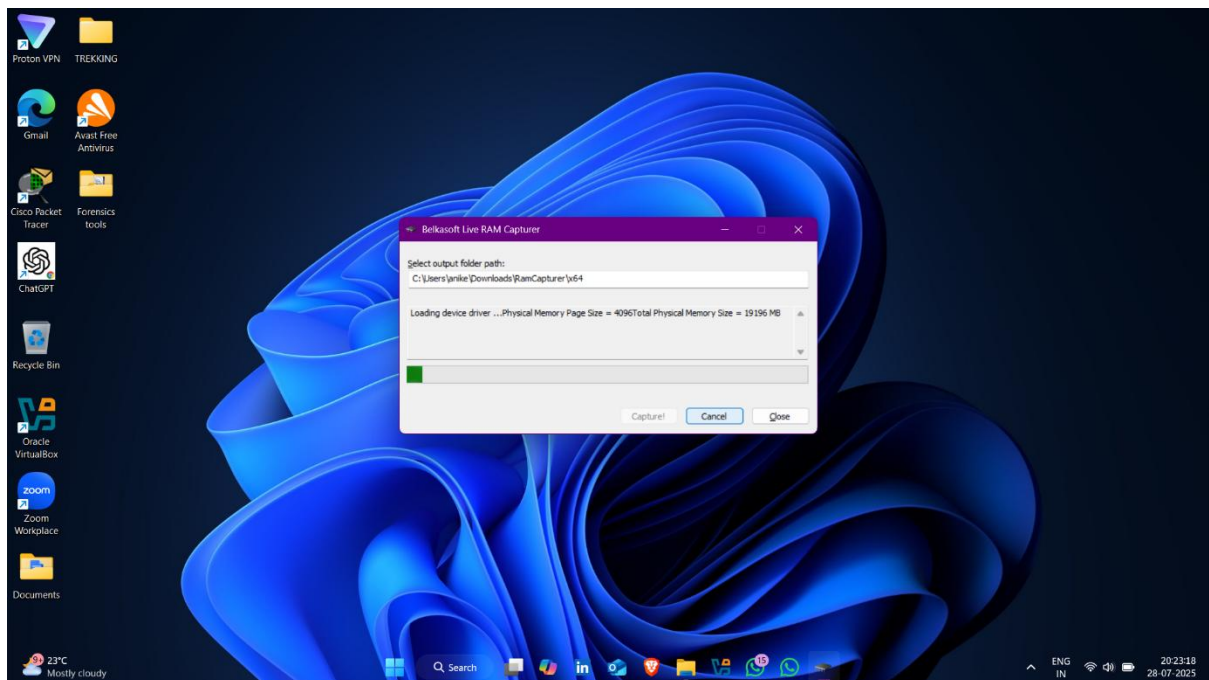
- Select folder output path



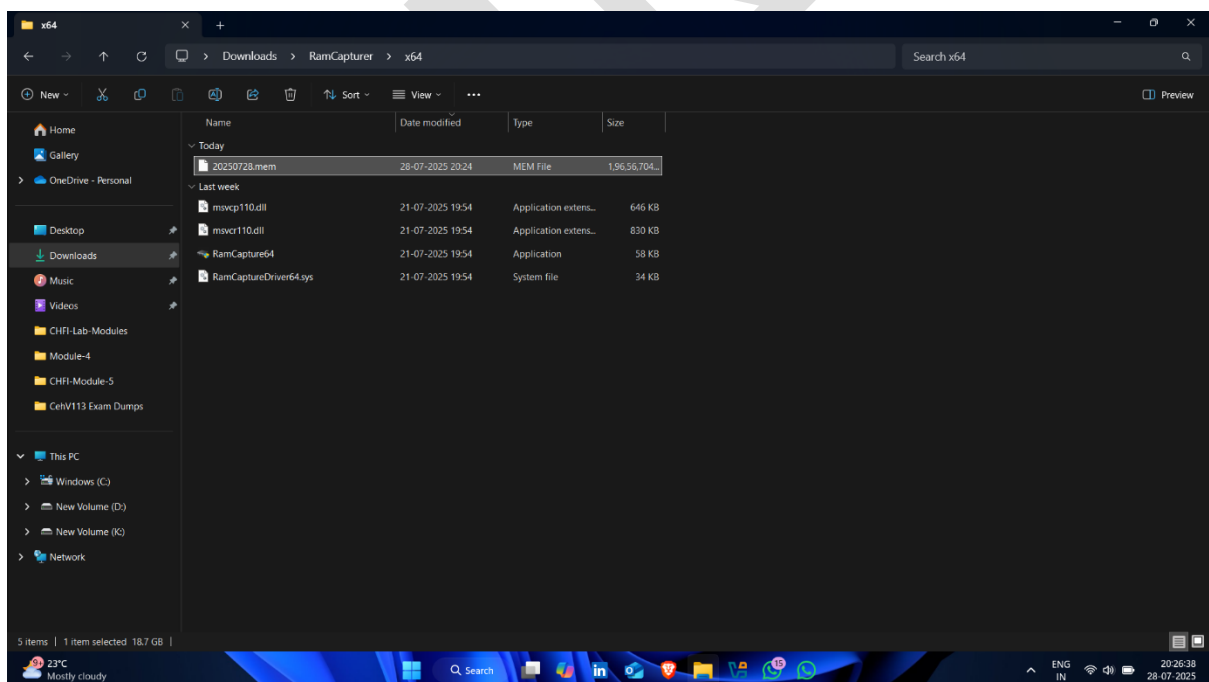
- Click on **capture**



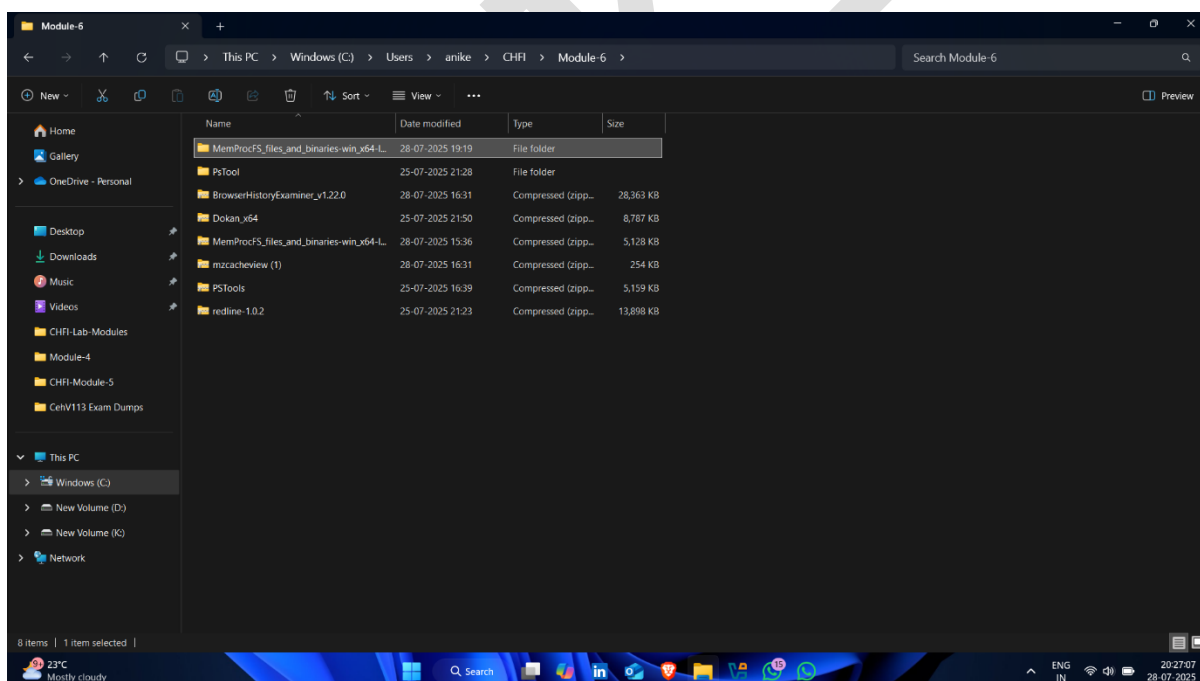
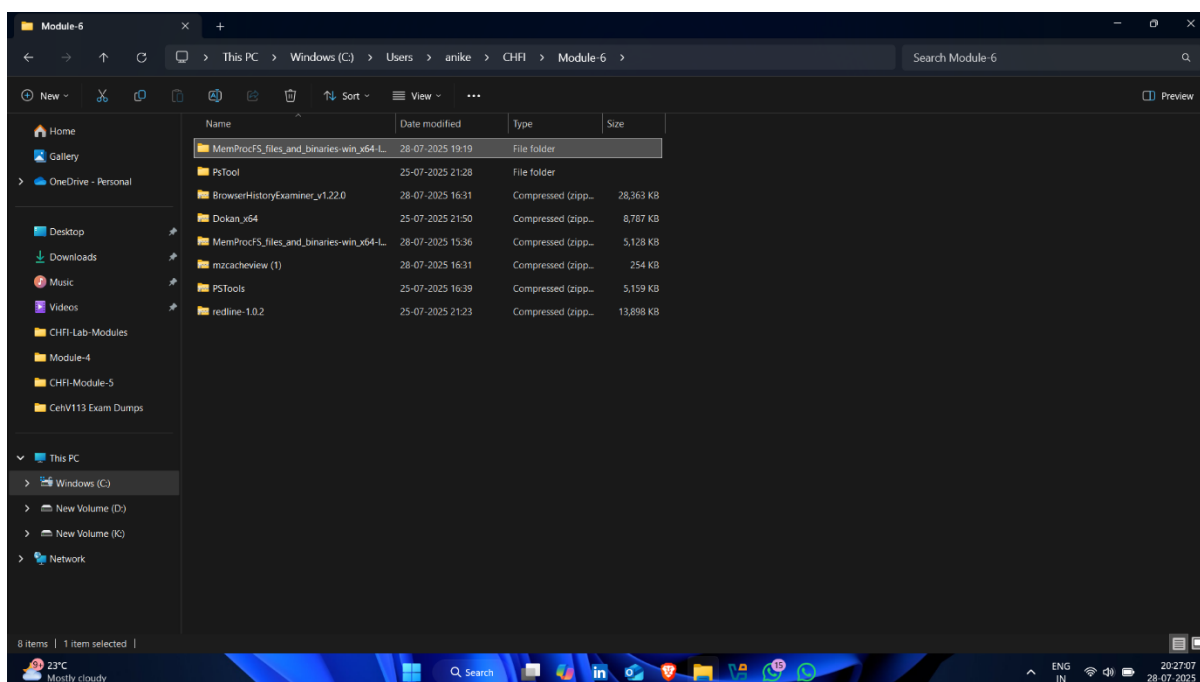
- **RamCapture Started** 🖱️ ✅



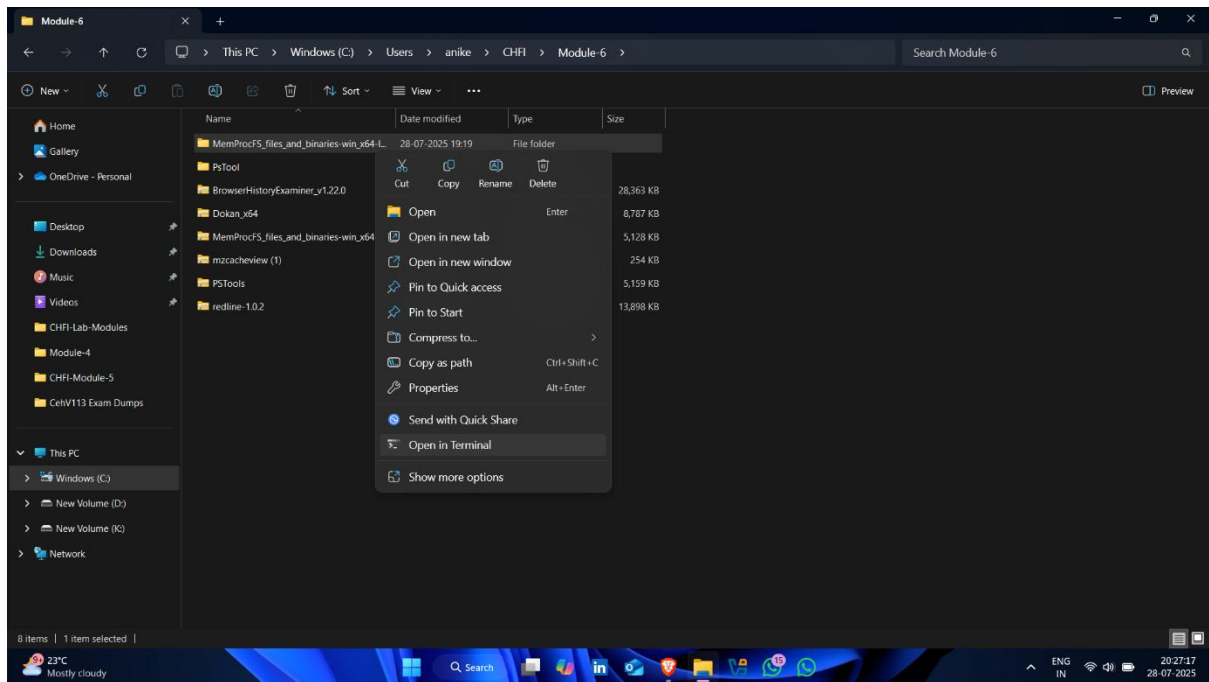
- **RamCapture File Created**



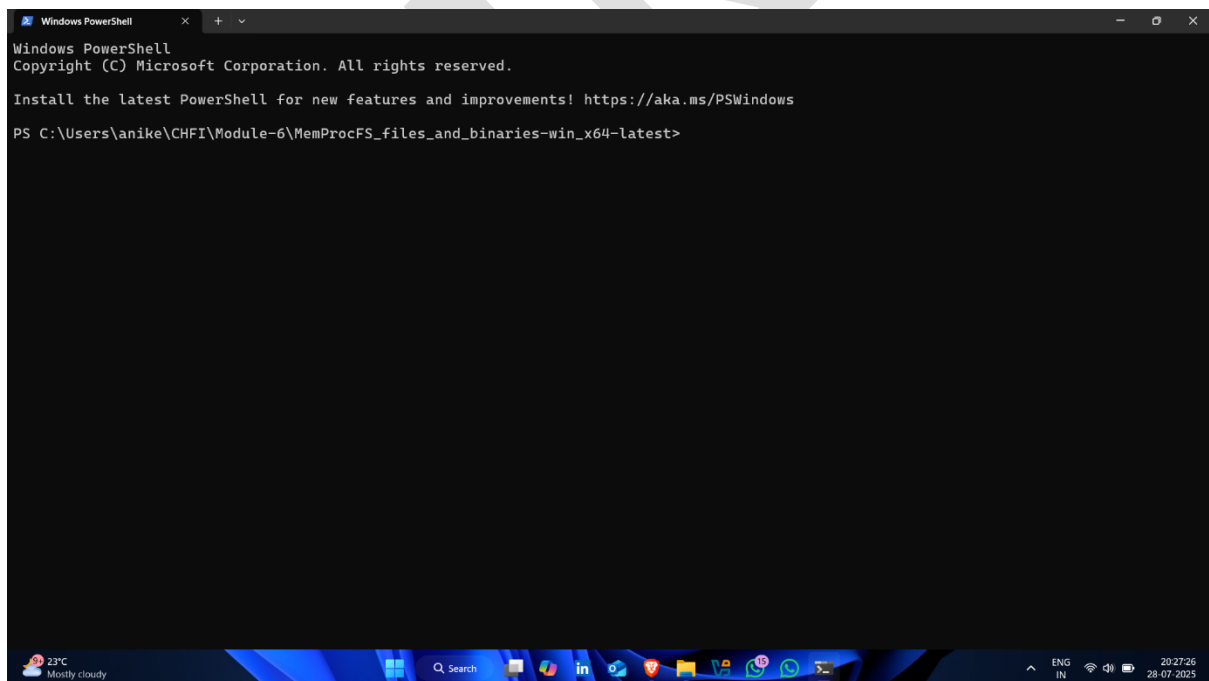
- Right click on **MemProcFs** Folder



- Click on **Open in Terminal**



- Folder open in Command Prompt



- Now enter following command

Command -: `.\MemProcFS.exe -device`

C:\Users\anike\Downloads\RamCapturer\x64\20250728.mem -forensic l

Explanation -:

Component	Explanation
.\MemProcFS.exe	Executes the MemProcFS tool from the current directory. This tool is used to mount memory dumps as a virtual file system.
-device C:\Users\anike\Downloads\RamCapturer\x64\20250728.mem	Specifies the path to the memory dump file that you want to analyze (in this case, a memory image captured on July 28, 2025).
-forensic l	Enables forensic mode. This ensures that all data is accessed in a read-only, non-intrusive manner—ideal for digital forensics investigations.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anike\CHFI\Module-6\MemProcFS_files_and_binaries-win_x64-latest> .\MemProcFS.exe -device C:\Users\anike\Downloads\RamCapturer\x64\20250728.mem -forensic 1
```

- Mounting Started 🙌 ✅

```
Windows PowerShell

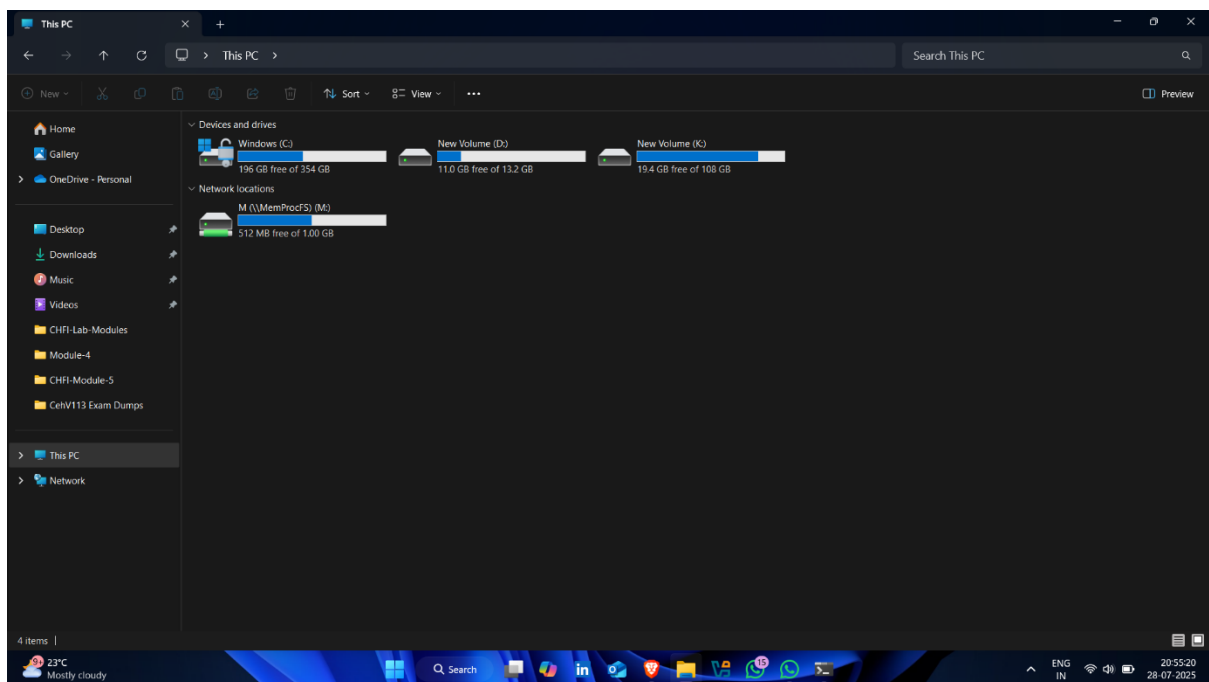
-licen...-accept-elastic-license-2-0 : accept the Elastic License 2.0 to enable built-in yara rules from Elastic.
-forensic-process-skip : comma-separated list of process names to skip.
-forensic-yara-rules : perform a forensic yara scan with specified rules. Full path to source or compiled yara rules should be specified. Example: -forensic-yara-rules "C:\Temp\my_yara_rules.yar"
-forensic : start a forensic scan of the physical memory immediately after startup if possible. Allowed parameter values range from 0-4. Note! forensic mode is not available for live memory.
0 = not enabled (default value)
1 = forensic mode with in-memory sqlite database.
2 = forensic mode with temp sqlite database deleted upon exit.
3 = forensic mode with temp sqlite database remaining upon exit.
4 = forensic mode with static named sqlite database (vmm.sqlite3).
default: 0 Example -forensic 4

PS C:\Users\anike\CHFI\Module-6\MemProcFS_files_and_binaries-win_x64-latest> .\MemProcFS.exe -device C:\Users\anike\Downloads\RamCapturer\x64\20250728.mem -forensic 1
[SYMBOL] Functionality may be limited. Extended debug information disabled.
[SYMBOL] Offline symbols unavailable - ID: SE6287D11450000
[SYMBOL] Reason: Unable to download kernel symbols to cache from Symbol Server.

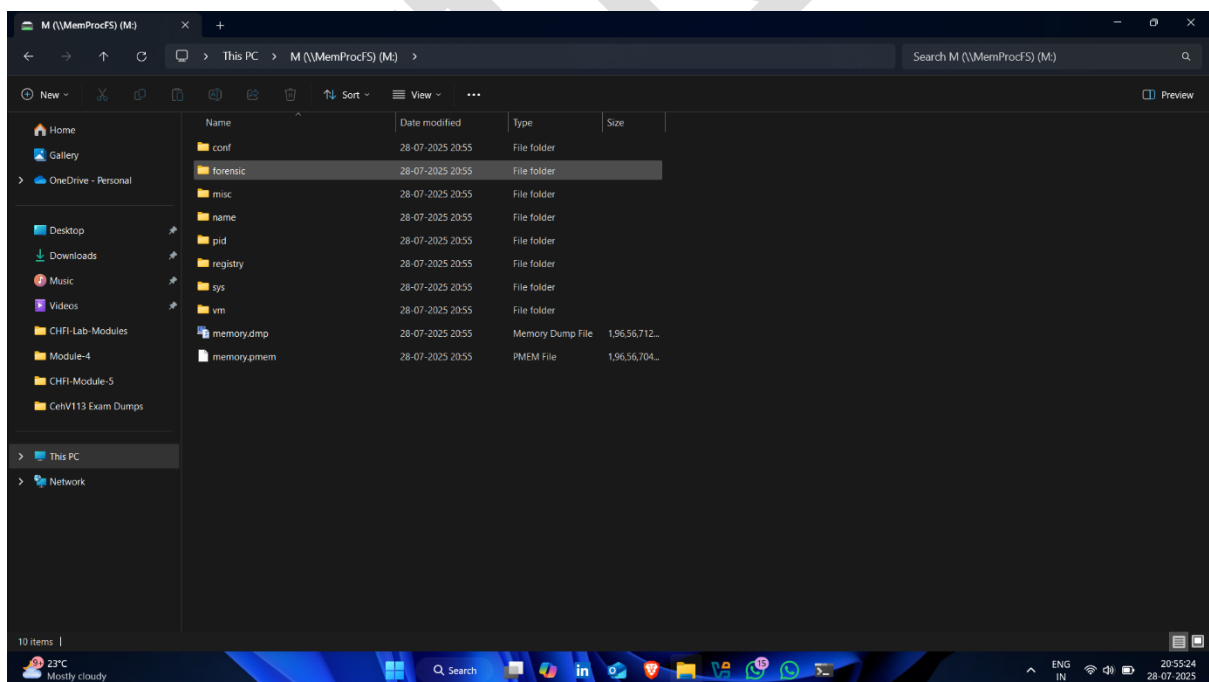
Initialized 64-bit Windows 10.0.26100
[FORENSIC] Built-in Yara rules from Elastic are disabled. Enable with: -license-accept-elastic-license-2-0
[PLUGIN] Python initialization failed. Python 3.6 or later not found.

===== MemProcFS =====
- Author:      Ulf Frisk - pcileech@frizk.net
- Info:        https://github.com/ulfriisk/MemProcFS
- Discord:     https://discord.gg/pcileech
- License:     GNU Affero General Public License v3.0
=====
MemProcFS is free open source software. If you find it useful please become a sponsor at: https://github.com/sponsors/ulfriisk Thank You :)
=====
- Version:     5.15.2 (Windows)
- Mount Point: M:\
- Tag:         26100_23a6f7a7
- Operating System: Windows 10.0.26100 (X64)
=====
```

- Image Mounted ✅👉



- Data inside image file ✅



Lab-4-: Capture And Examine Windows registry Files On Live System

Capturing and examining the Windows registry on a live system is a crucial task in **digital forensics** and **incident response**, where investigators retrieve and analyze registry hives **while the system is running** — without shutting it down or altering its state too much.

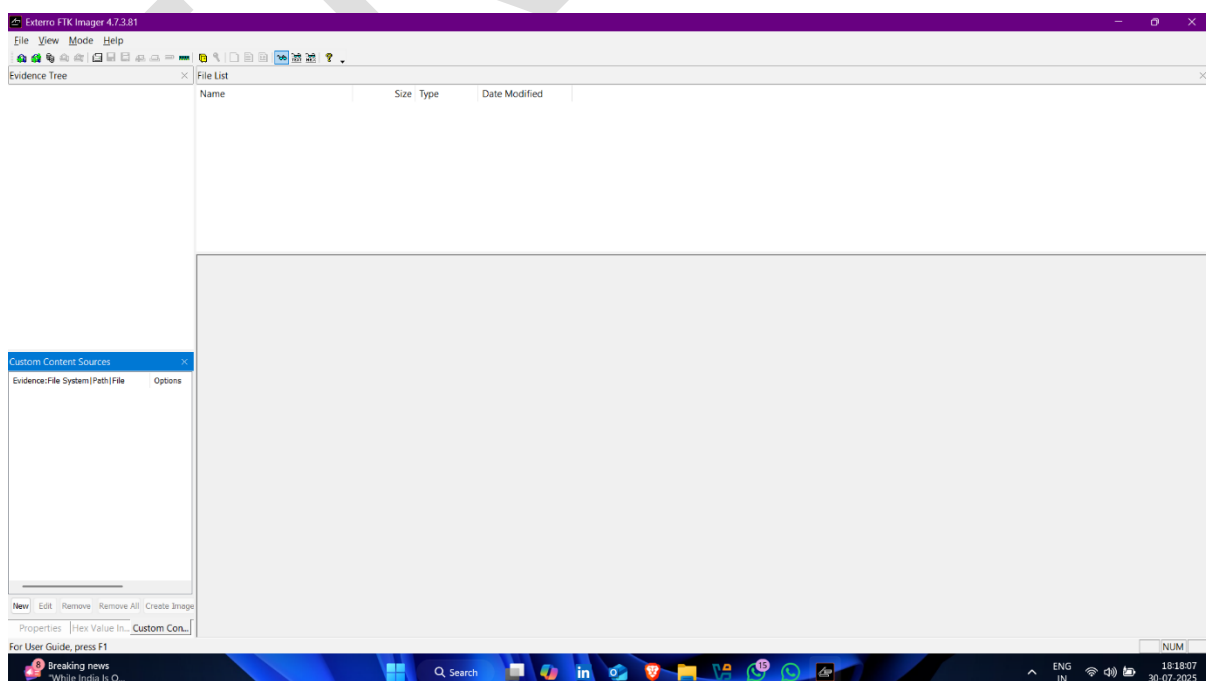
A) Capture And Examine Windows registry Files On Live System Using FTK Imager

FTK Imager is a free forensic imaging tool developed by **Exterro** (formerly **AccessData**). It allows you to:

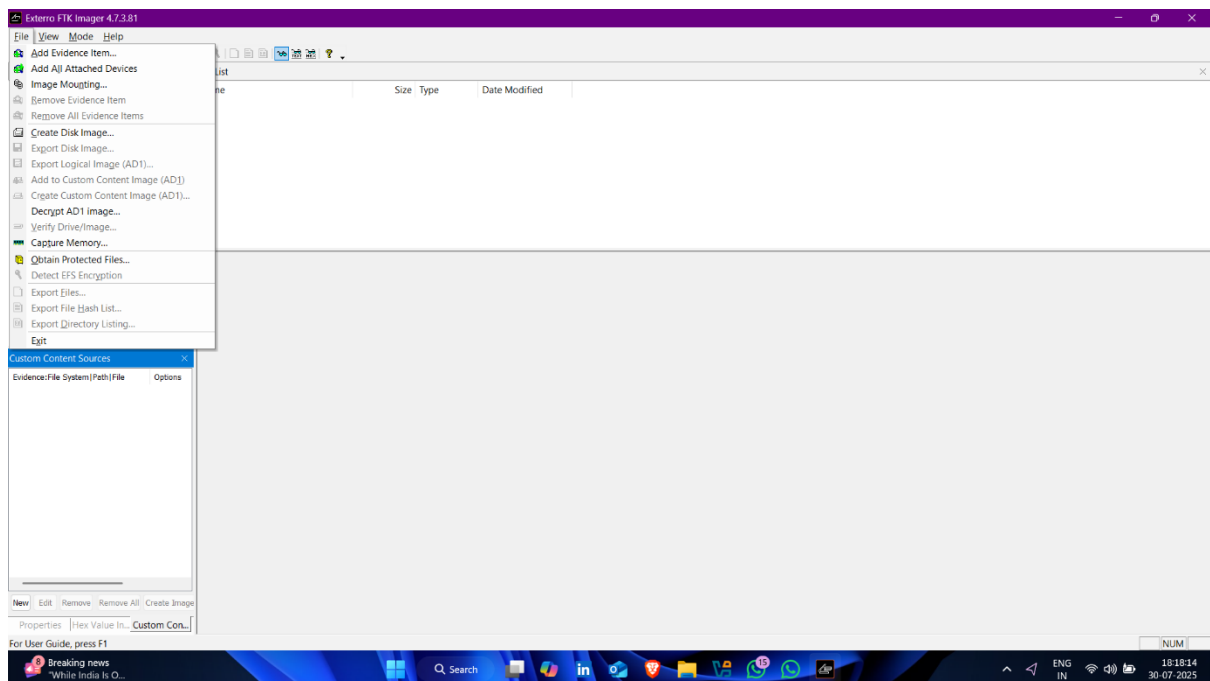
- Capture memory or disk images (live or dead systems)
- View files, including **locked system files**
- Export critical artifacts like **Windows registry hives** without altering the system

How to use it -:

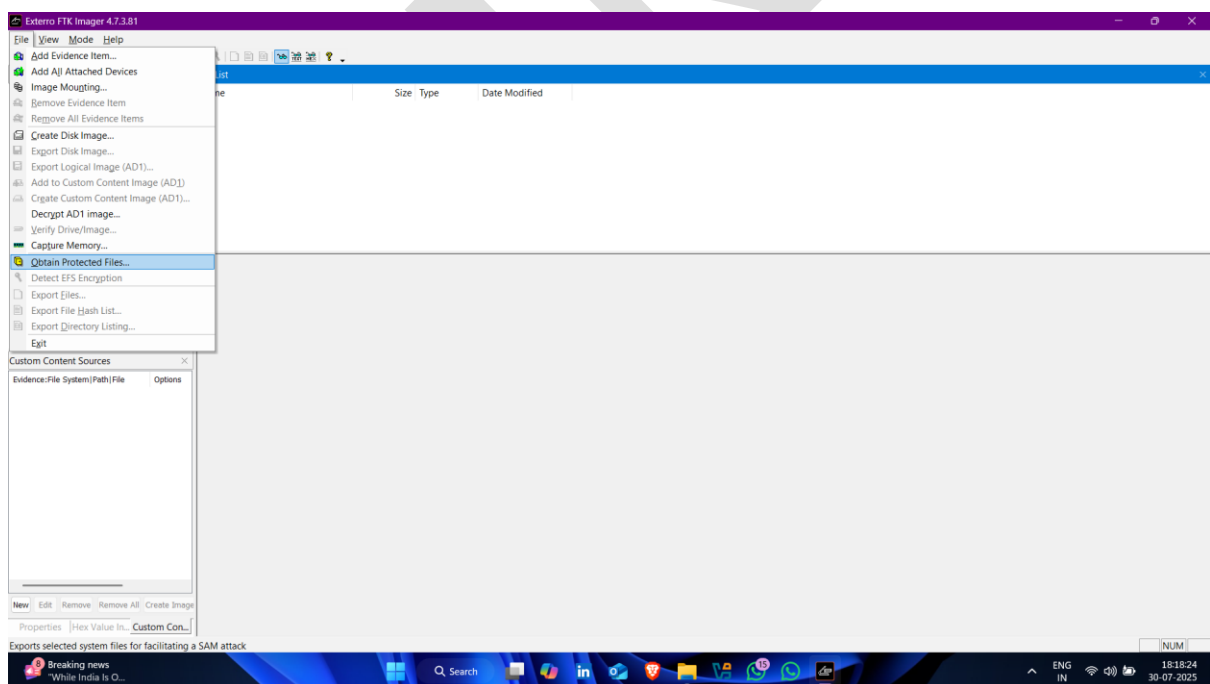
- Open FTK Imager



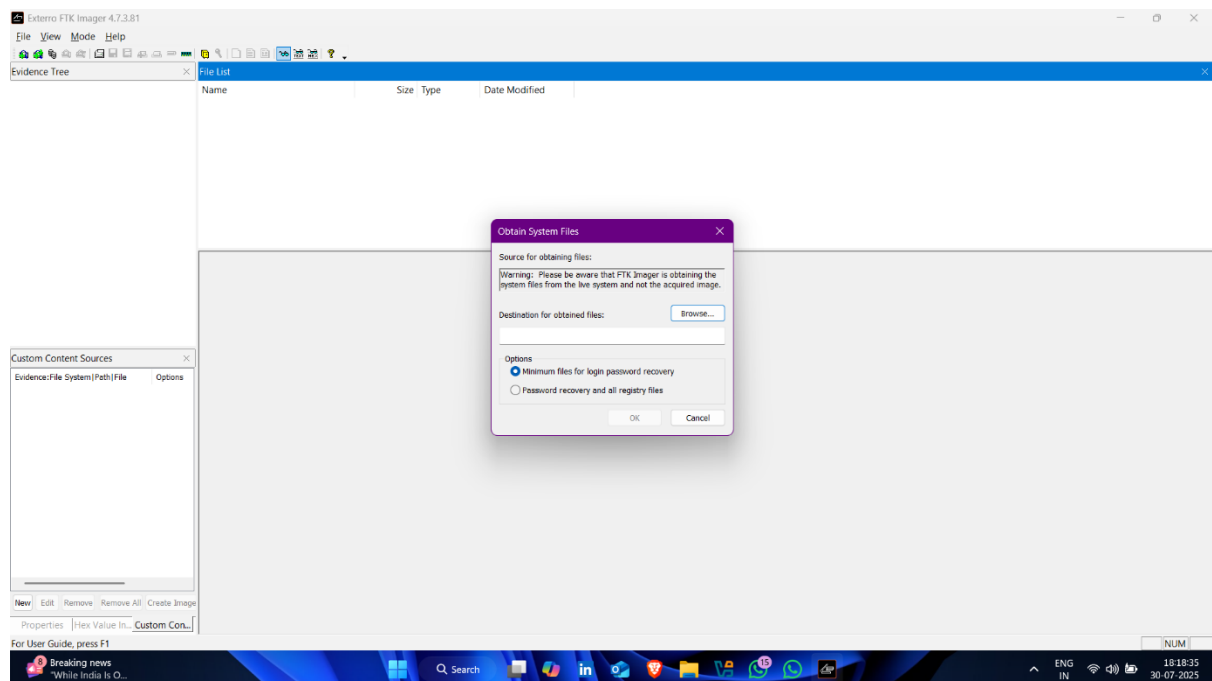
- Click on **File** section



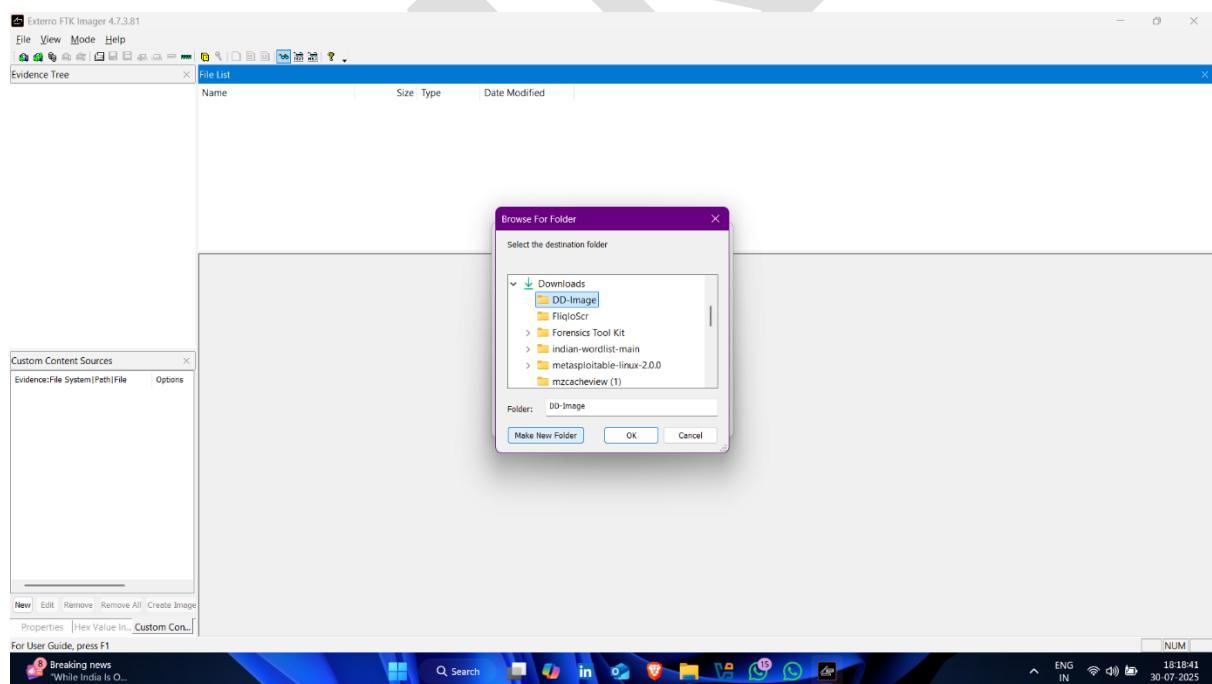
- Click on **Obtain Protected Files**



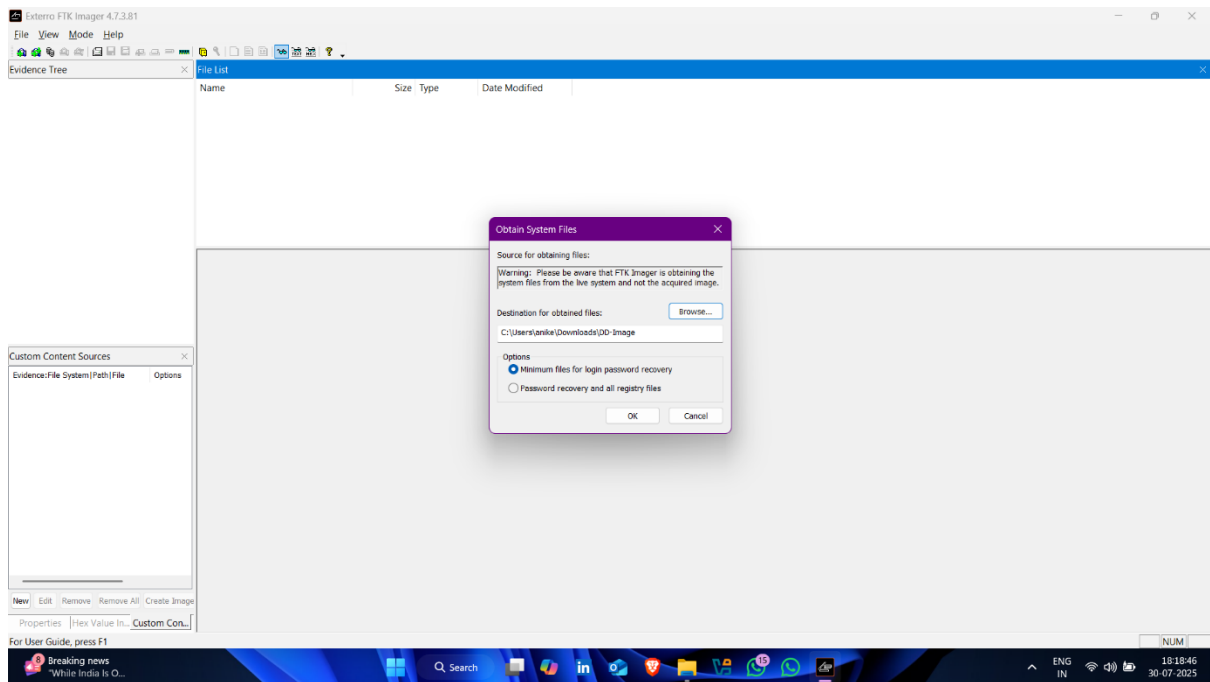
- Click on **Browse** to set destination path



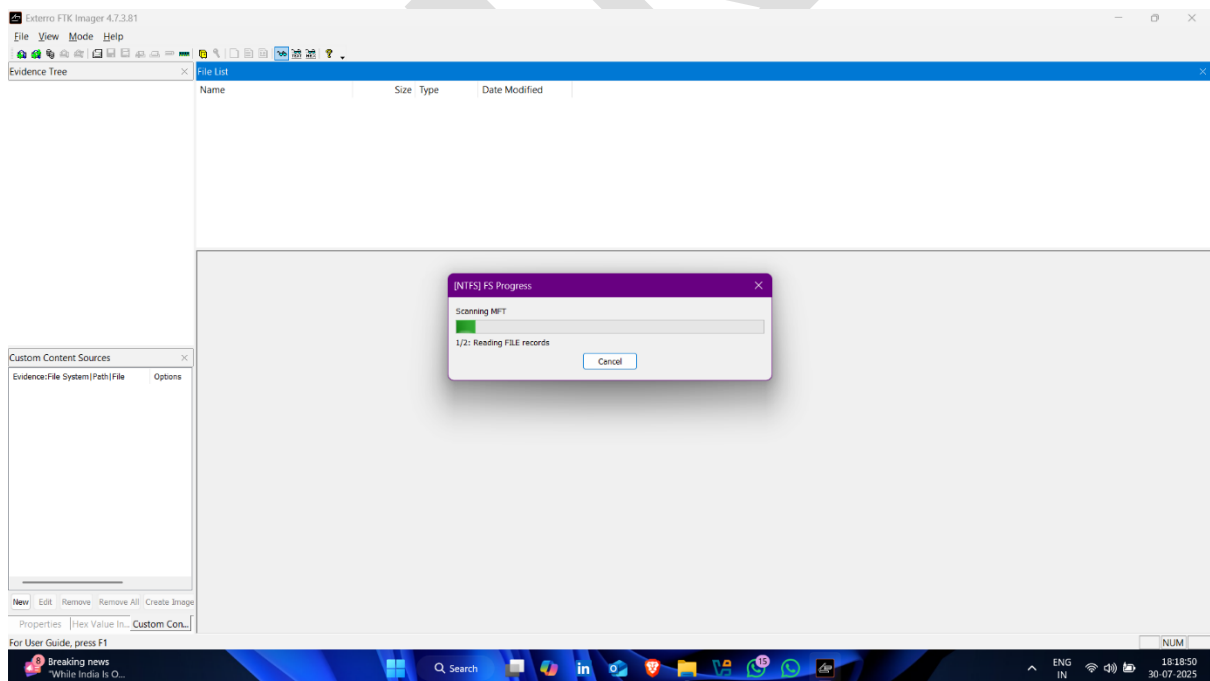
- Select destination and click on ok



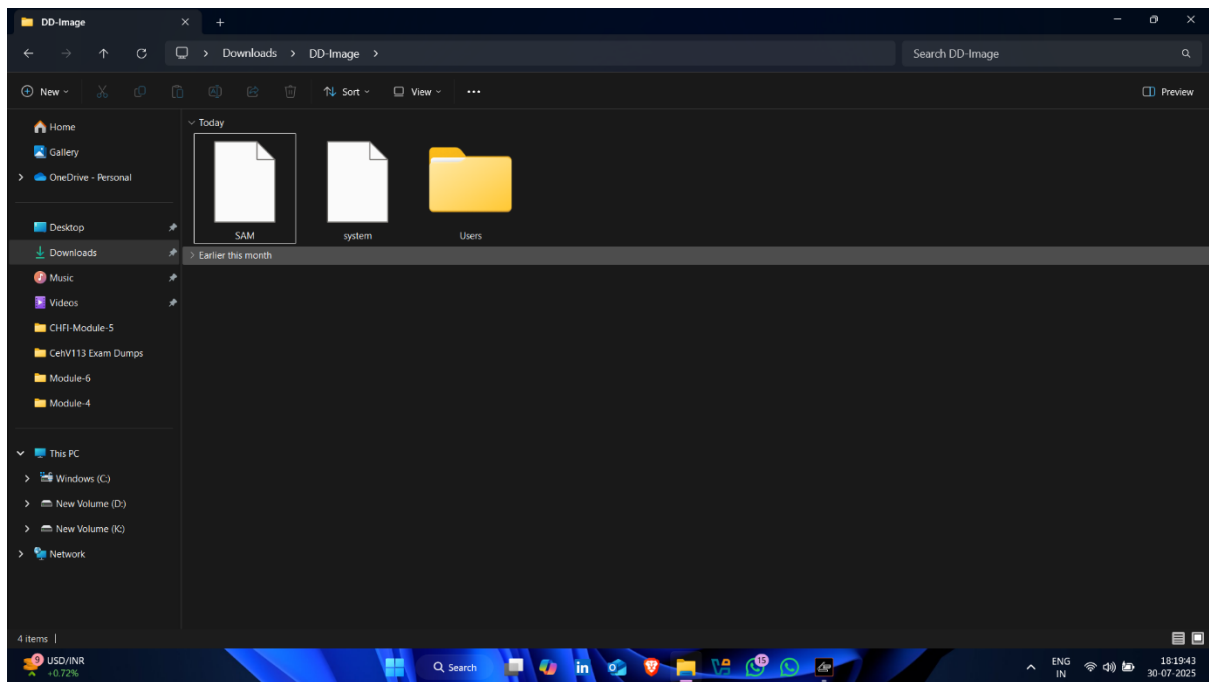
- click on ok



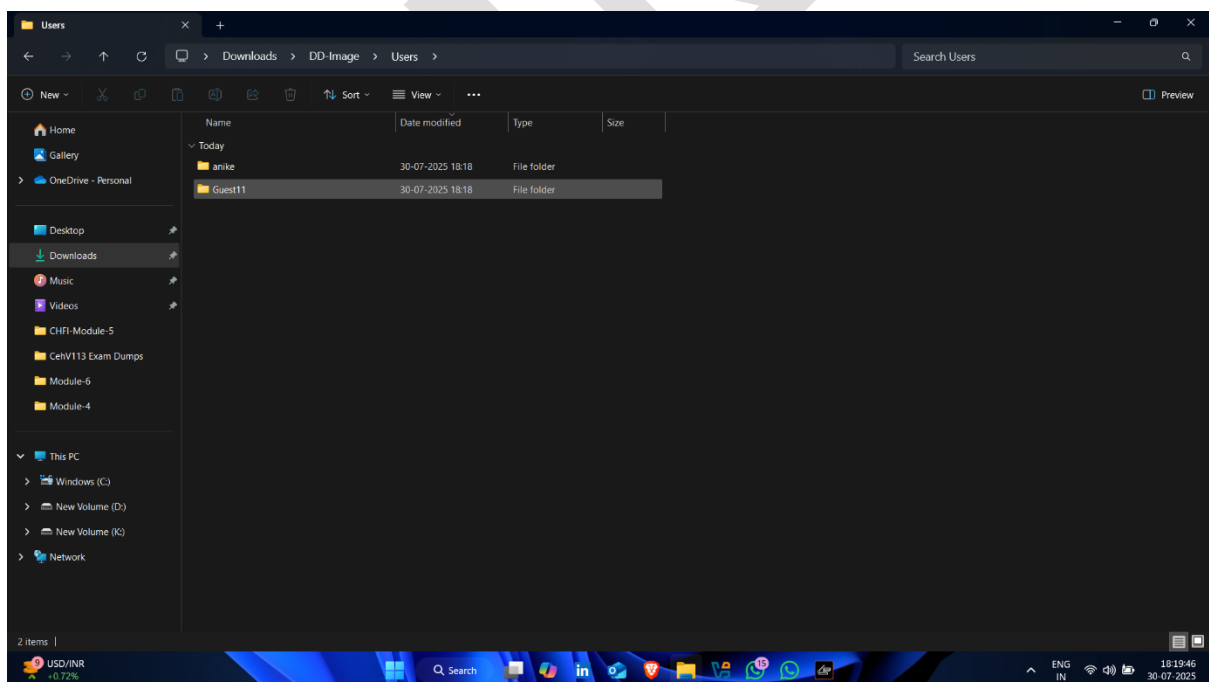
- Process Started



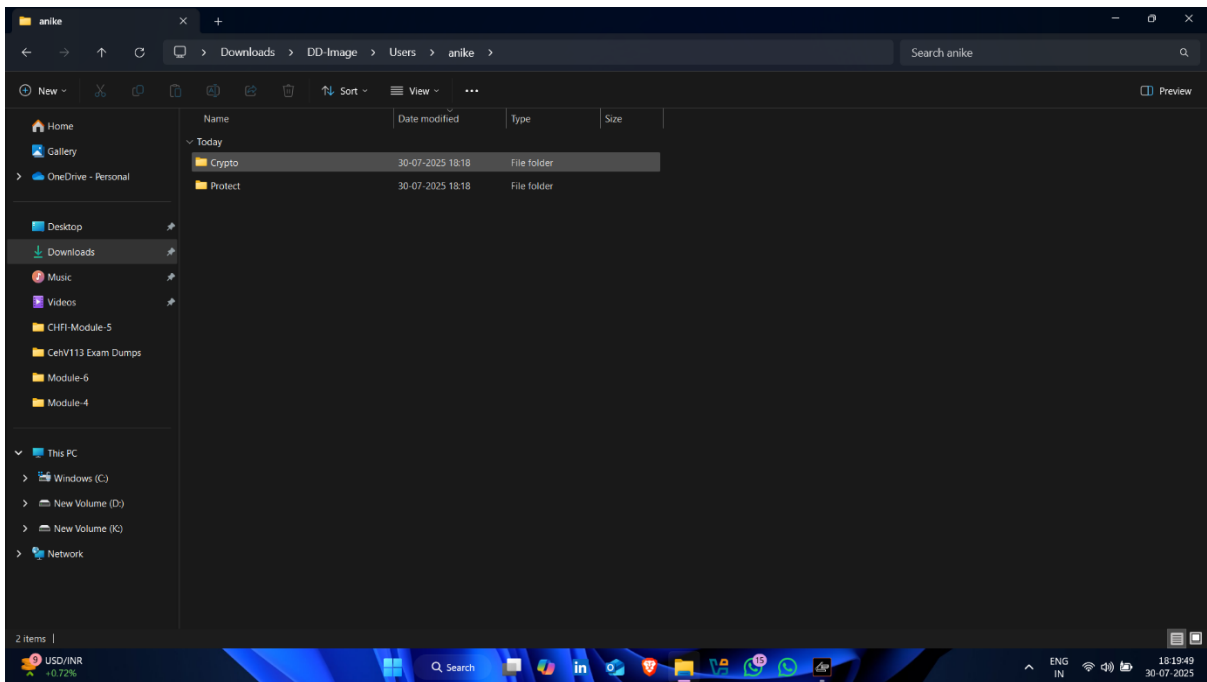
- Files are Created ✓👉



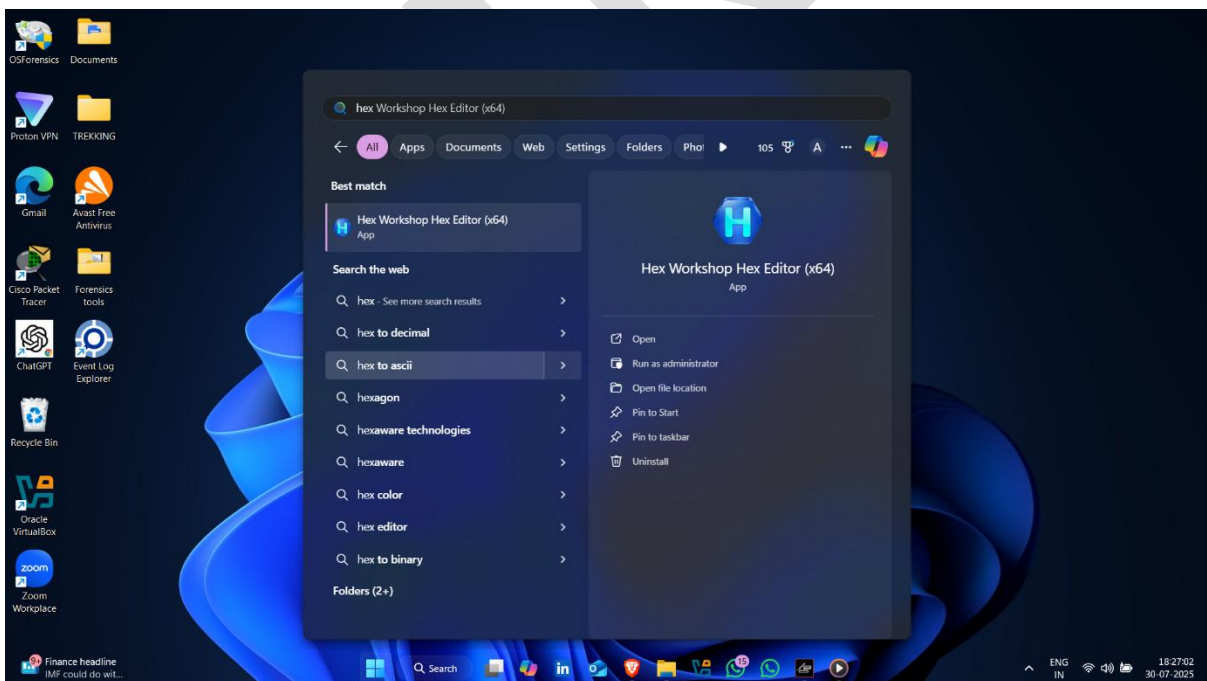
- Click on **guest** account



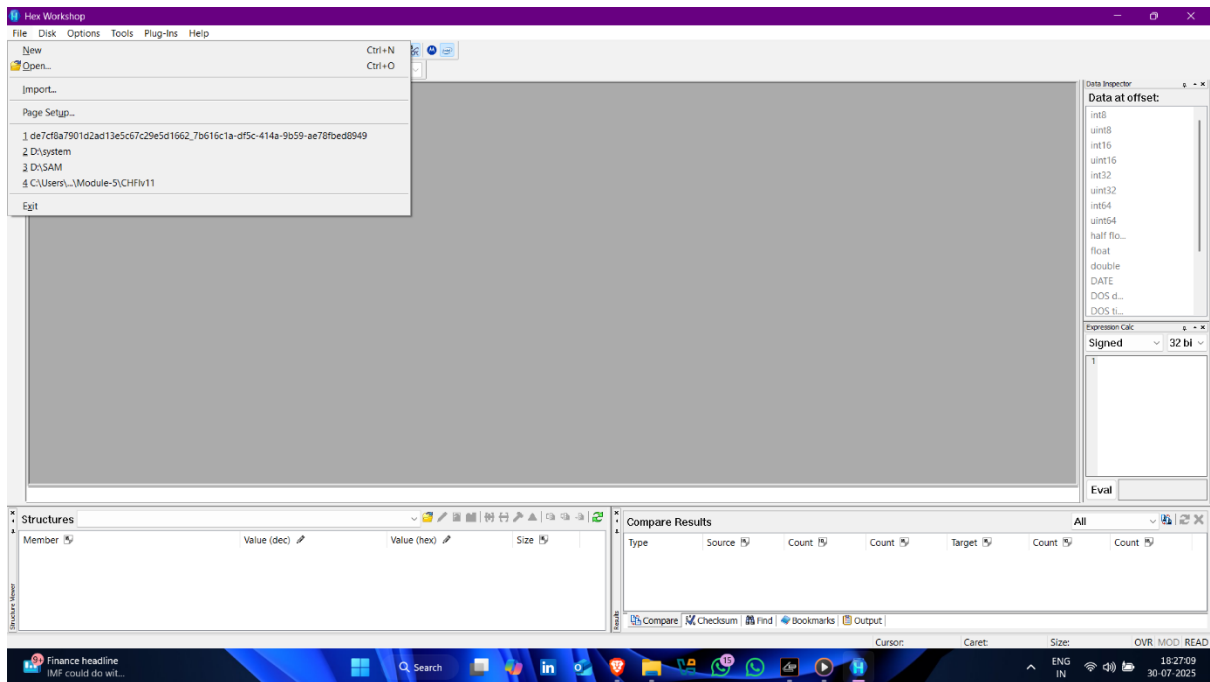
- Open any folder



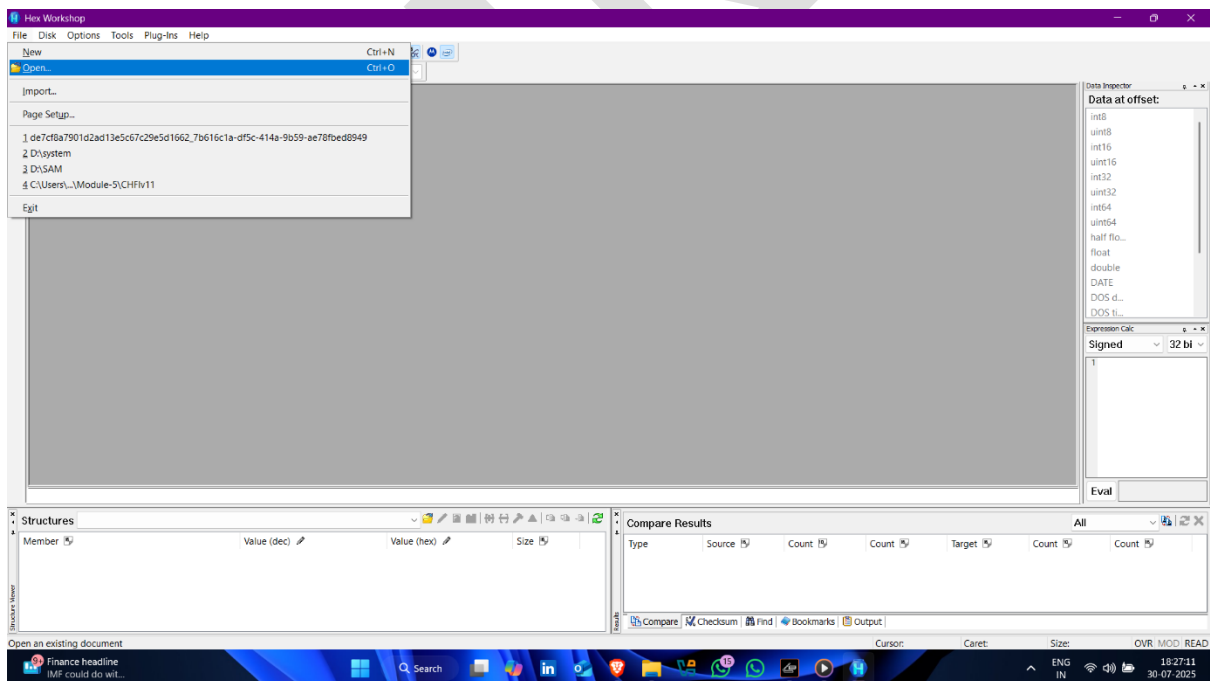
- now open **Hex workshop**



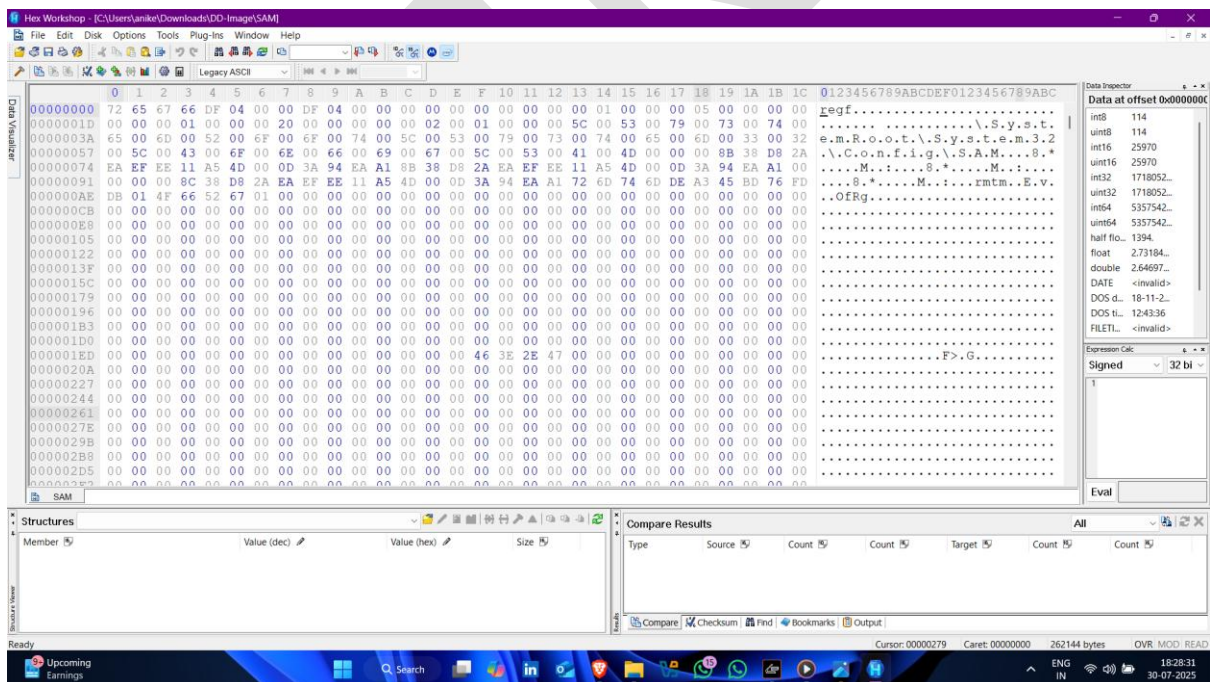
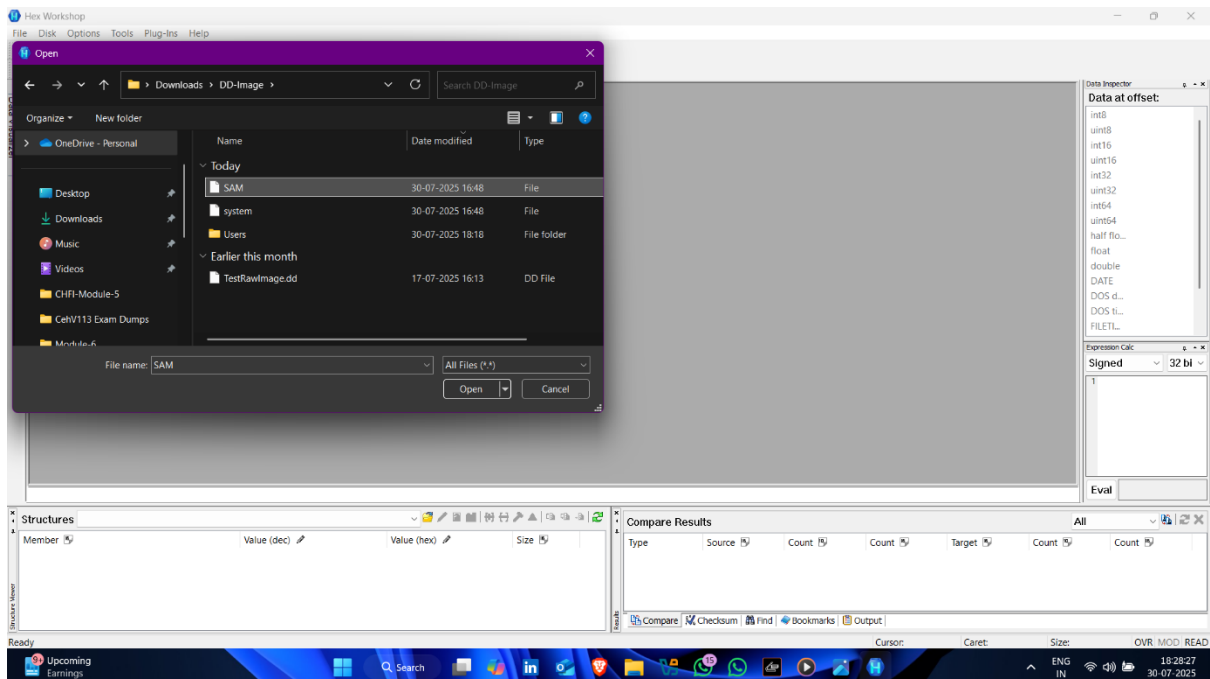
- click on **file** section



- click on **open** option



- select file and click on open



Lab-5-: Examine Web Browser Artifacts

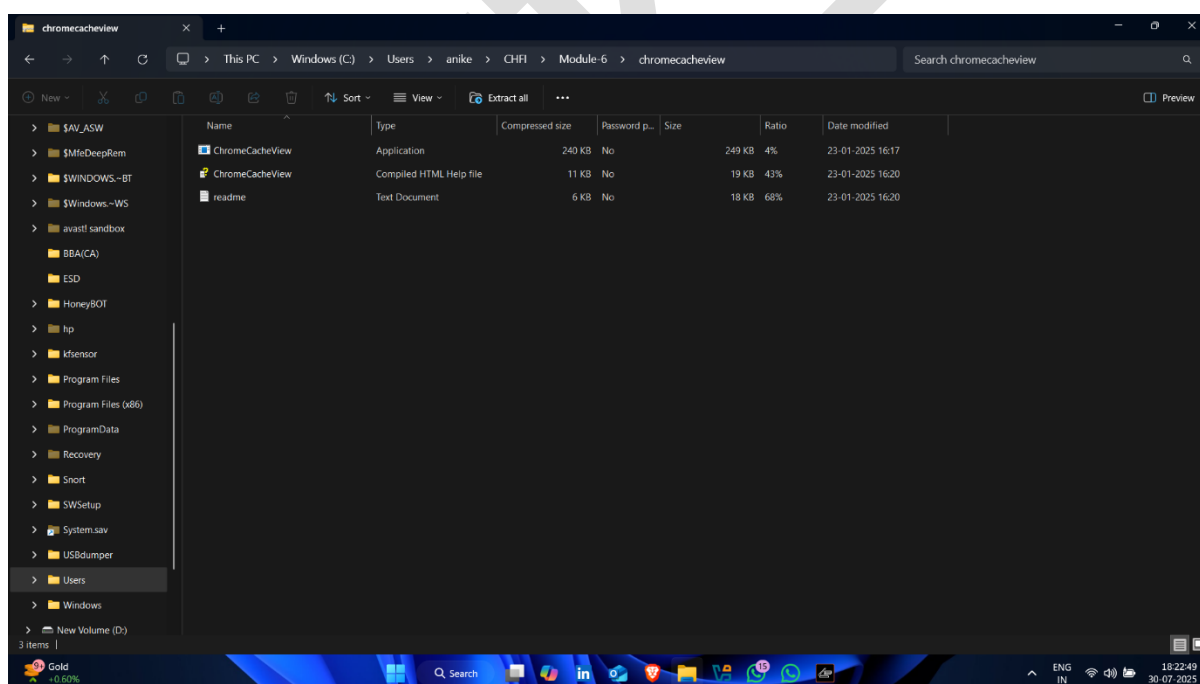
Examining web browser artifacts means **retrieving and analyzing evidence** left behind by a user's **online activity** through their **web browser** (like Chrome, Firefox, Edge, etc.).

A) Examine Web Browser Artifacts Using ChromeCacheView

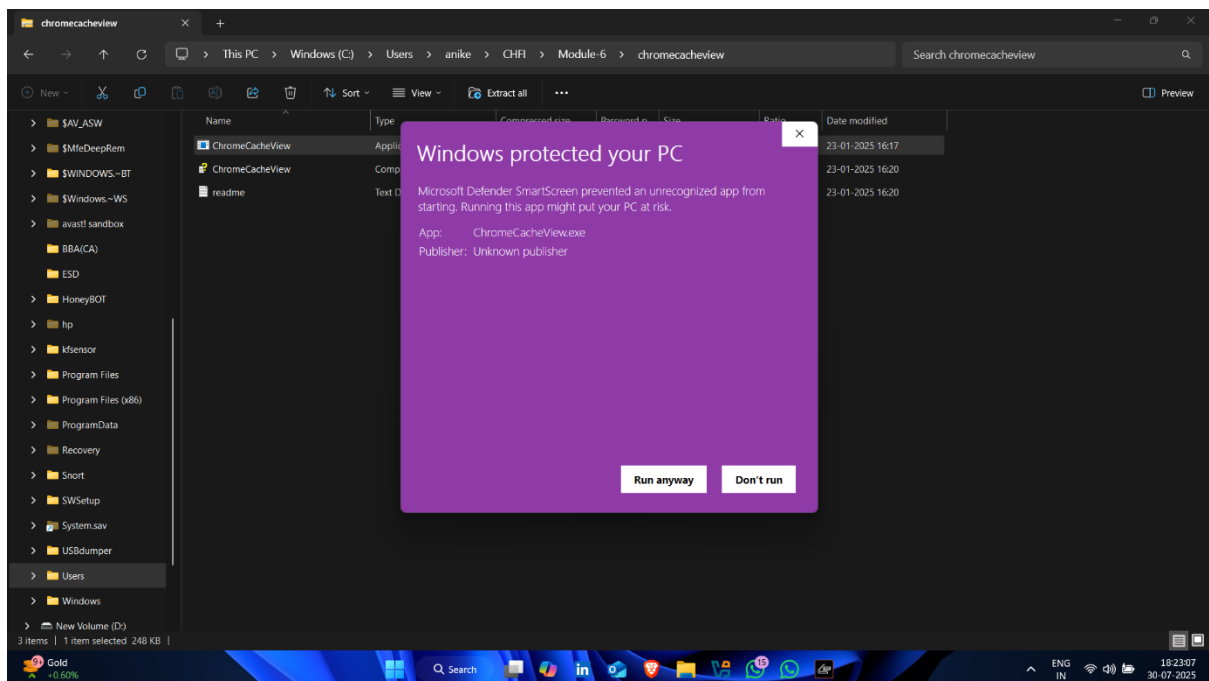
ChromeCacheView is a **free forensic utility** developed by **NirSoft** that allows you to **view and extract files** stored in the **Google Chrome web browser cache**.

How to use it -:

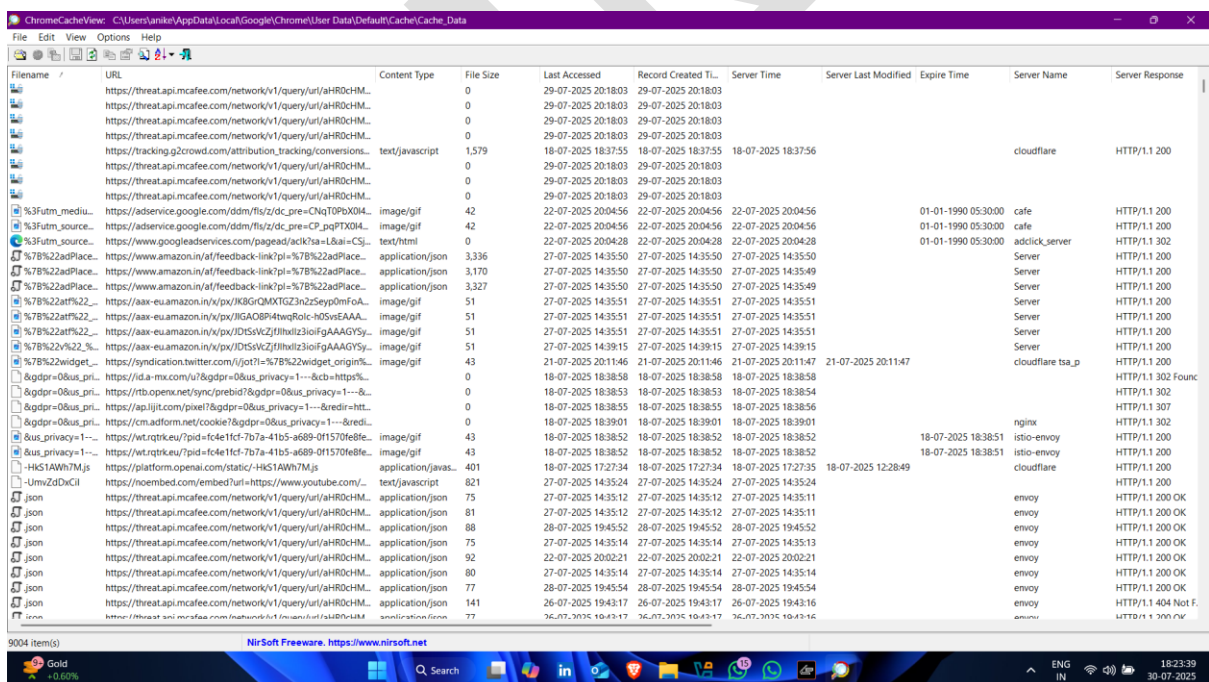
- click on **ChromeCacheView.exe** file and install application



- click on **run anyway**



- retrieve all history of browsers 🖱️✅



Lab-6 Extract And Rebuild Cache Web Pages Of Google Chrome

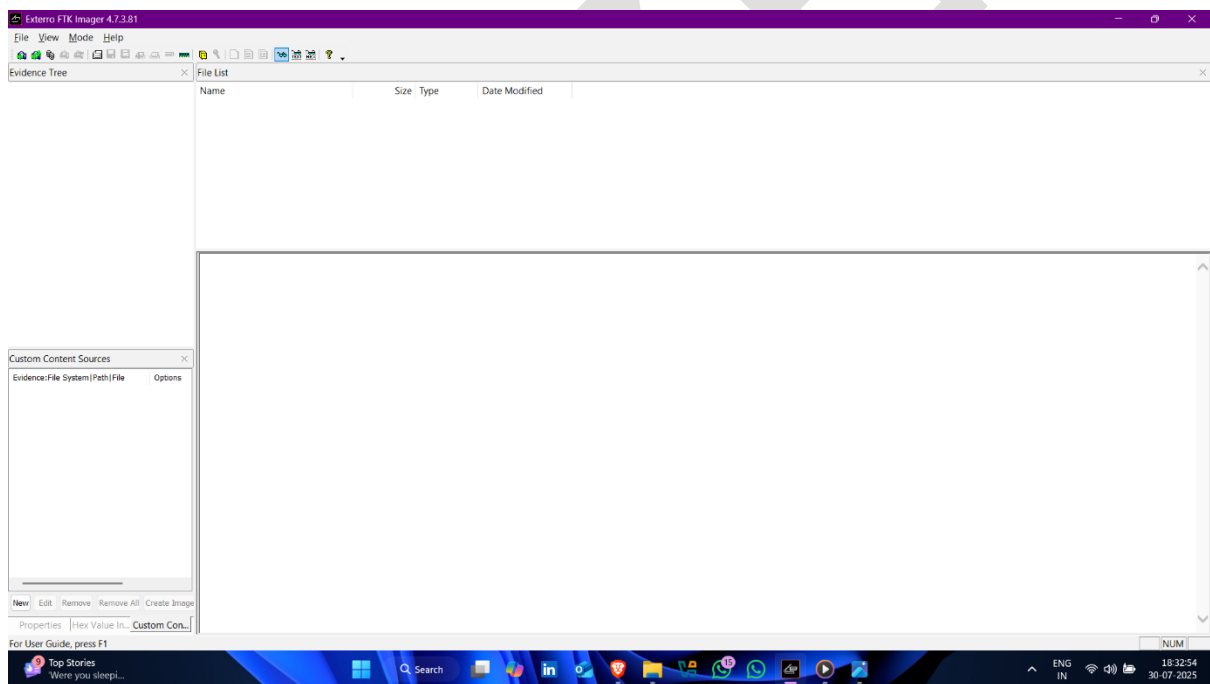
Extract cached content (HTML, images, scripts, etc.) from Chrome's local cache storage

🔑 **Rebuild** or **reconstruct** those web pages to see what the user visited — even if the browser history was deleted.

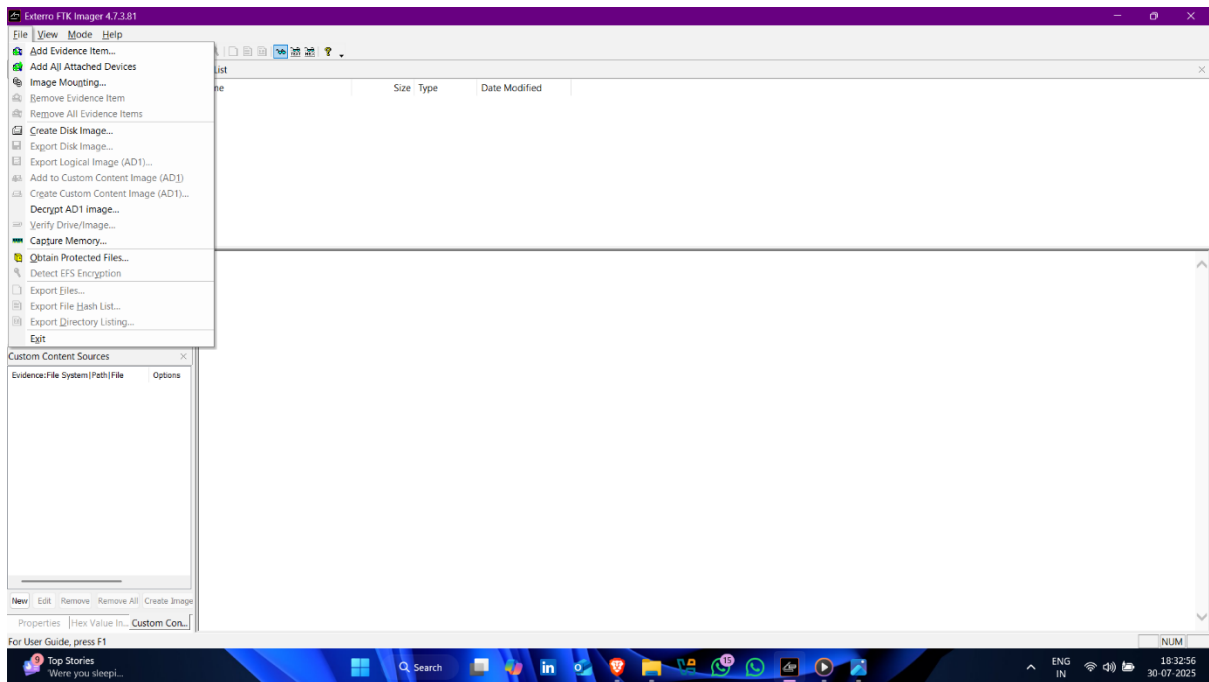
A) Extract And Rebuild Cache Web Pages Of Google Chrome Using FTK Imager

How to use it -:

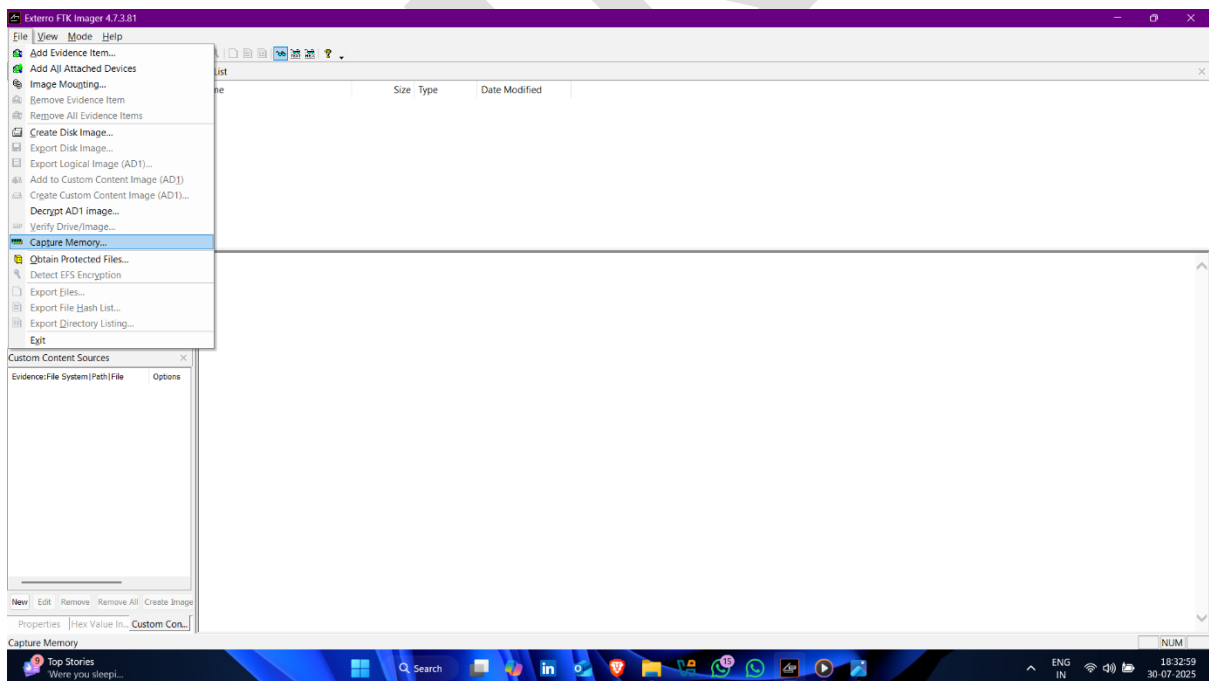
- Open FTK Imager



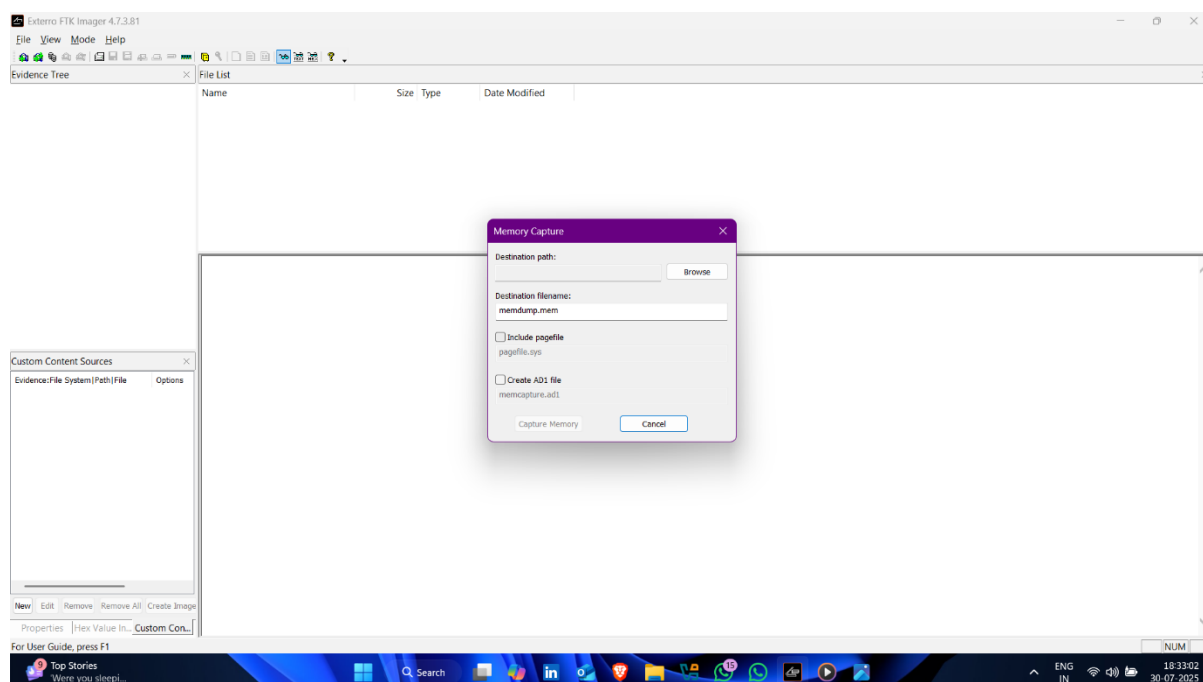
- Click on **File** section



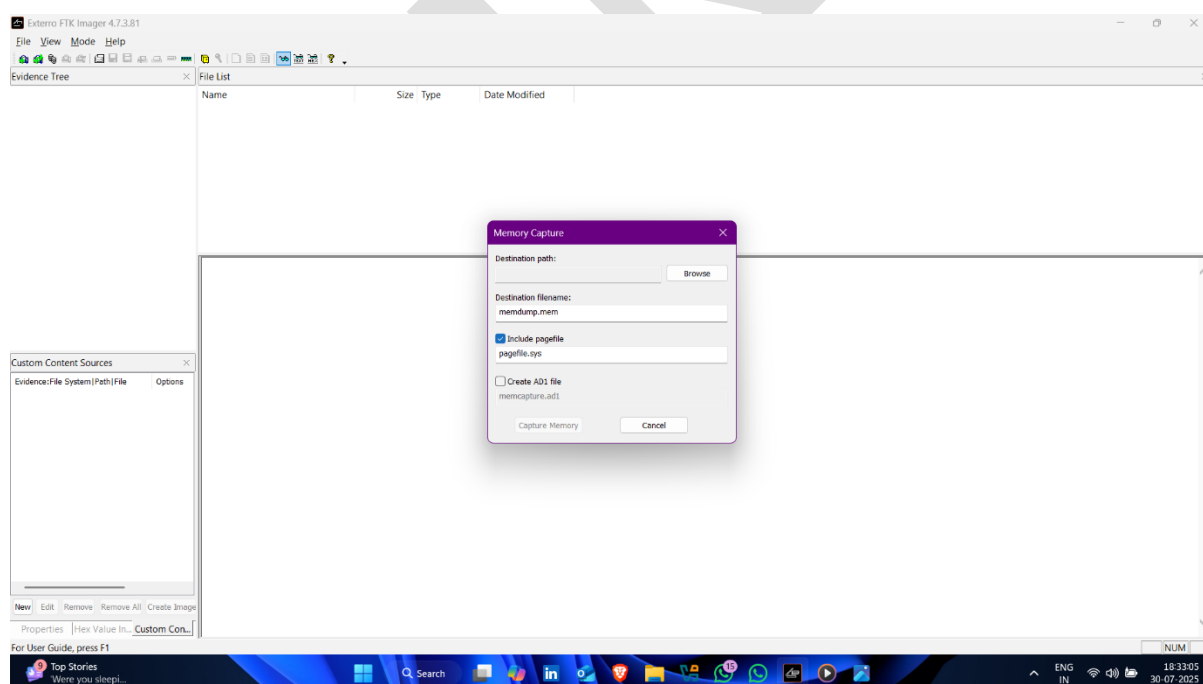
- Click on **capture memory**



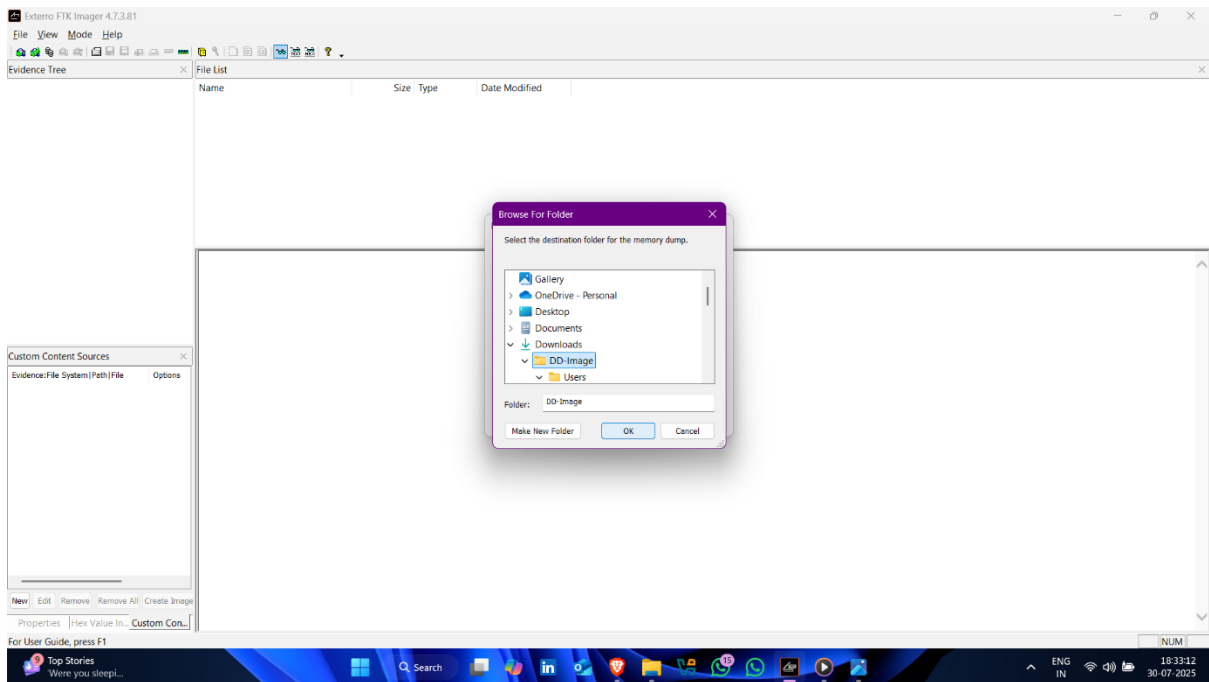
- **Select Include Pagefile**



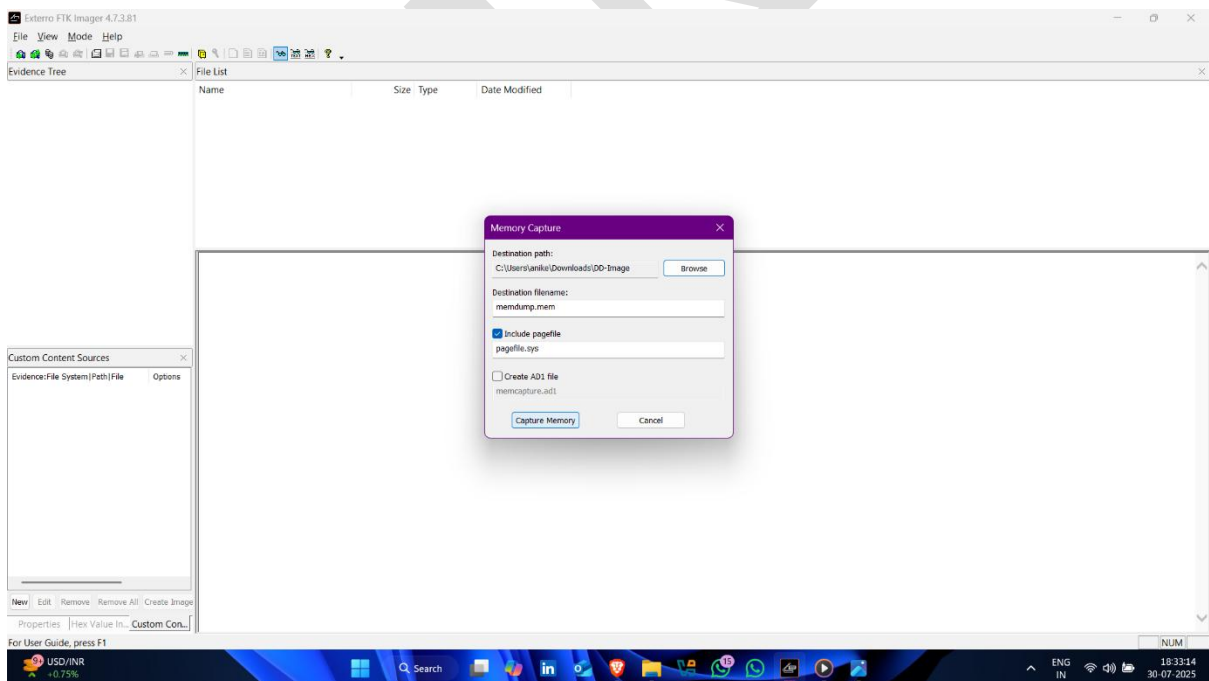
- **Click on **Browse** To set Destination**



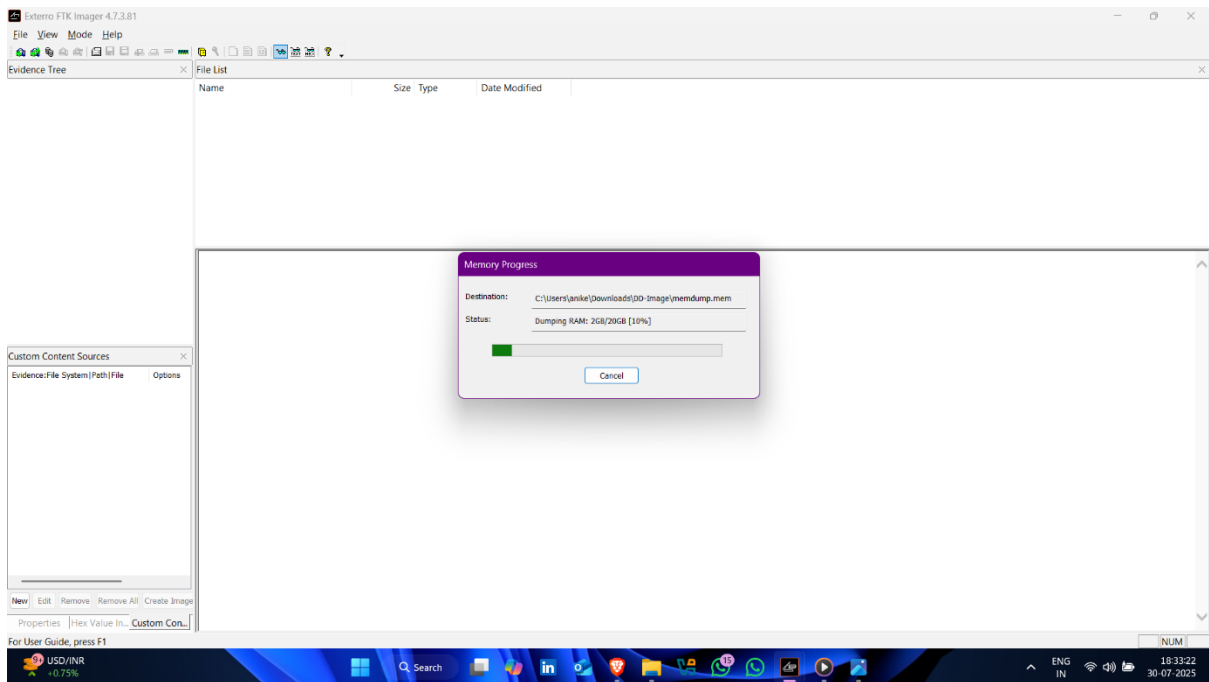
- Select destination and click on **ok**



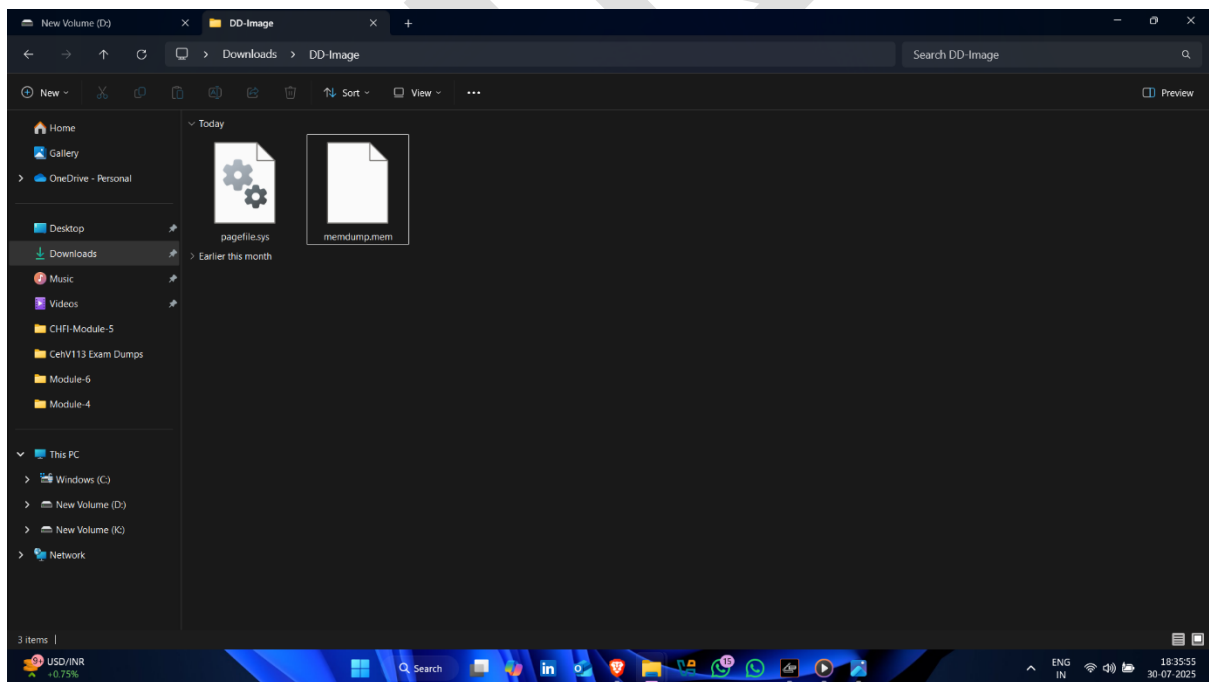
- Click on **Capture Memory**



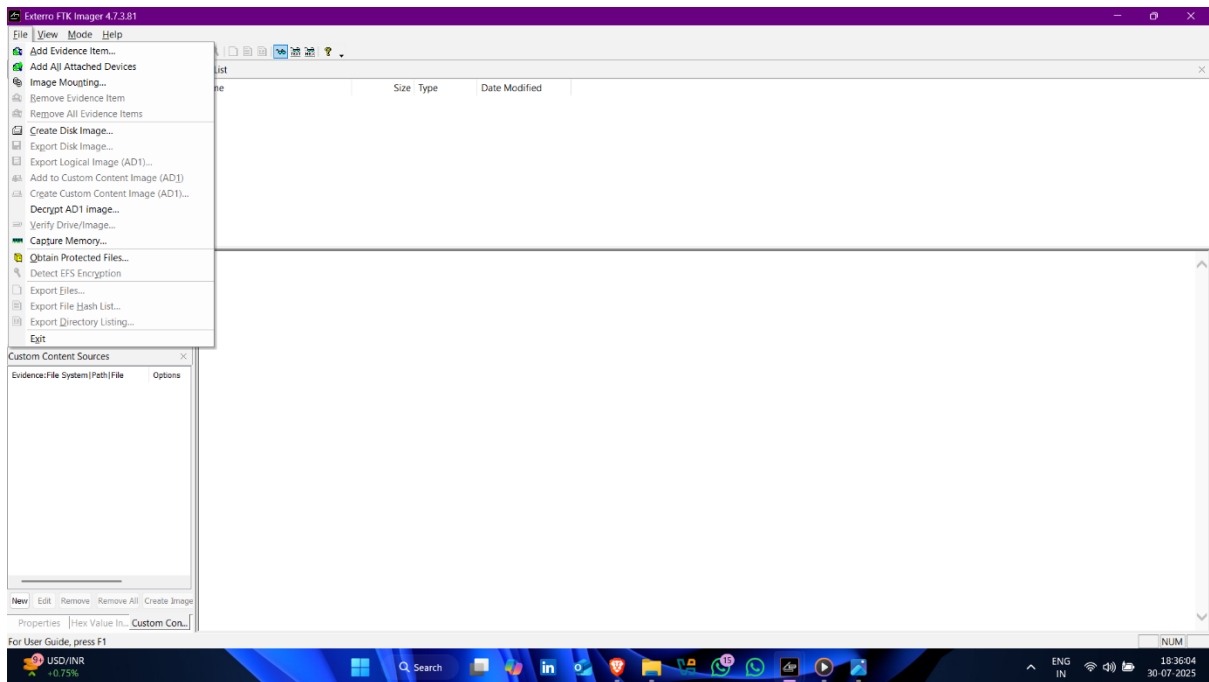
- RamCapture Started



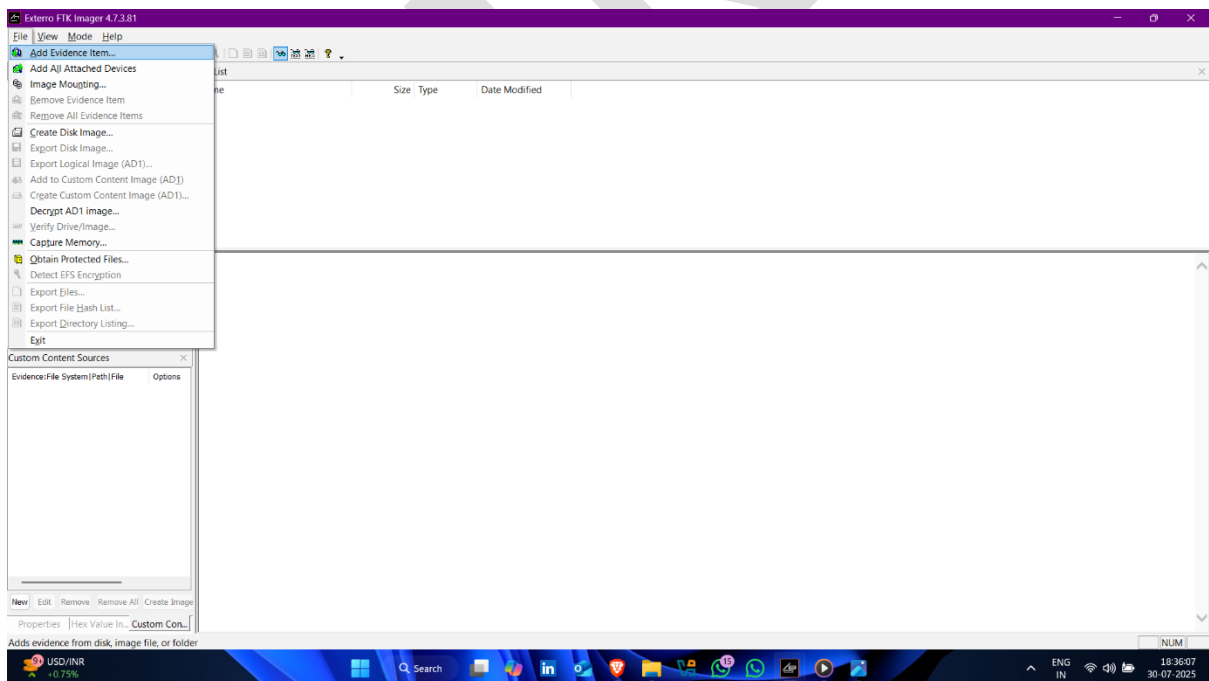
- Ram Image Created 🖱️



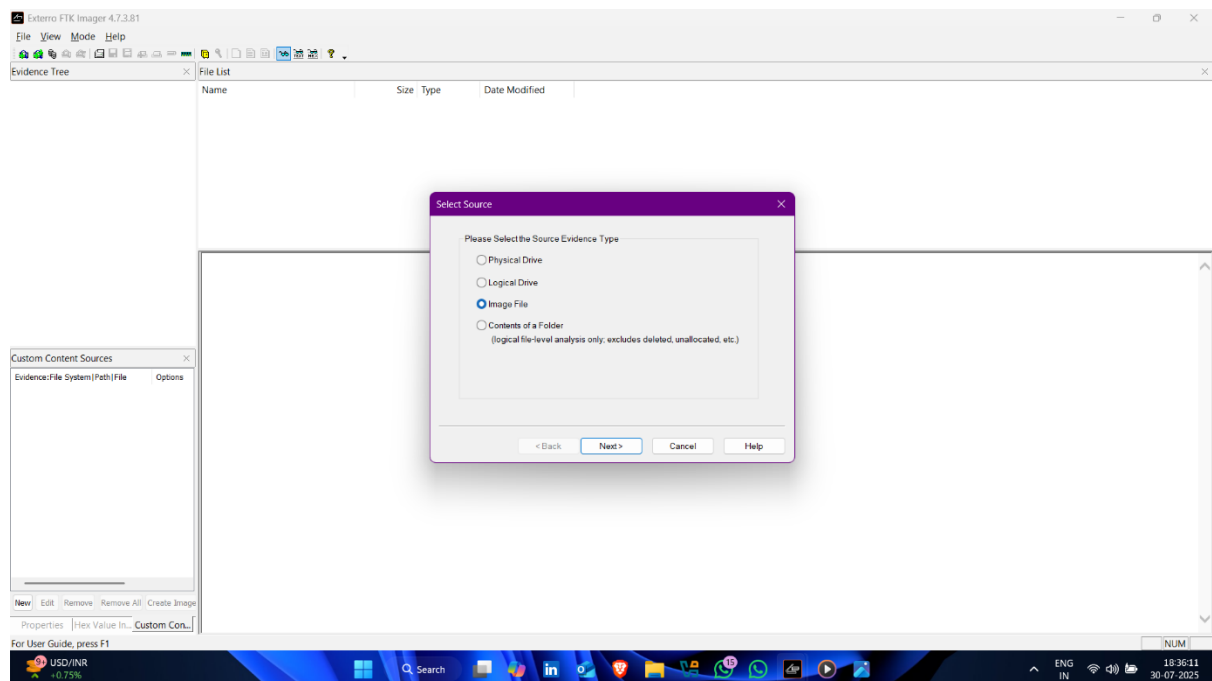
- Now Once again Open **FTK Imager** and click on **file** section



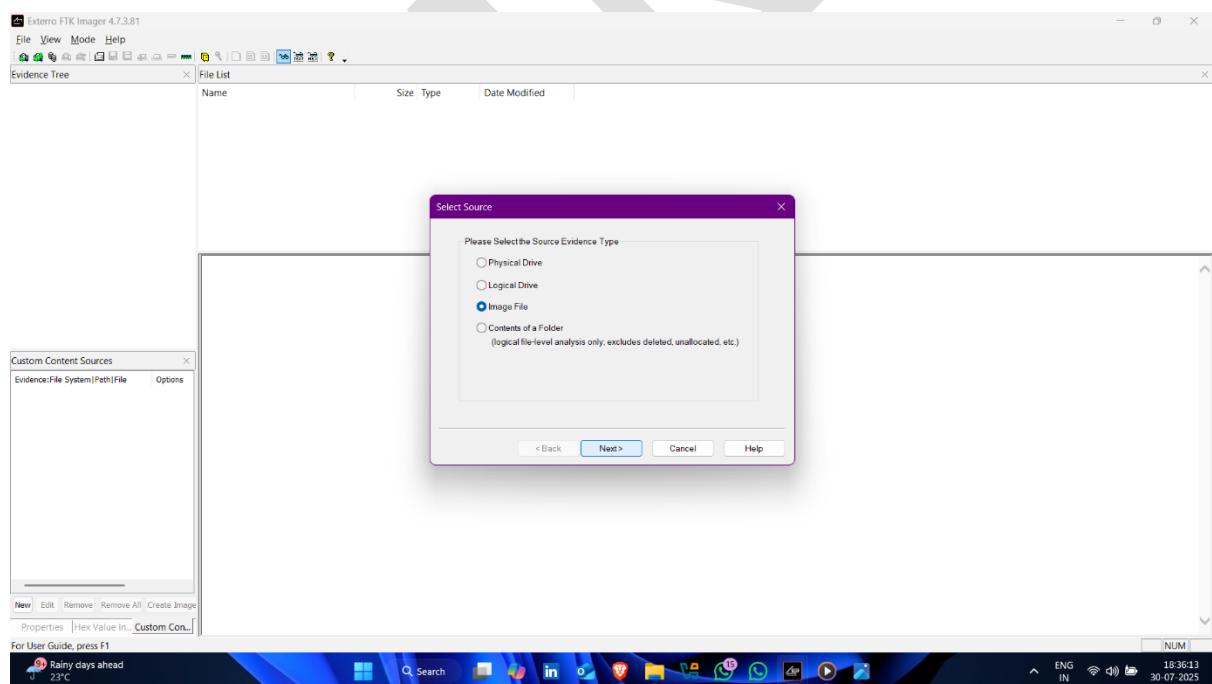
- Click on **Add Evidence Item**



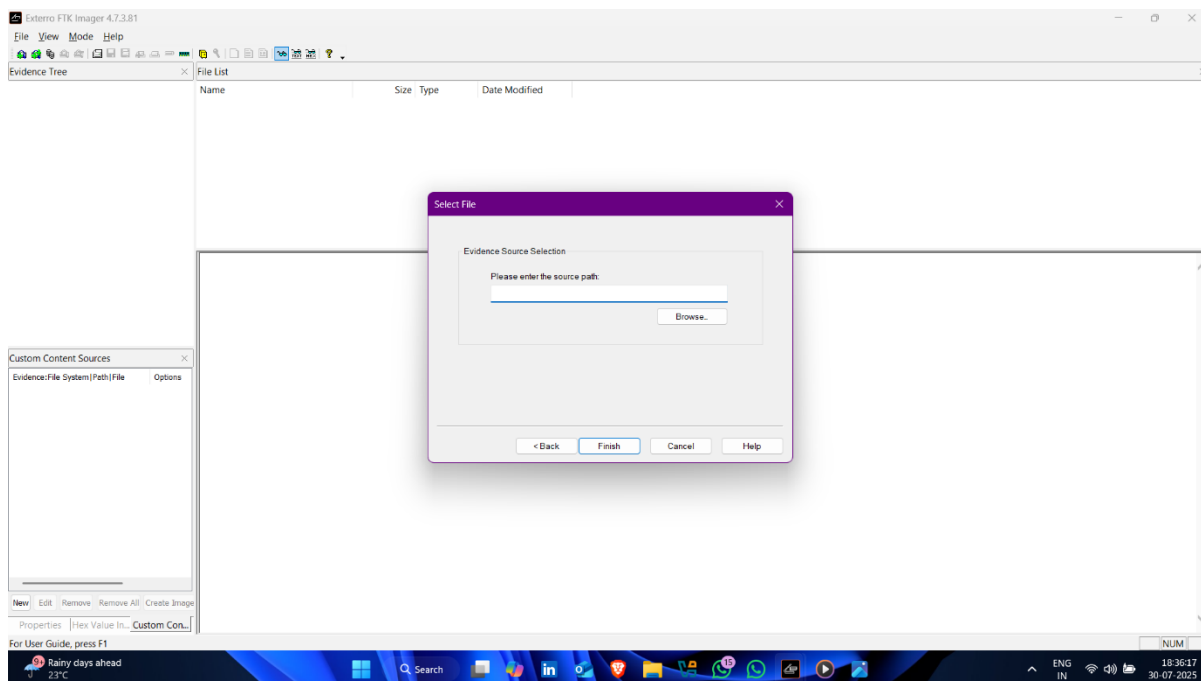
- Click on **Image File**



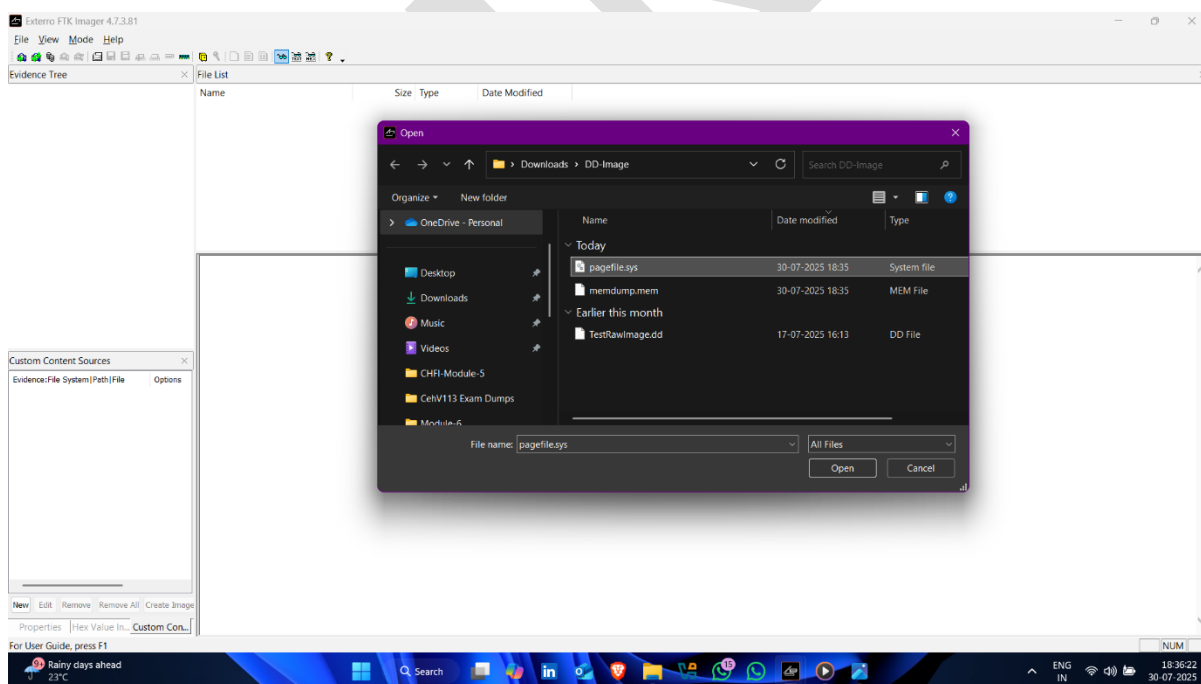
- click on **next**



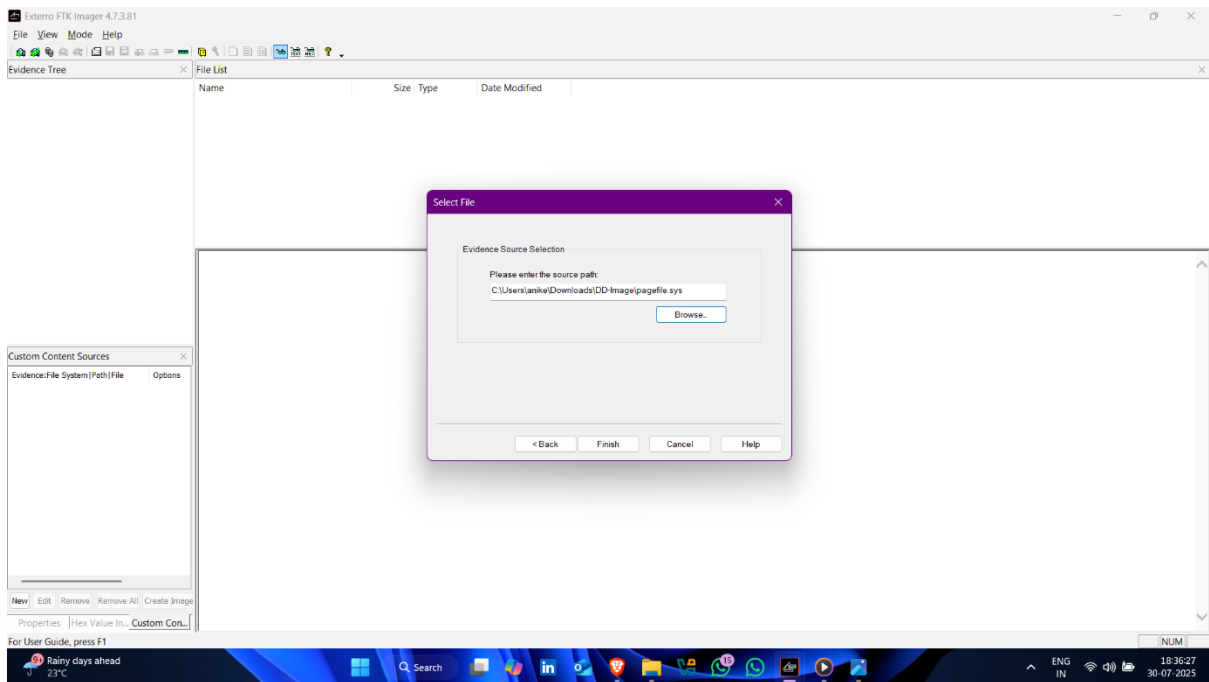
- click on **browse** Option



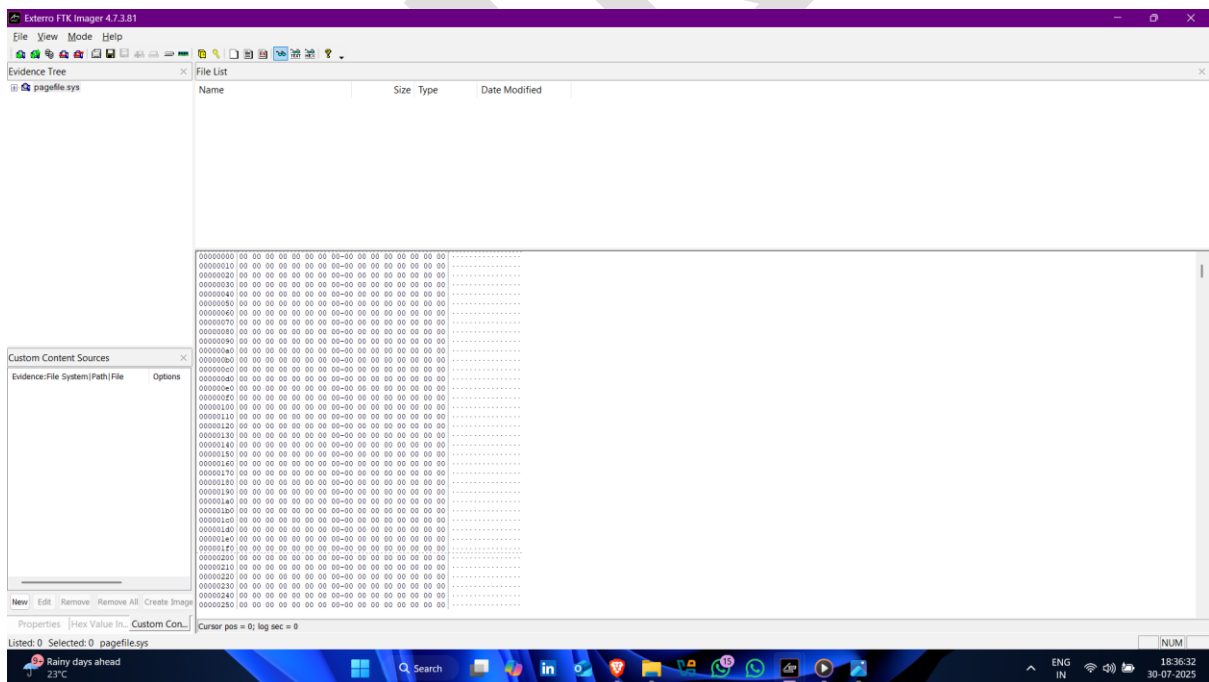
- **Select Image File**

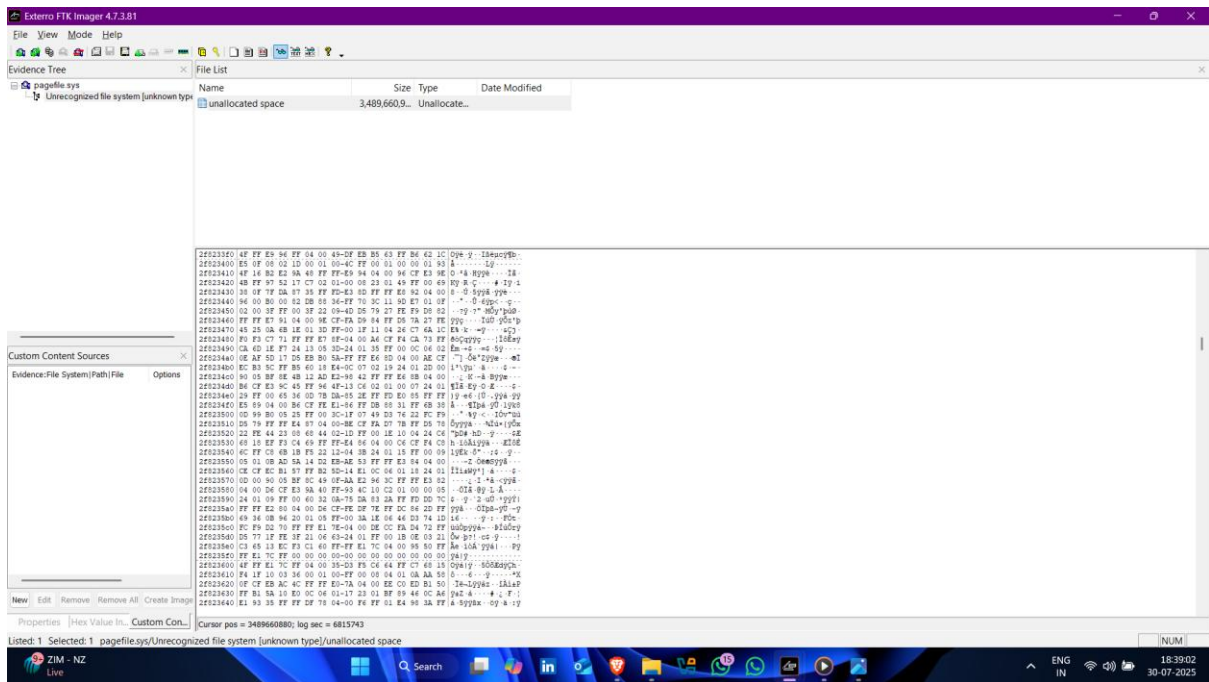


- click on **Finish**



- **Result**  





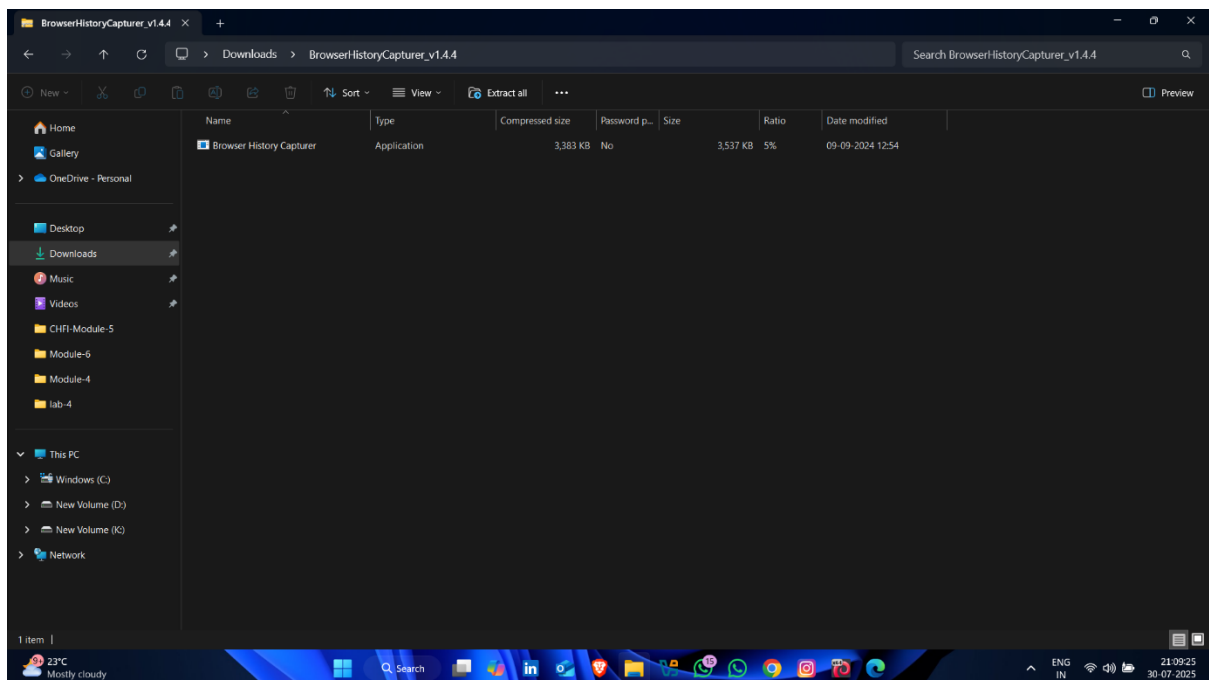
B) Extract And Rebuild Cache Web Pages Of Google Chrome Using Browser History Examiner

Browser History Examiner (BHE) is a **digital forensic tool** developed by **Foxton Forensics** that allows investigators to **extract, analyze, and report** on internet browsing history from popular web browsers like:

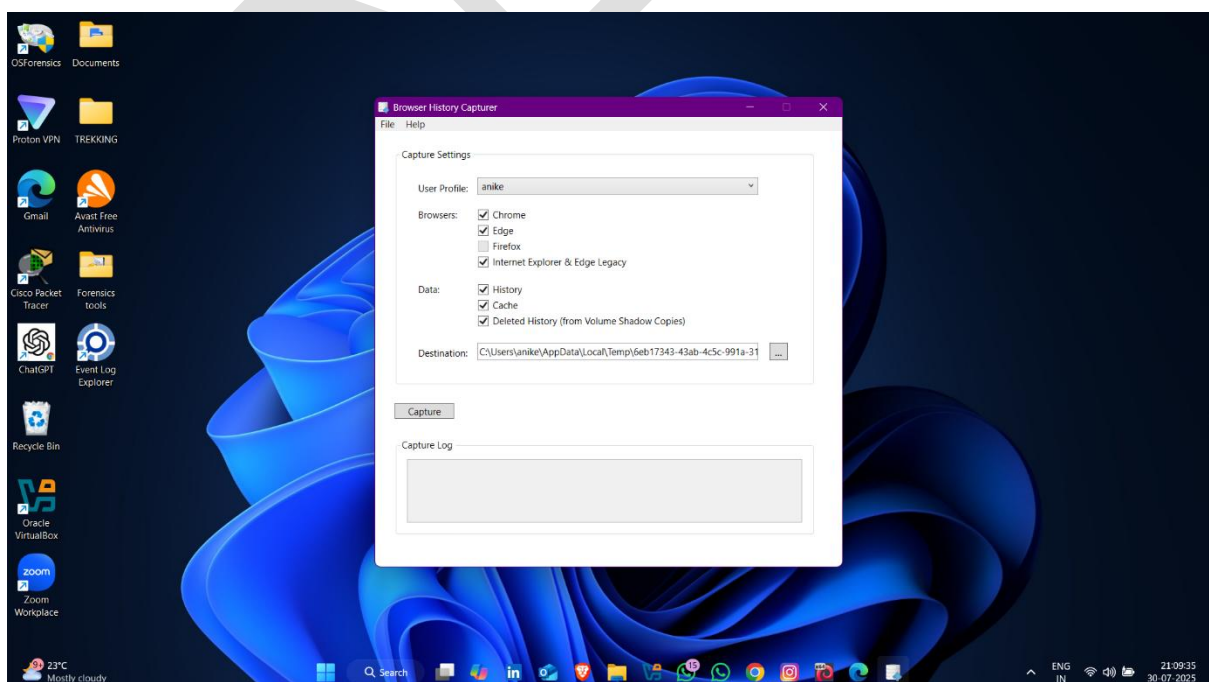
- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Internet Explorer
- Brave, Opera, Chromium-based browsers

How to use it -:

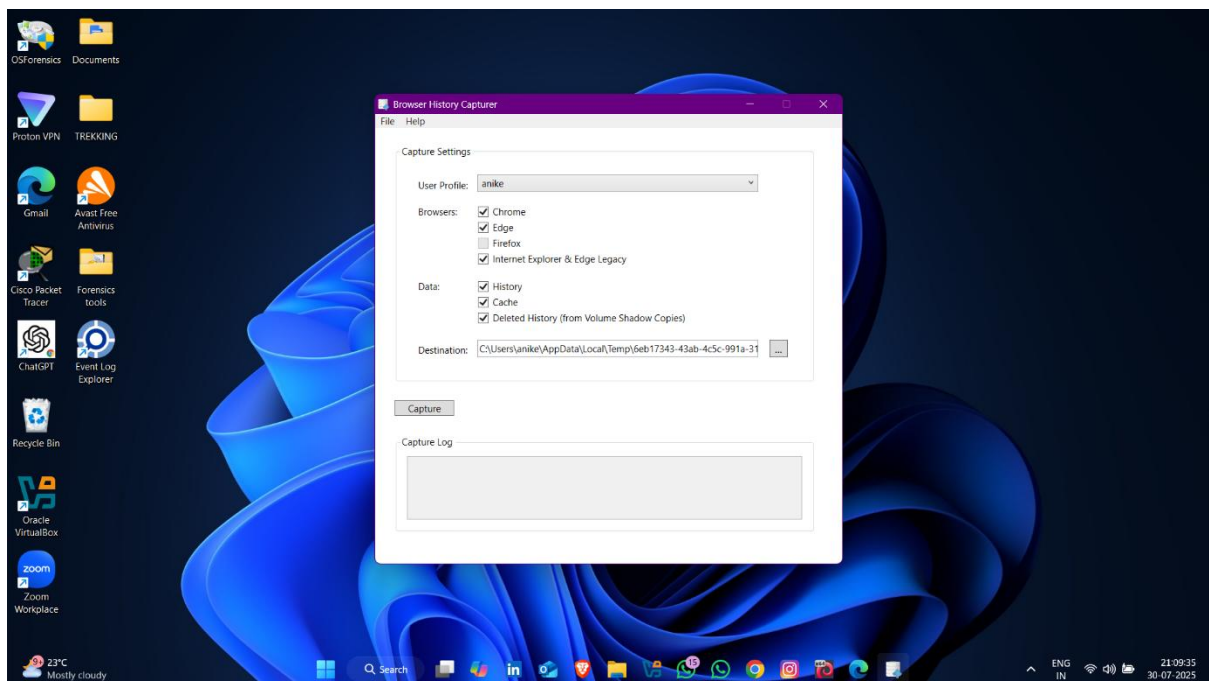
- Run **Browser History Examiner** Application



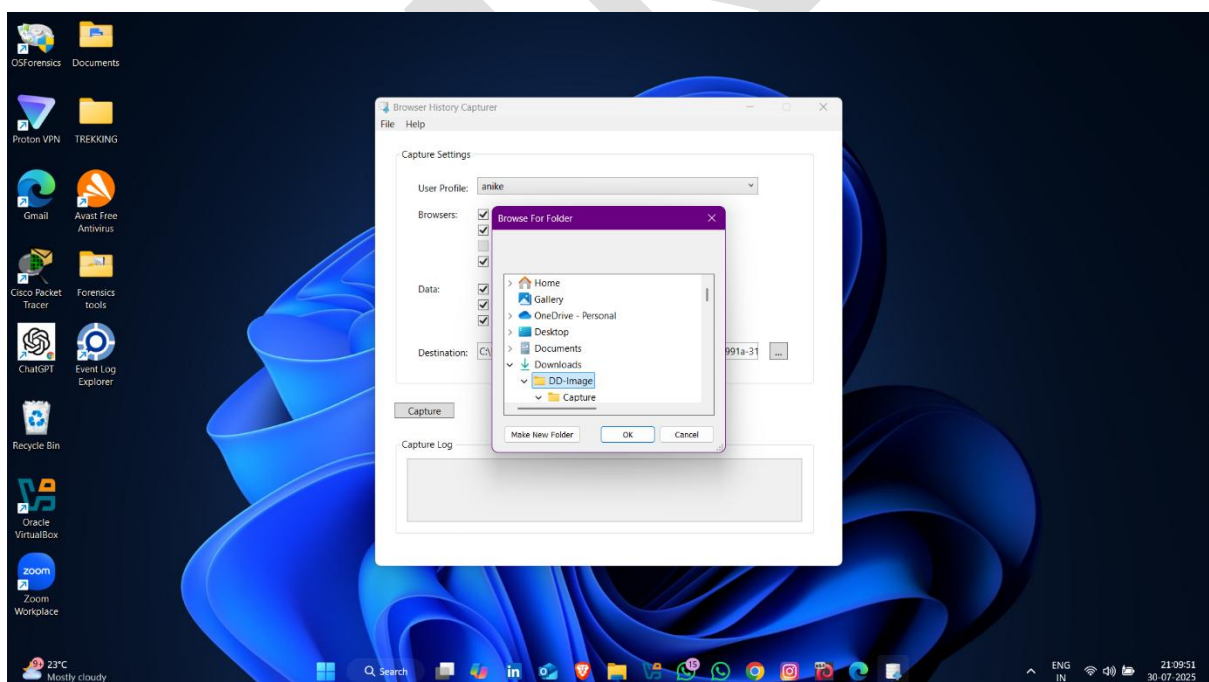
- Application Open 🖱️
- Now select **checkboxes** that you want to retrieve history of browser



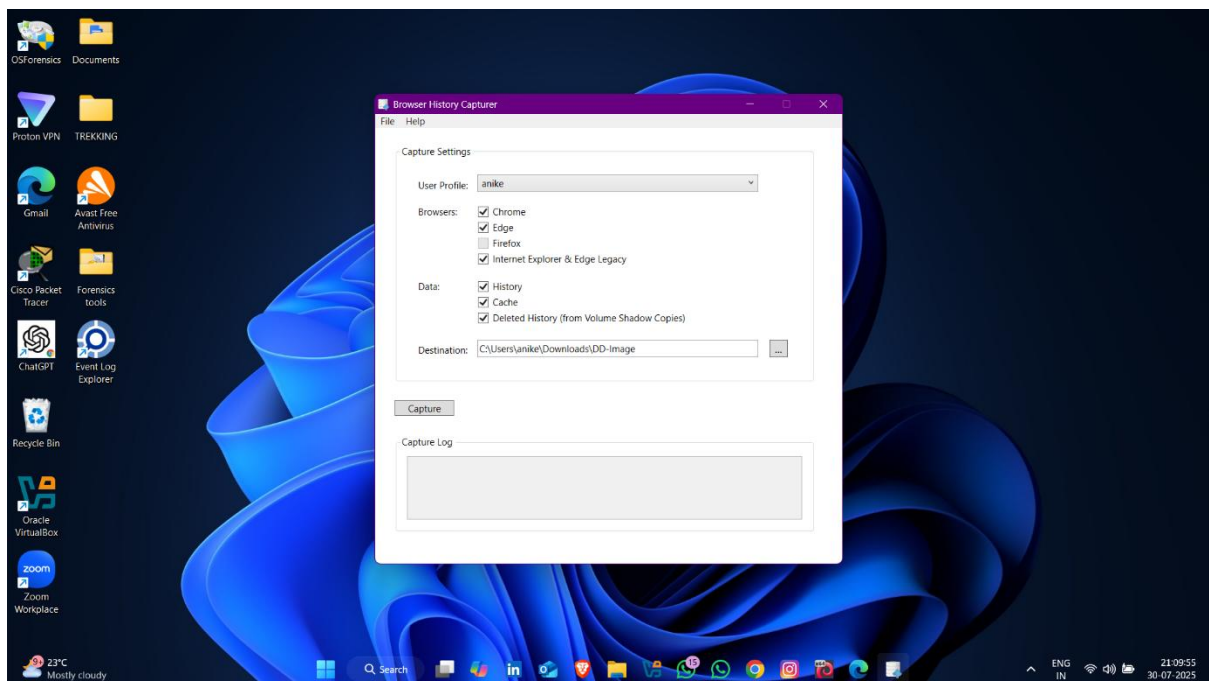
- Then click on **Three Dots (...)** – to select destination path



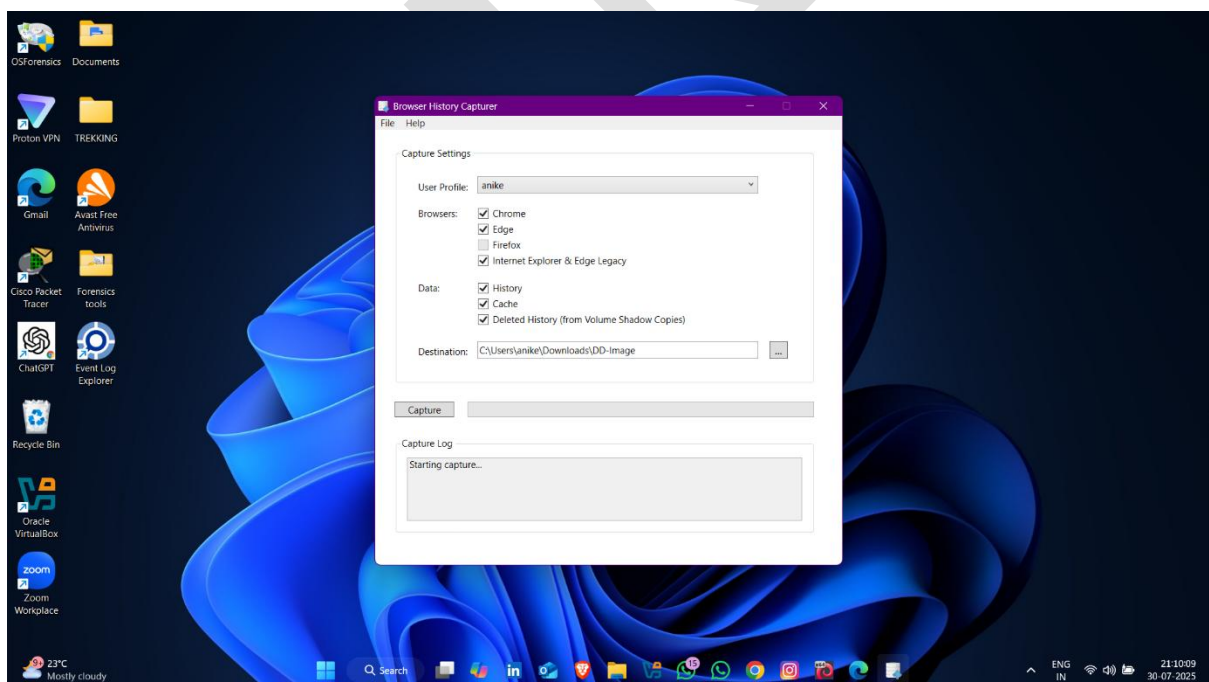
- Select Destination and click on **ok**

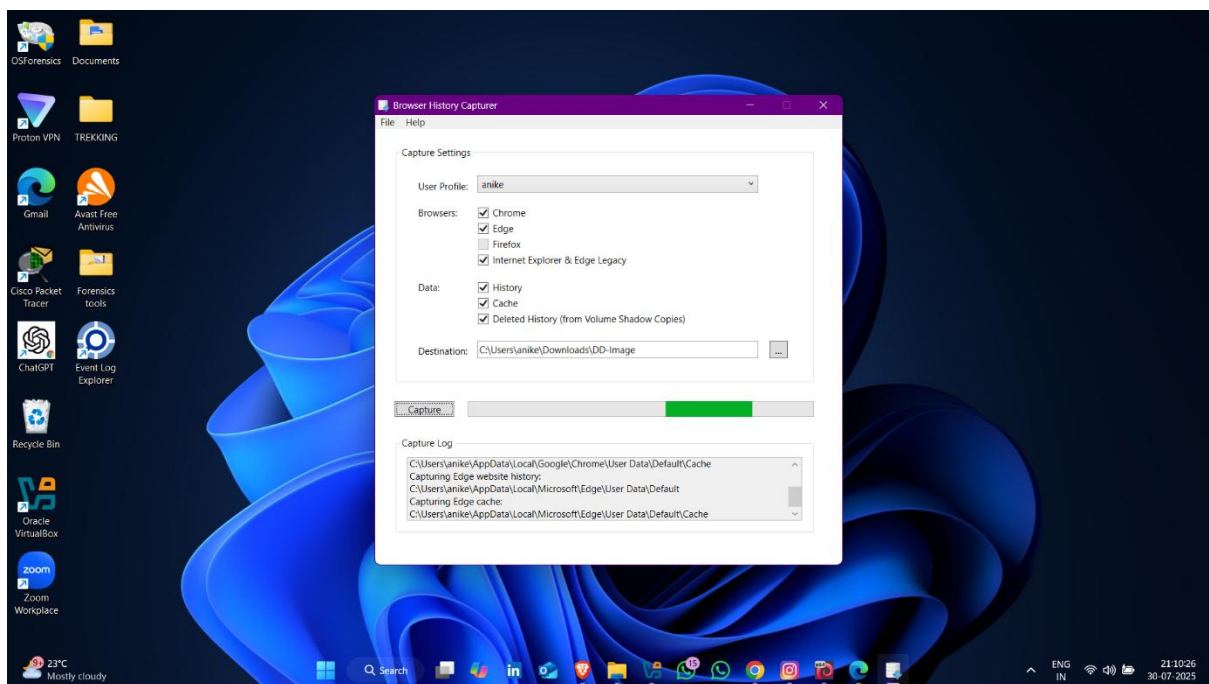


- Click on capture

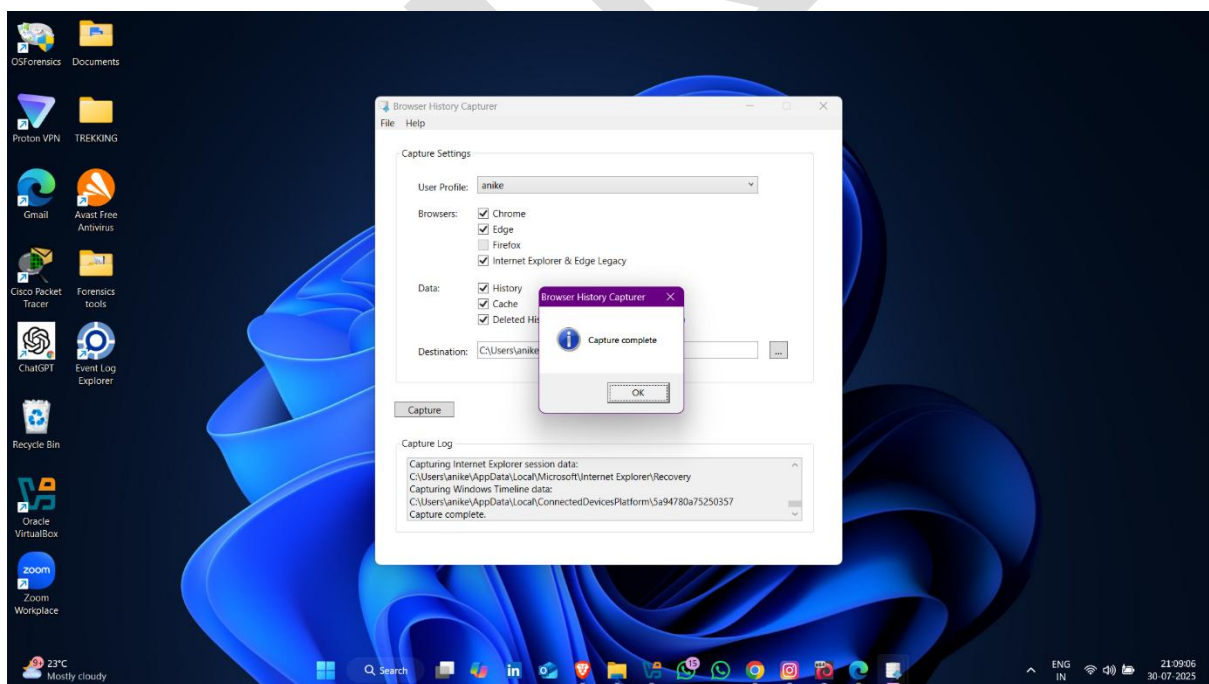


- History Capturing started 🖱️

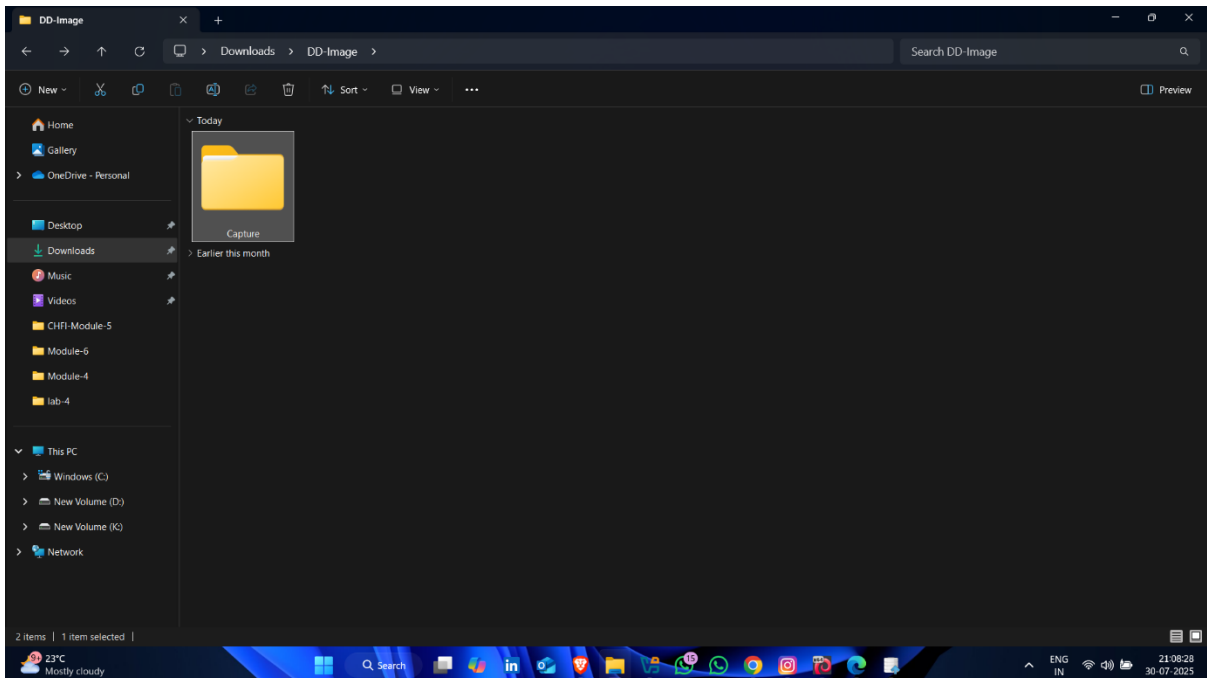




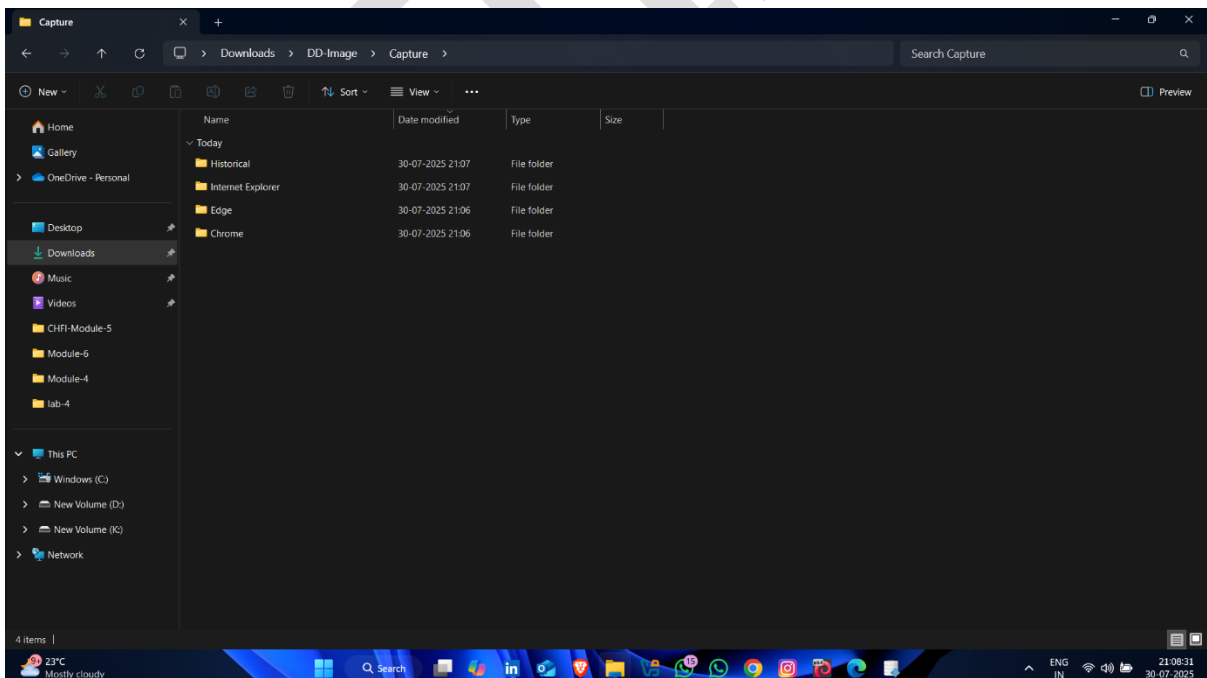
• Capture Complete 🖱️✅

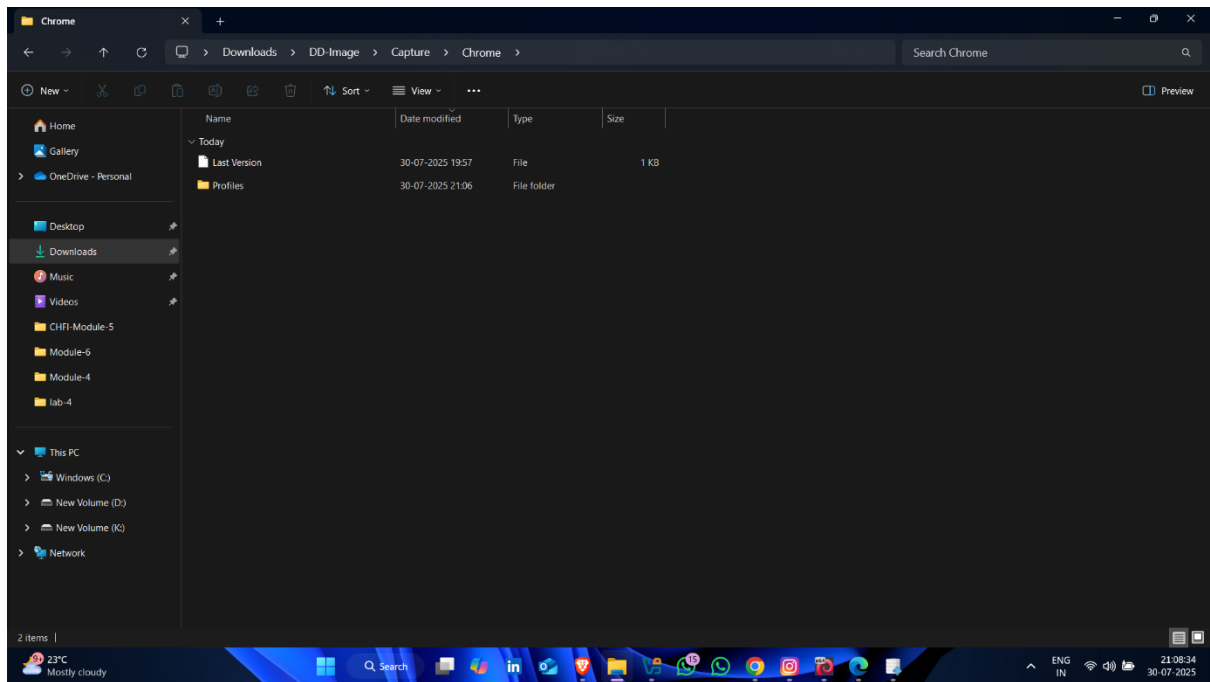


- Go to Destination Path
- Here is created separate folder , open it

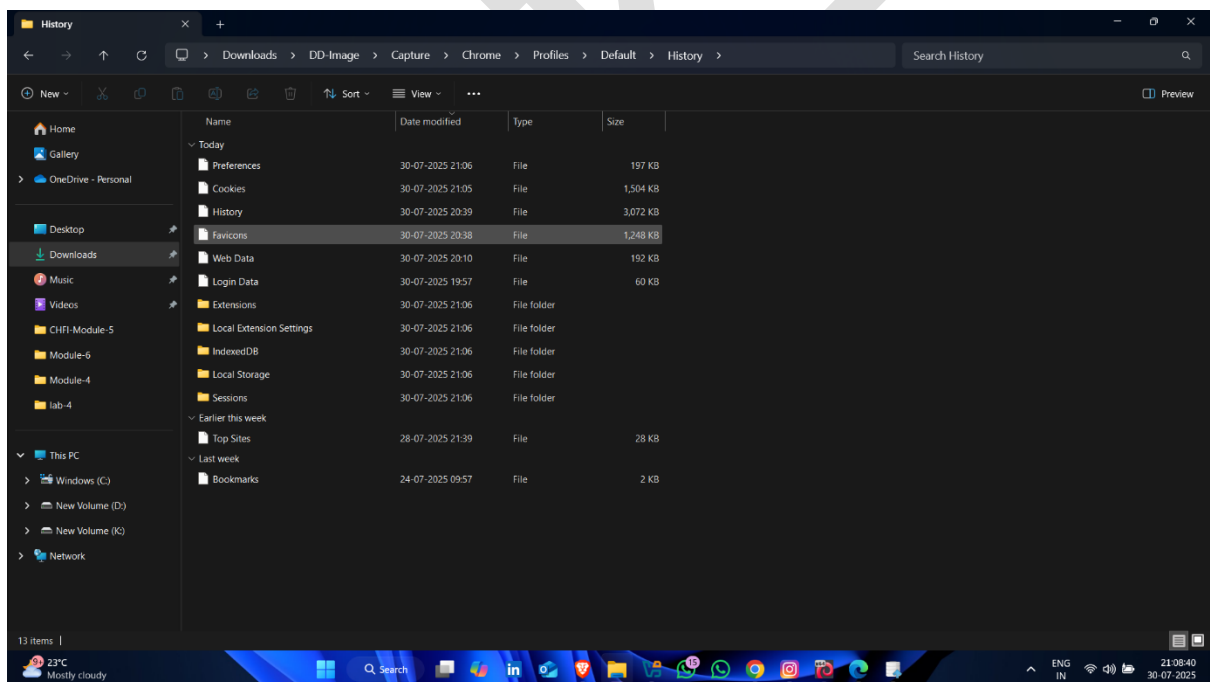


- All Browser history and cache folder





• Result



Lab-7 Discover And Extract Forensics Data From Computer

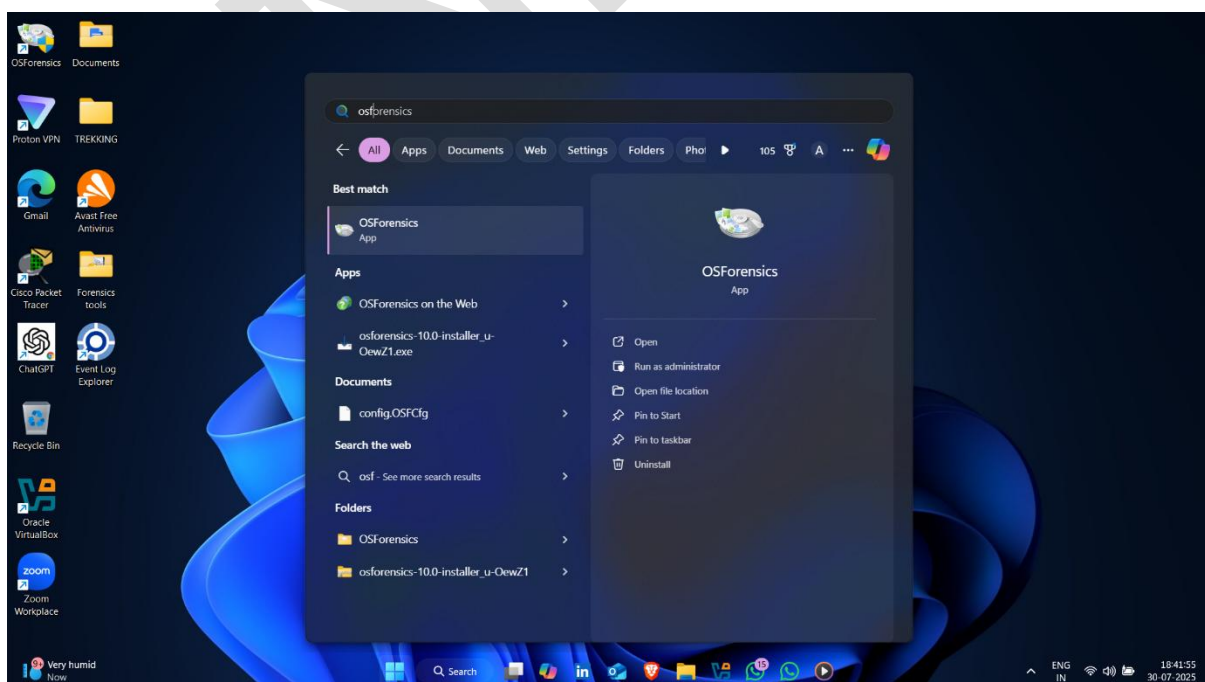
Discover and Extract Forensics Data from a Computer is a crucial process in **computer forensics** where investigators identify, recover, and preserve digital evidence from a computer system in a forensically sound manner. This process is used in cybercrime investigations, legal cases, incident response, and internal audits.

A) Discover And Extract Forensics Data From Computer Using OSForensic Tool

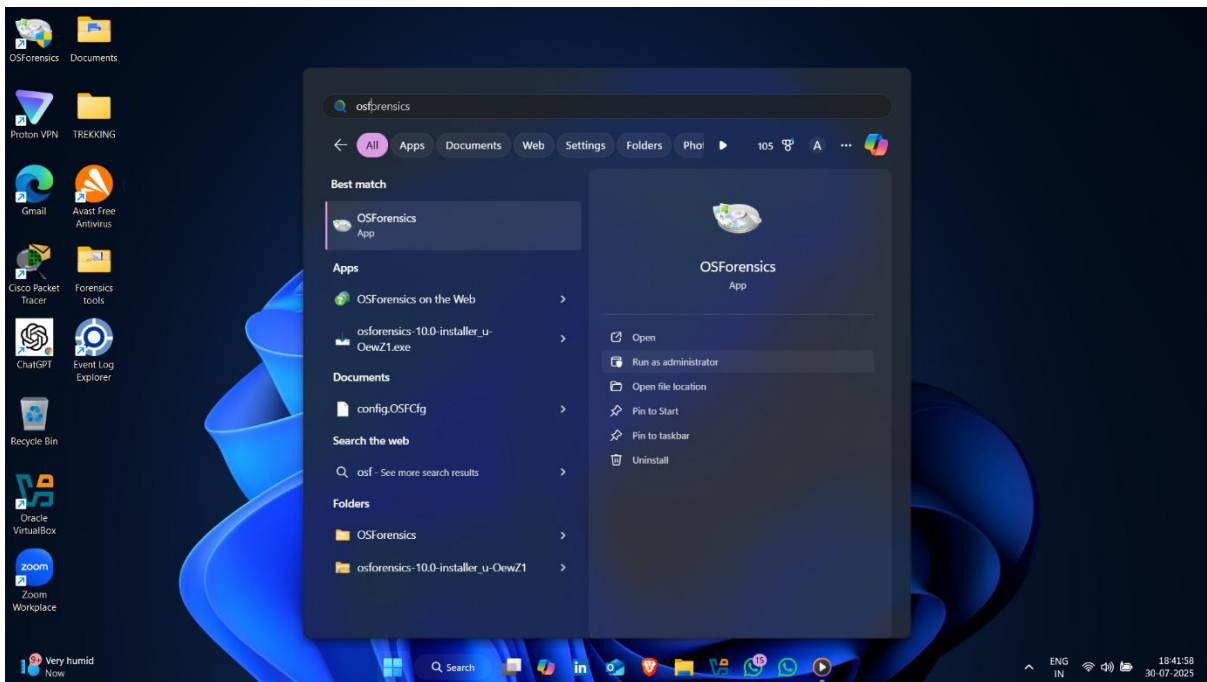
OSForensics is a powerful digital forensics tool developed by **PassMark Software** used by forensic investigators to **identify, collect, analyze, and preserve digital evidence** from computers. It is widely used in law enforcement, cybersecurity, and corporate investigations.

How to use it -:

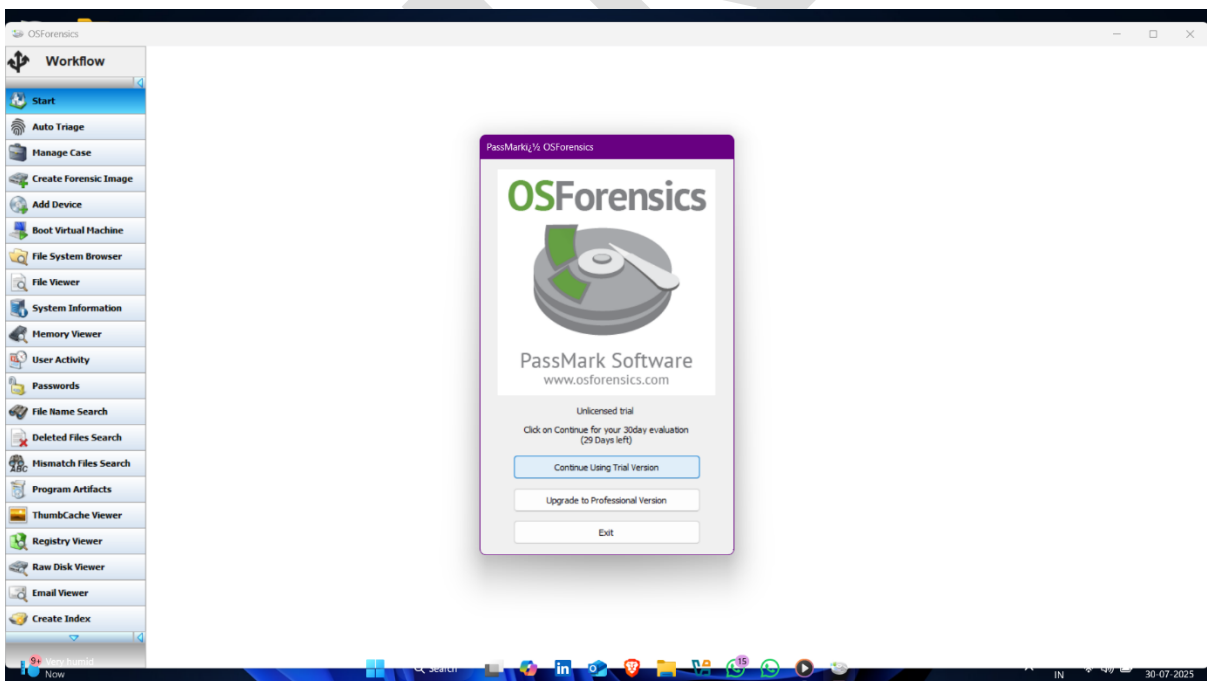
- Open **OSForensics** Application



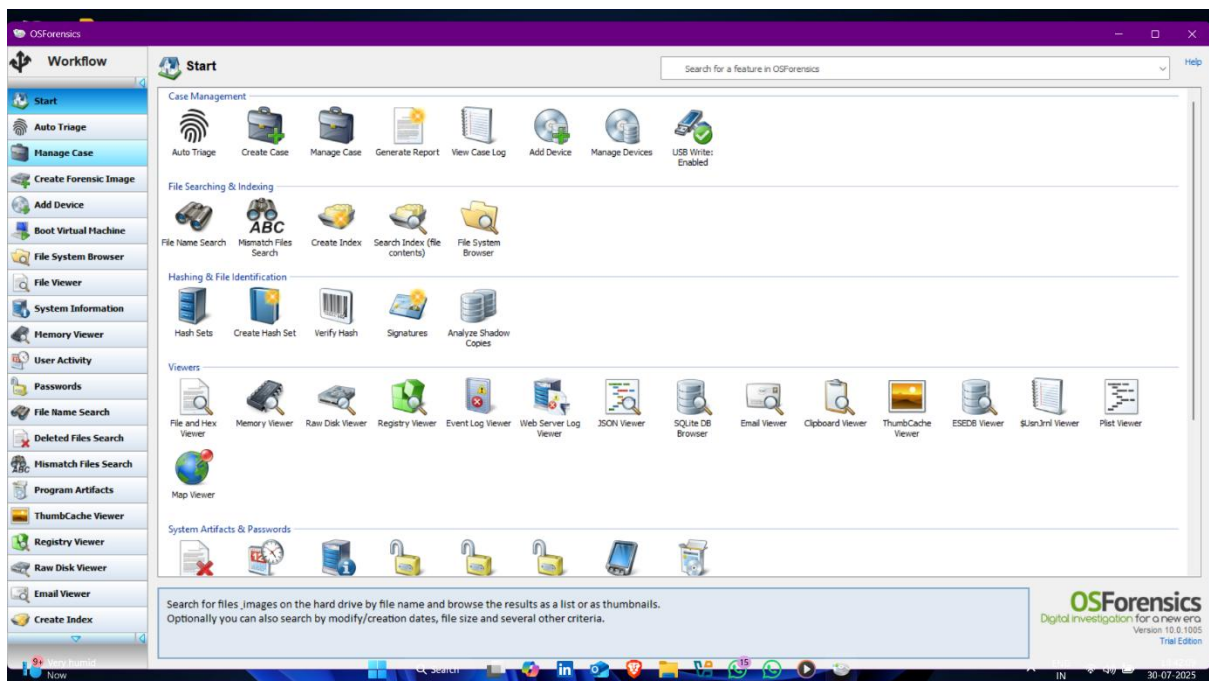
- Click on **Run as administrator**



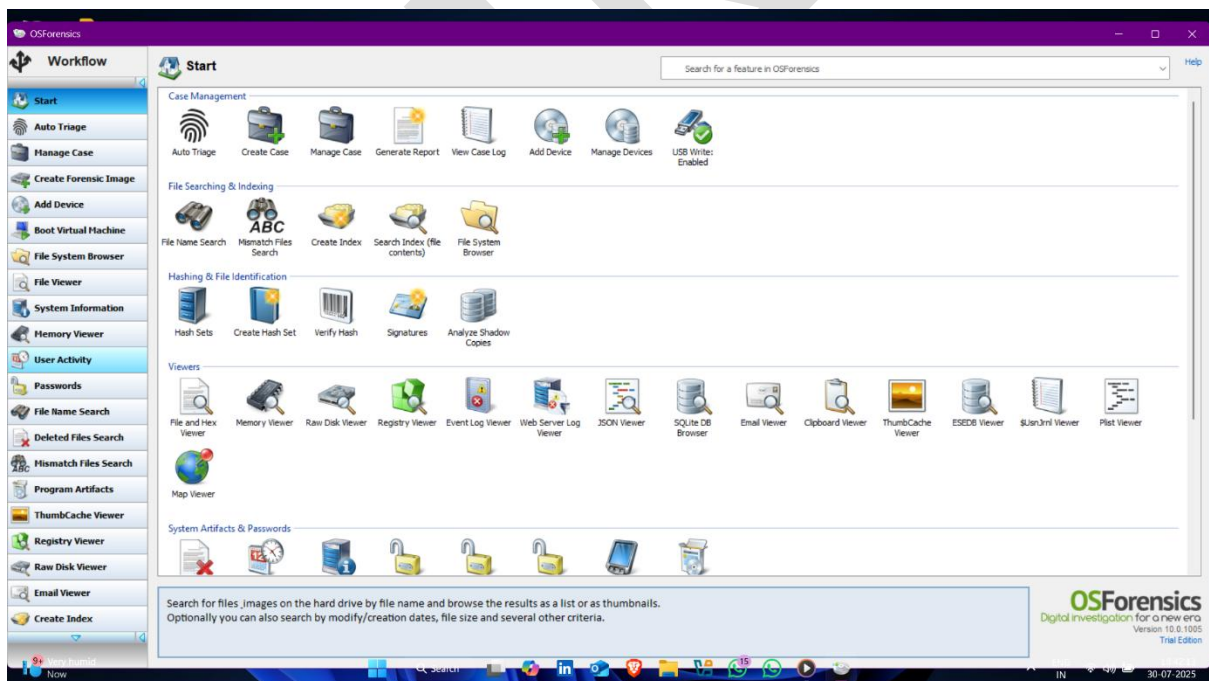
- Click on **Continue Using Trial Version**



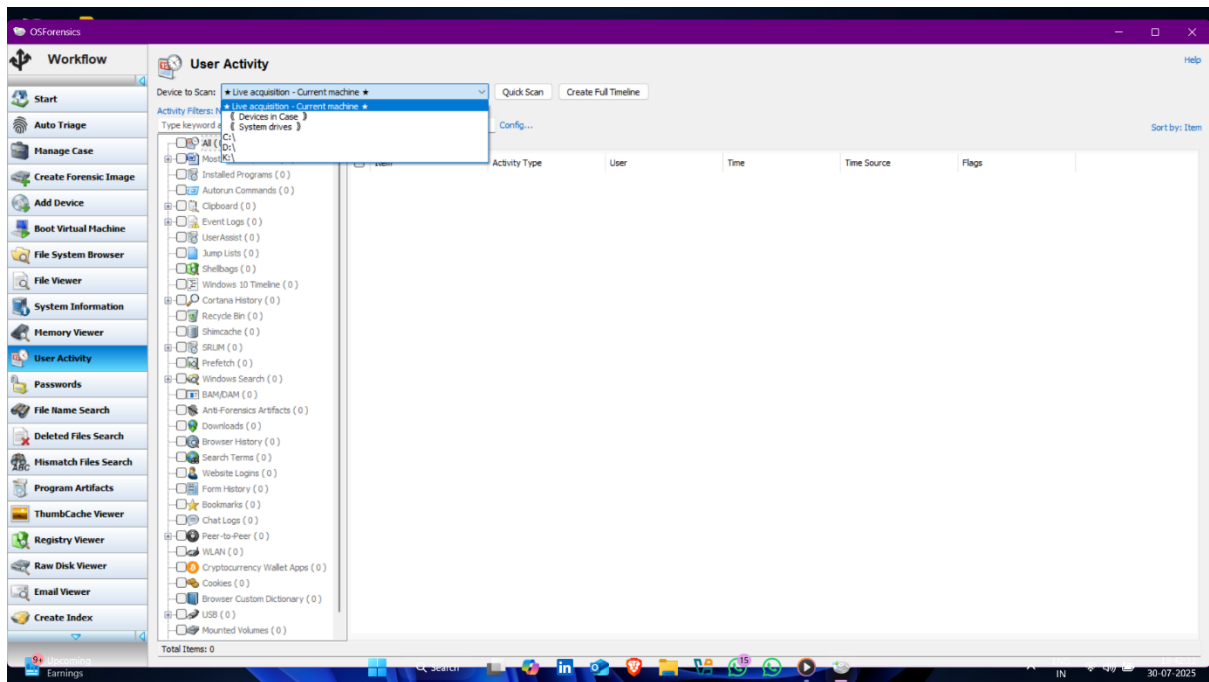
- **OsForensics Application Started**



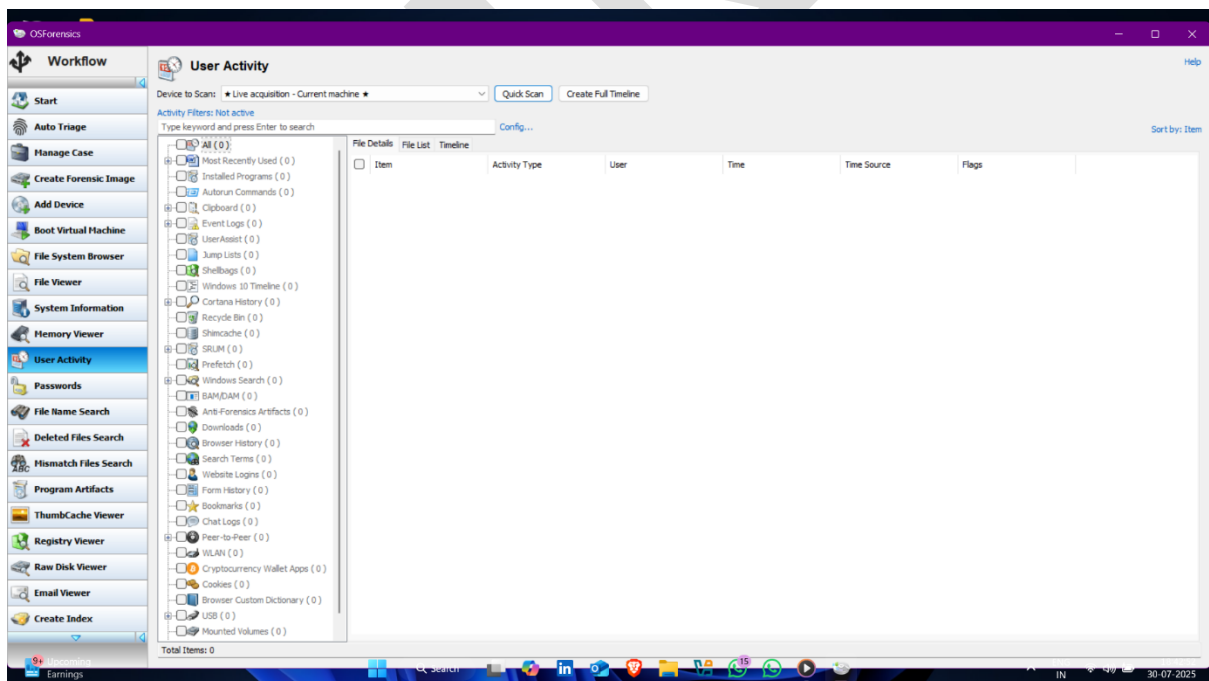
- **Click on User Activity**



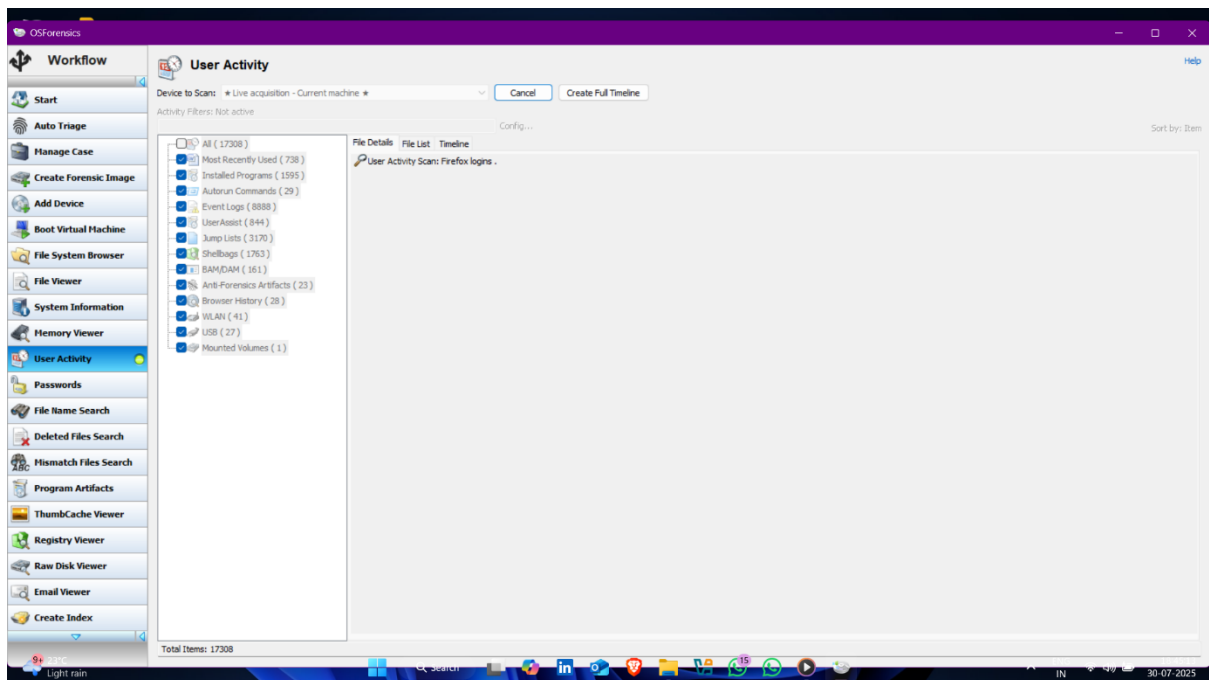
- **Select Live acquisition -current machine**



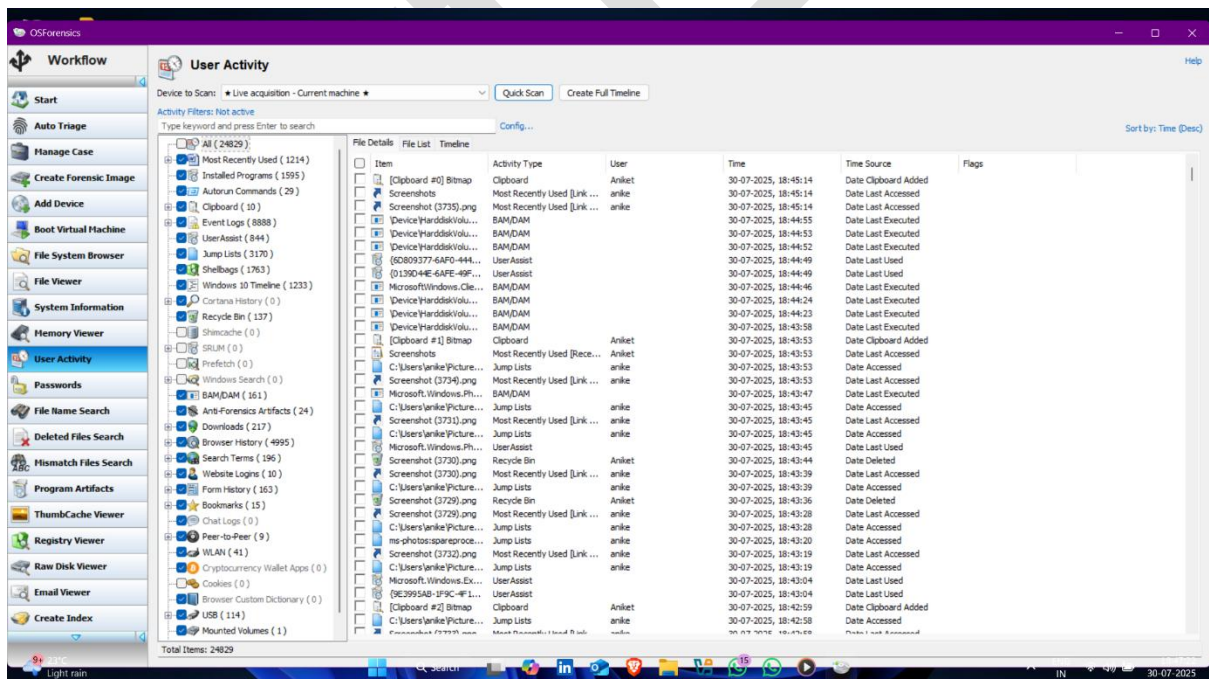
- **Click on Quick Scan**



- Analysis Started



- Retrieve all computer data



• Browser History

The screenshot shows the OSForensics application interface. The 'User Activity' tab is selected, displaying a list of browser history items. The left sidebar shows various workflow options, and the main pane displays a table of activity data.

File Name	File List	Timeline	Config...
File Details	File List	Timeline	Config...
Title	URL	Date Last Accessed	Visit Count
Internet Explorer <td>Username</td> <td>Profile</td> <td></td>	Username	Profile	
https://www.softpedia.com/...	https://www.softpedia.com/...	30-07-2025, 18:33:16	7045
https://mandiant.redline.com/...	https://mandiant.redline.com/...	30-07-2025, 18:27:26	19
https://learn.microsoft.com/...	https://learn.microsoft.com/...	30-07-2025, 18:27:19	43
https://www.openwall.com/...	https://www.openwall.com/...	30-07-2025, 18:26:21	44
https://www.r-studio.com/...	https://www.r-studio.com/...	30-07-2025, 18:20:54	7040
https://www.runtime.org/...	https://www.runtime.org/...	30-07-2025, 18:16:41	4625
https://github.com/504en/...	https://github.com/504en/...	30-07-2025, 18:16:37	4625
https://www.softpedia.com/...	https://www.softpedia.com/...	30-07-2025, 18:15:48	22
https://winhex.en.softonl...	https://winhex.en.softonl...	30-07-2025, 18:15:47	21
https://www.autopsy.co...	https://www.autopsy.co...	30-07-2025, 18:15:47	2
https://www.exterro.com/...	https://www.exterro.com/...	30-07-2025, 18:15:47	1
https://www.easeus.com/...	https://www.easeus.com/...	30-07-2025, 18:15:47	42
https://sourceforge.net/bp...	https://sourceforge.net/bp...	30-07-2025, 18:15:47	4624
https://jcypted.software...	https://jcypted.software...	30-07-2025, 18:15:47	4624
https://www.sensiguard.c...	https://www.sensiguard.c...	30-07-2025, 18:15:47	4648
https://jascrypt.net/downl...	https://jascrypt.net/downl...	30-07-2025, 18:15:46	7034
http://testfire.net/	http://testfire.net/	30-07-2025, 18:15:31	6013
https://github.com/beefpr...	https://github.com/beefpr...	30-07-2025, 18:15:31	6005
https://github.com/Securi...	https://github.com/Securi...	30-07-2025, 18:15:31	4624
https://www.compulec.ch/...	https://www.compulec.ch/...	30-07-2025, 18:15:31	4624
https://www.kfsensor.net/	https://www.kfsensor.net/	30-07-2025, 18:15:31	4648
https://honeybot.softwar...	https://honeybot.softwar...	30-07-2025, 18:15:31	4624
https://www.snort.org/do...	https://www.snort.org/do...	30-07-2025, 18:15:31	4624
https://youtu.be/6U6Wp...	https://youtu.be/6U6Wp...	30-07-2025, 18:15:31	4648
https://github.com/halDe...	https://github.com/halDe...	30-07-2025, 18:15:31	4648
http://go.microsoft.com/f...	http://go.microsoft.com/f...	30-07-2025, 18:15:31	4624
http://go.microsoft.com/f...	http://go.microsoft.com/f...	30-07-2025, 18:15:31	4624
http://go.microsoft.com/f...	http://go.microsoft.com/f...	30-07-2025, 18:15:31	4624

The screenshot shows the OSForensics application interface. The 'User Activity' tab is selected, displaying a list of system events. The left sidebar shows various workflow options, and the main pane displays a table of activity data.

Item	Event Channel	Event Time	Event ID	Event Record ID	User ID (Linux)	Event Information
Service Installed	System	30-07-2025, 18:33:16	7045	151885		AccessData Driver 10, Ser...
Windows Update Succ...	System	30-07-2025, 18:27:26	19	151885		Security Intelligence Upda...
Windows Started Inst...	System	30-07-2025, 18:27:19	43	151884		Security Intelligence Upda...
Windows Started Dow...	System	30-07-2025, 18:26:21	44	151883		Security Intelligence Upda...
Service Start Type Ch...	System	30-07-2025, 18:20:54	7040	151880		Background Intelligent Tra...
Failed Logon Attempts	Security	30-07-2025, 18:16:41	4625	17068343		Logon Type: 3 (Network)...
Failed Logon Attempts	Security	30-07-2025, 18:16:37	4625	17068317		Logon Type: 3 (Network)...
Remote Desktop Servi...	Microsoft-Windows-Termi...	30-07-2025, 18:15:48	22	268220		User: HP\Aniket, Session 1...
Remote Desktop Servi...	Microsoft-Windows-Termi...	30-07-2025, 18:15:47	21	268219		User: HP\Aniket, Session 1...
Finished Processing Us...	Microsoft-Windows-User P...	30-07-2025, 18:15:47	2	4731		Session: 1
Received User Logon ...	Microsoft-Windows-User P...	30-07-2025, 18:15:47	1	4727		Session: 1
End Session Arbitration	Microsoft-Windows-Termi...	30-07-2025, 18:15:47	42	268218		User: HP\Aniket, Session 1...
Successful Logon	Security	30-07-2025, 18:15:47	4624	17066990		Logon Type: 2 (Interactiv...
Successful Logon	Security	30-07-2025, 18:15:47	4624	17066989		Logon Type: 2 (Interactiv...
Logon Attempted Unsu...	Security	30-07-2025, 18:15:47	4648	17066988		Account Name: HP\$, Acco...
Service Terminated Un...	System	30-07-2025, 18:15:46	7034	151854		VMware NAT Service serv...
System Update	System	30-07-2025, 18:15:31	6013	151720		11 seconds, ~330 Indx St...
Event Log Service Sta...	System	30-07-2025, 18:15:31	6005	151719		Logon Type: 2 (Interactiv...
Successful Logon	Security	30-07-2025, 18:15:31	4624	17065899		Logon Type: 2 (Interactiv...
Successful Logon	Security	30-07-2025, 18:15:31	4624	17065898		Account Name: HP\$, Acco...
Logon Attempted Unsu...	Security	30-07-2025, 18:15:31	4648	17065897		Logon Type: 2 (Interactiv...
Successful Logon	Security	30-07-2025, 18:15:31	4624	17065875		Logon Type: 2 (Interactiv...
Logon Attempted Unsu...	Security	30-07-2025, 18:15:31	4648	17065874		Account Name: HP\$, Acco...
Logon Attempted Unsu...	Security	30-07-2025, 18:15:31	4648	17065873		Account Name: HP\$, Acco...
Successful Logon	Security	30-07-2025, 18:15:31	4624	17065857		Logon Type: 0 (System)...
Windows Update Succ...	System	30-07-2025, 18:15:30	19	151784		
Service Status Chang...	System	30-07-2025, 18:15:25	7036	151761		
Service Status Chang...	System	30-07-2025, 18:15:25	7036	151762		
Device Connected (Dis...	Microsoft-Windows-Parti...	30-07-2025, 18:15:21	1006	806		Connected Ntme device ?...
Operating System Sta...	System	30-07-2025, 18:15:20	12	151725		Start time 30-07-2025, 18...
Operating System Sta...	System	30-07-2025, 18:15:13	13	151724		Stop time 30-07-2025, 16...
Event Log Service Sta...	System	30-07-2025, 16:47:56	6006	151712		

Lab-8 Extract Information About Loaded Process On A Computer

It refers to **gathering detailed data about all currently running processes** (also called **active or loaded processes**) on a system — like what programs are running in memory, their associated files, parent-child relationships, and more. This is critical in **digital forensics, malware analysis, and incident response**.

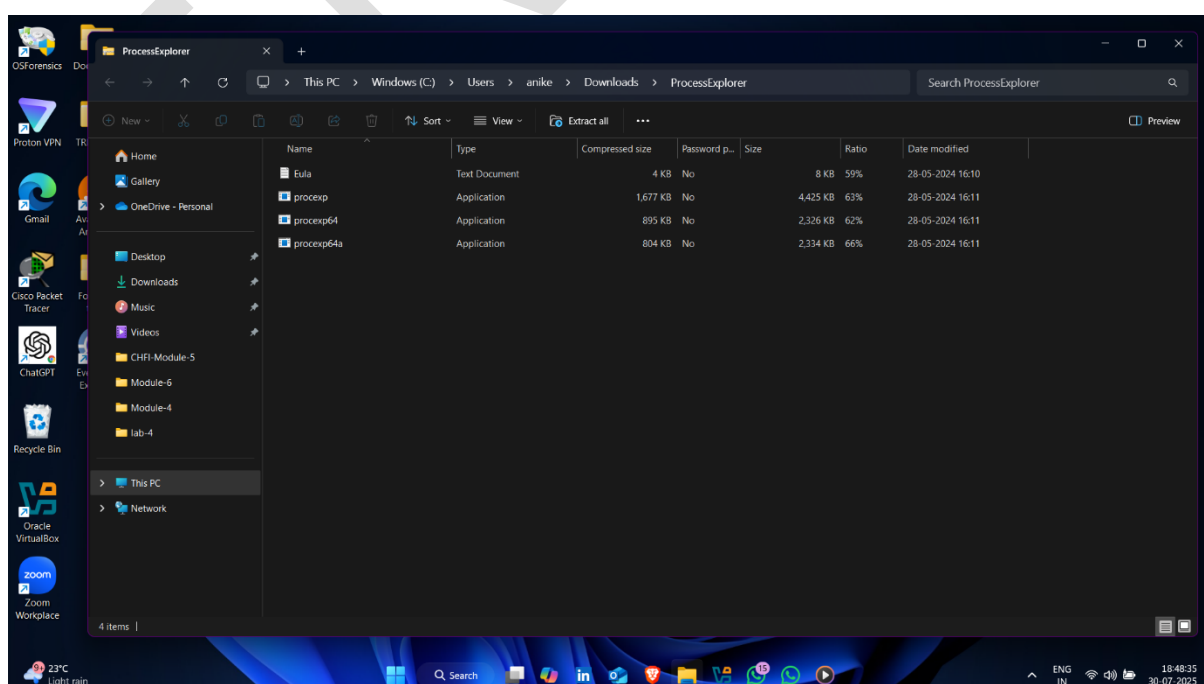
A) Extract Information About Loaded Process On A Computer Using ProcessExplorer Tool

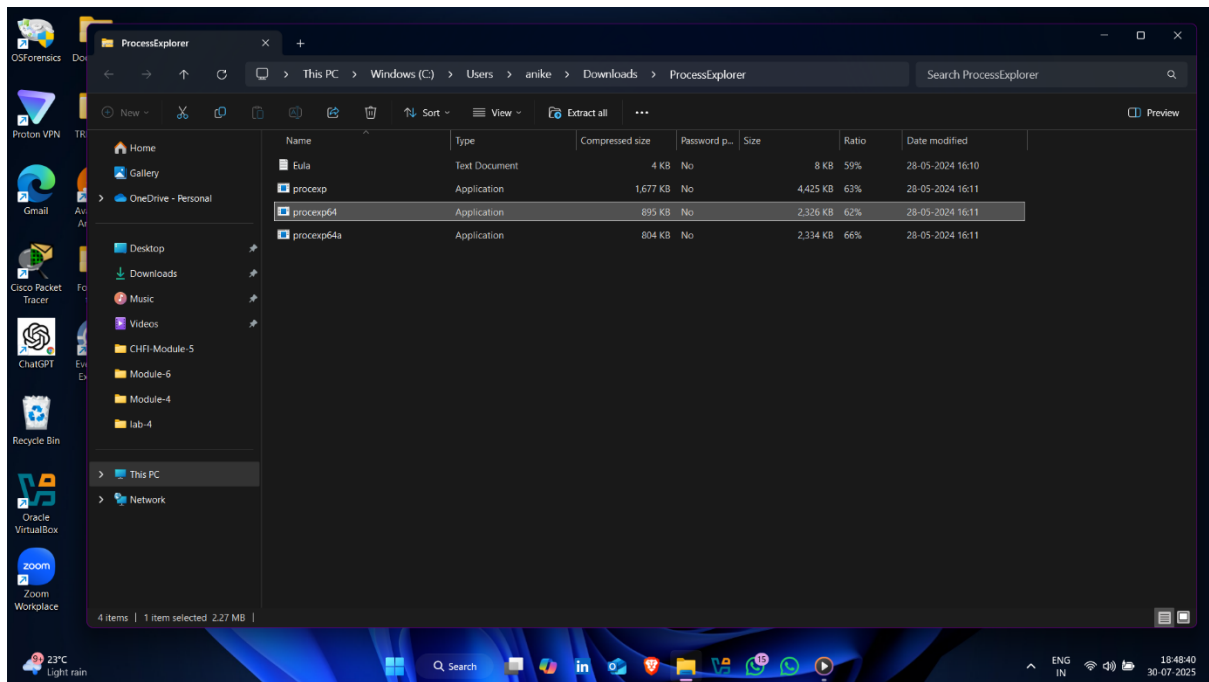
Process Explorer is an advanced system monitoring tool for **Windows**, developed by **Sysinternals (now part of Microsoft)**. It provides **detailed real-time information** about all running processes on a system — much more than Task Manager.

It is widely used in **digital forensics, malware analysis, reverse engineering, and system troubleshooting**.

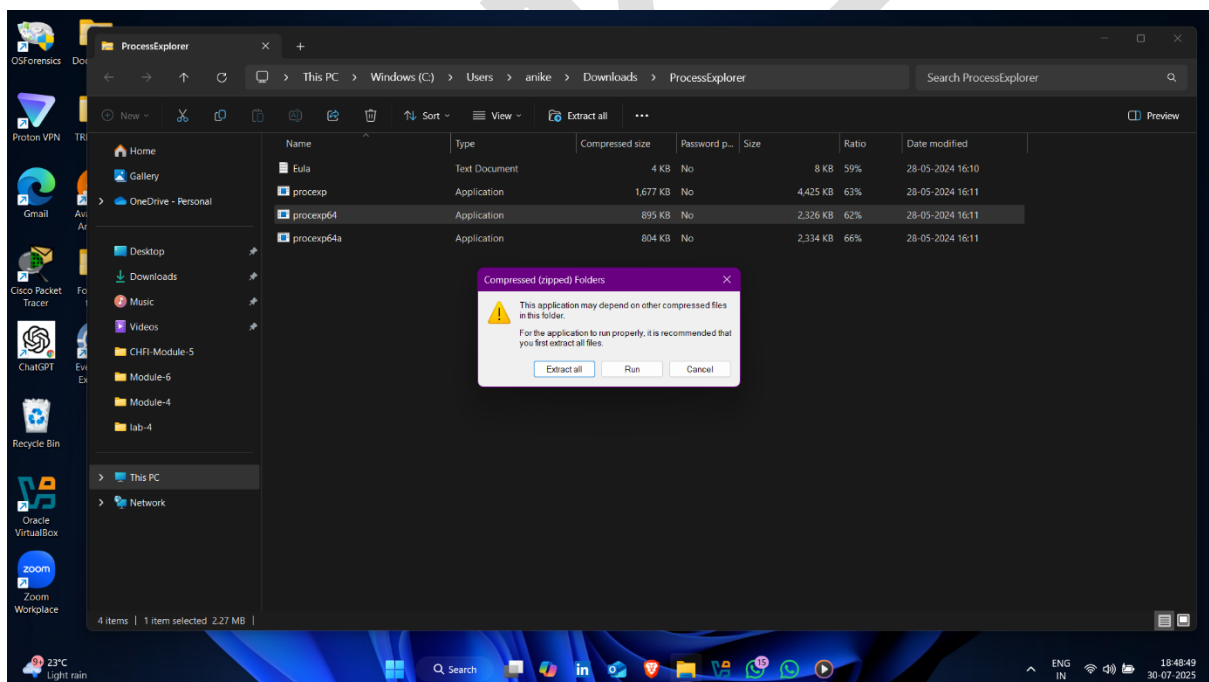
How to use it -:

- Run **procexp64.exe** File

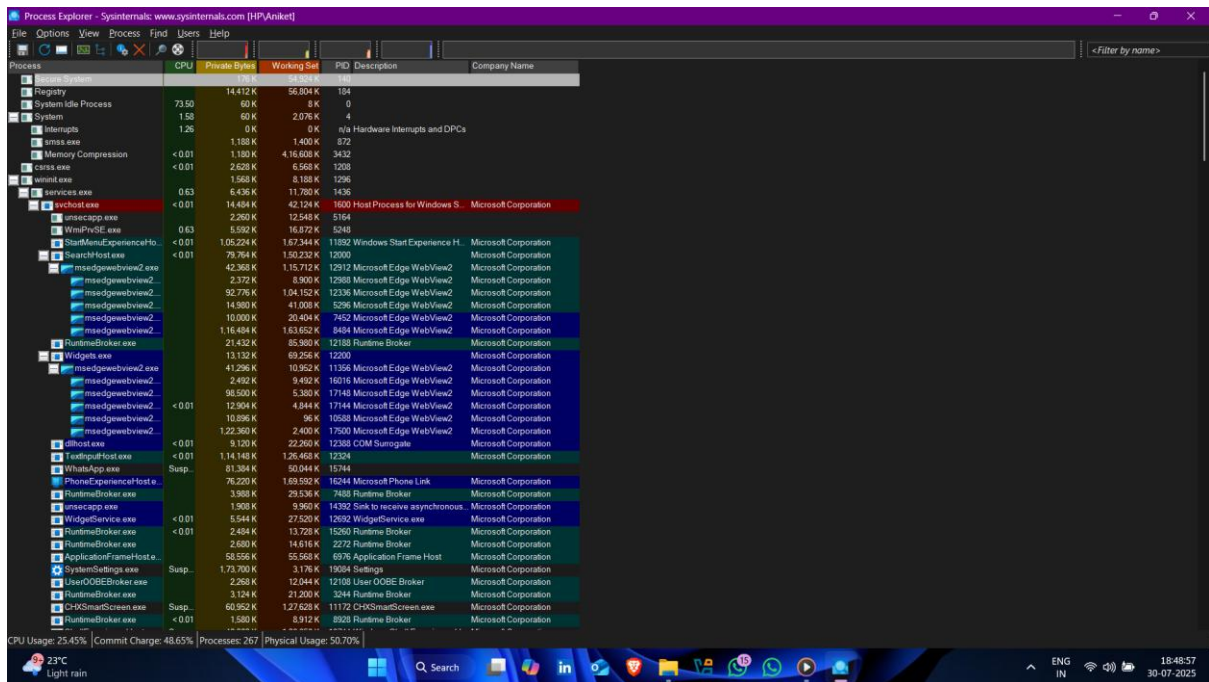




- Click on **Run** option



- It shows all process running on computer



Lab-9 View , Monitor And Analyze Events Occurred On A Winodws Machines

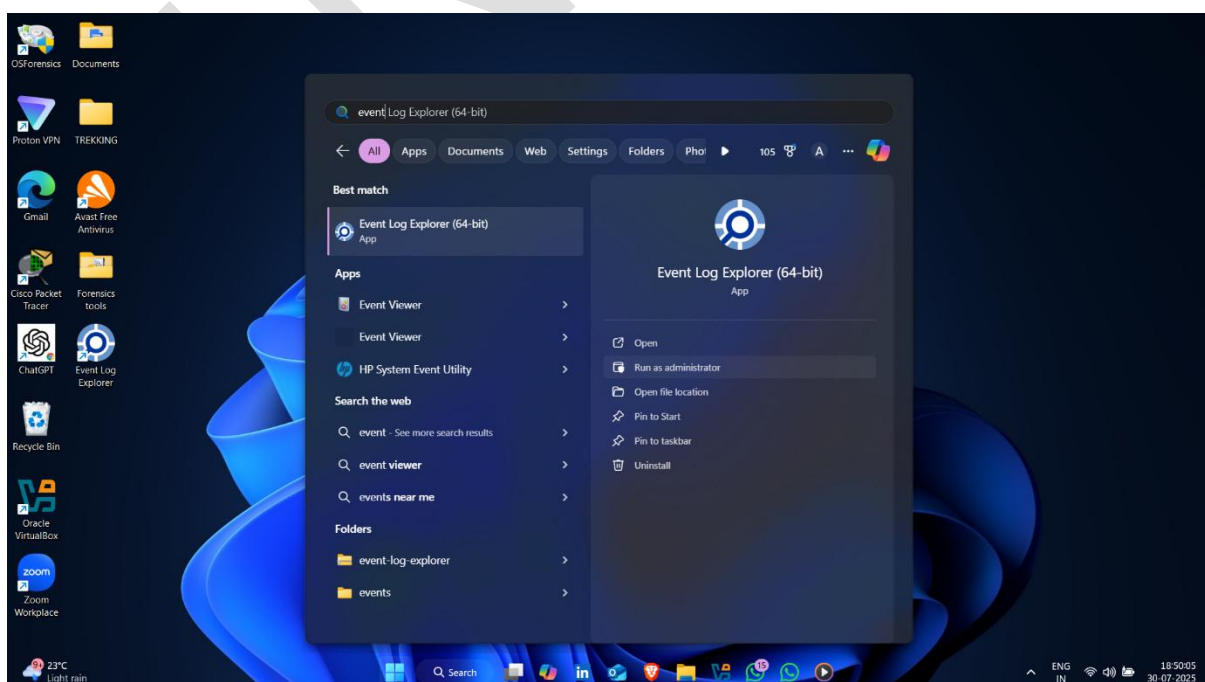
It means **collecting and examining event logs** that Windows automatically generates when users, software, or the system itself perform actions — like **logins, file accesses, software installation, errors, shutdowns, etc.** This is crucial in **cybersecurity, digital forensics, incident response, and troubleshooting.**

A) View , Monitor And Analyze Events Occurred On A Winodws Machines Using Event Explorer Tool

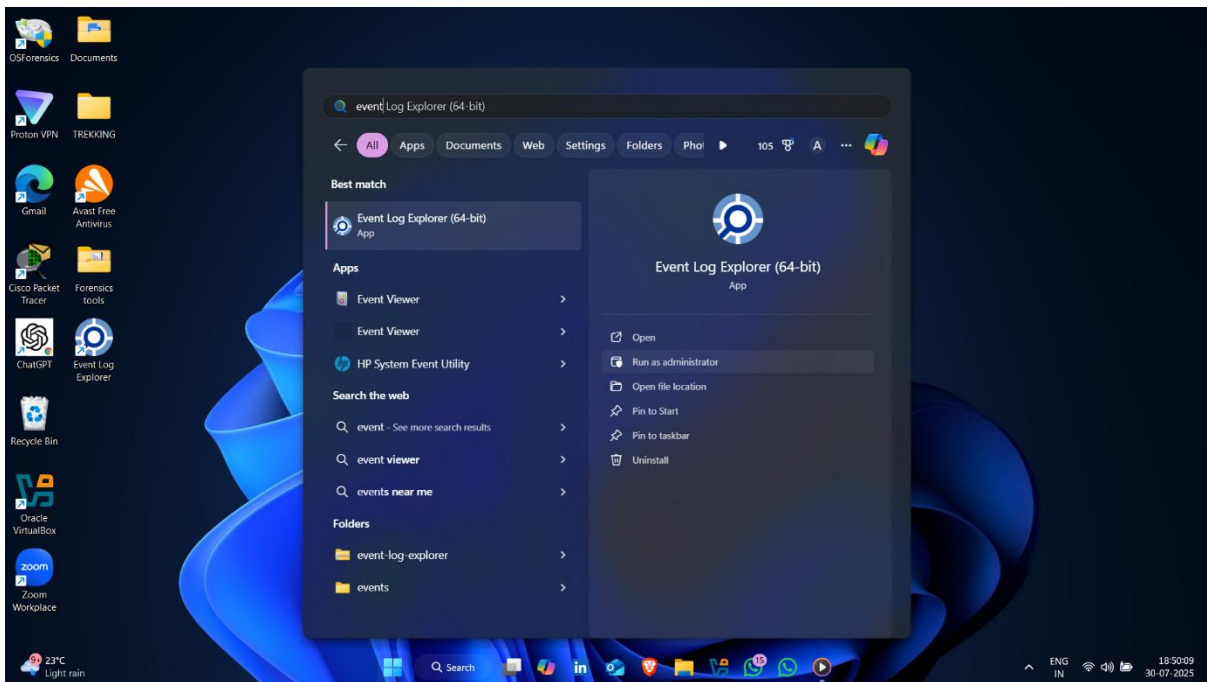
Event Explorer is a **Windows-based GUI tool** designed to help users and investigators **browse, search, and analyze Windows Event Logs** more easily and efficiently than the built-in Event Viewer. It is **not a default Windows tool** like Event Viewer, but a **third-party or enhanced forensic tool** used in **digital forensics, system auditing, and incident response.**

How to use it -:

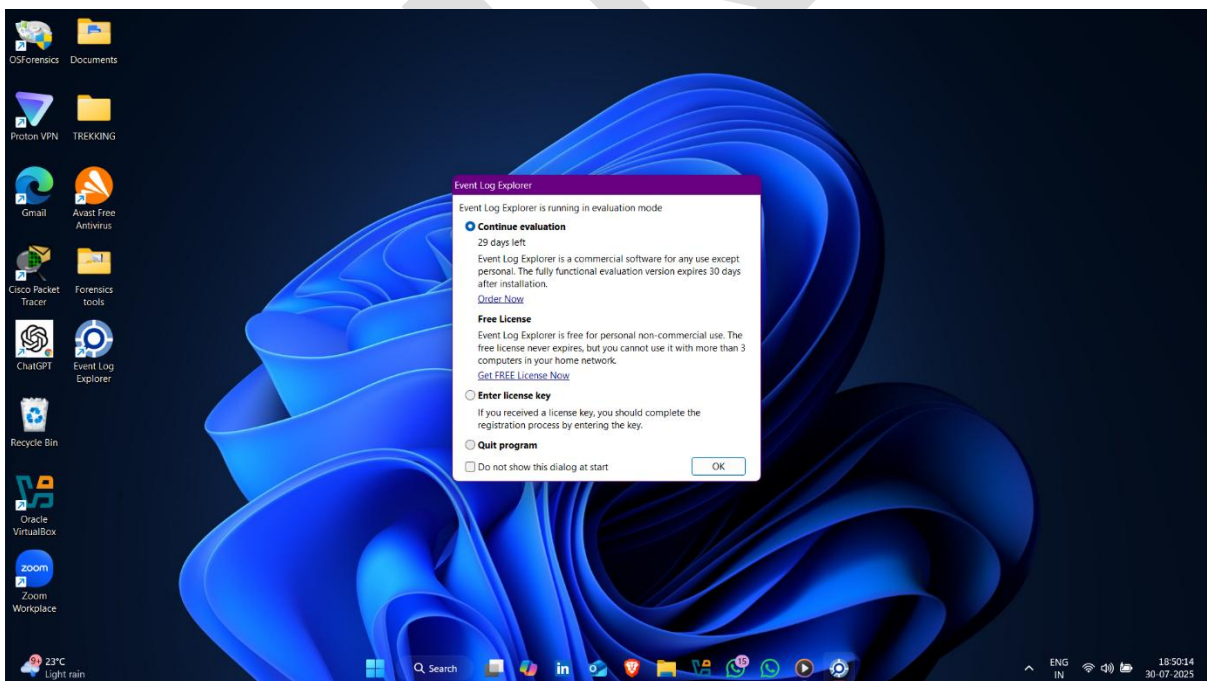
- Event Log Explorer Tool 🖱️

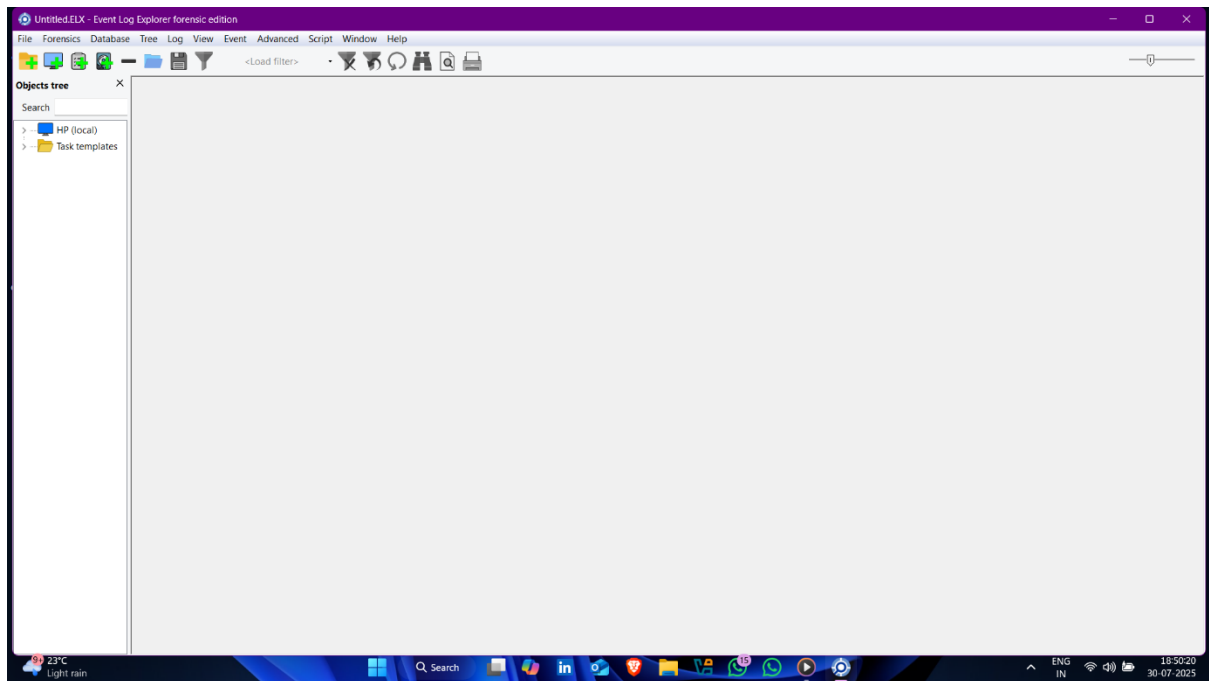


- Click on **Run As Administrator**

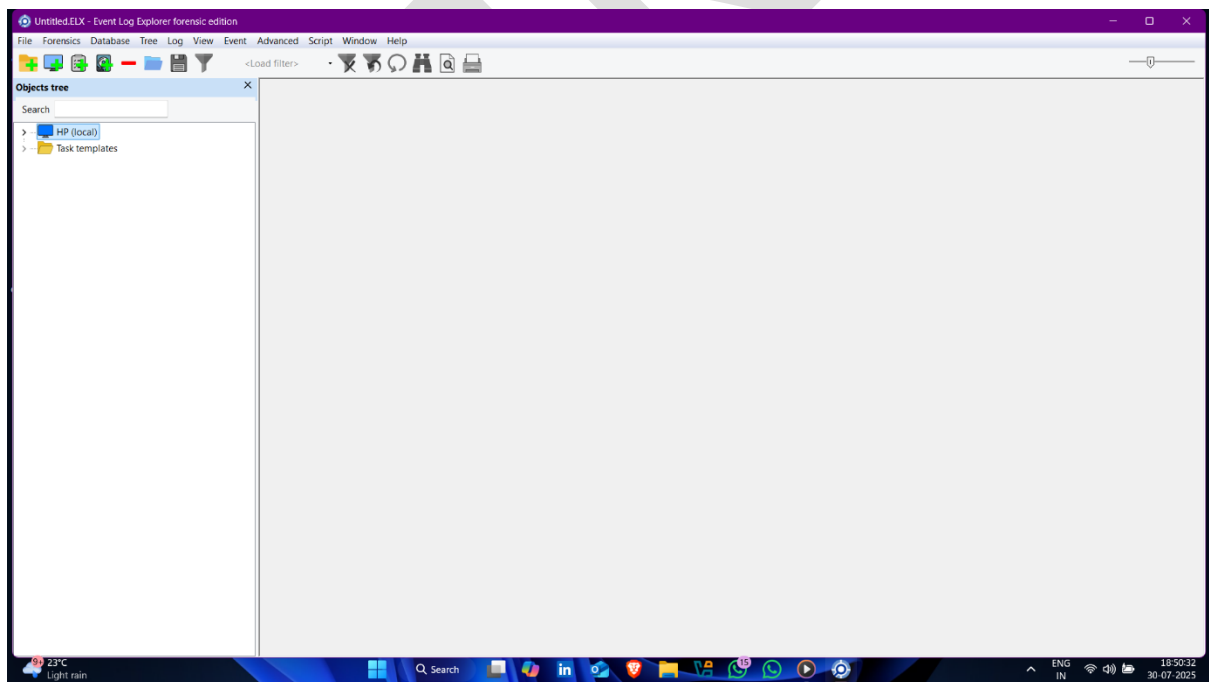


- Select First Option and click on ok

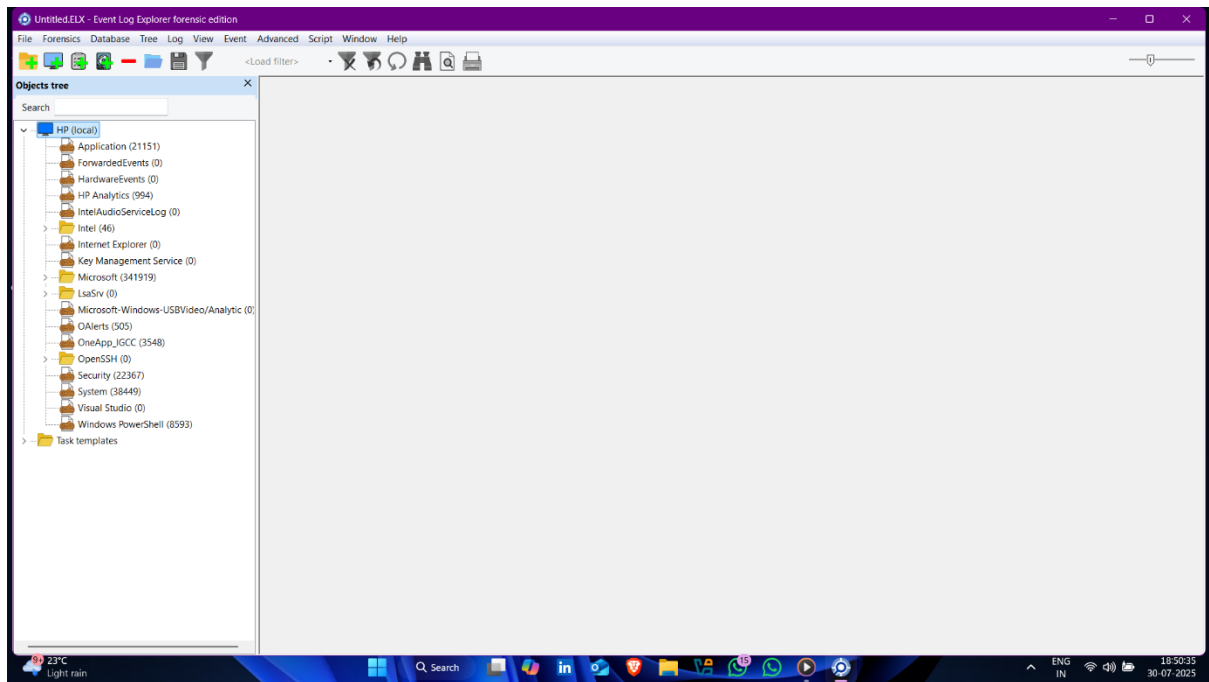




- Click on Hp Device



- click on Application Sections



- collect Logs

